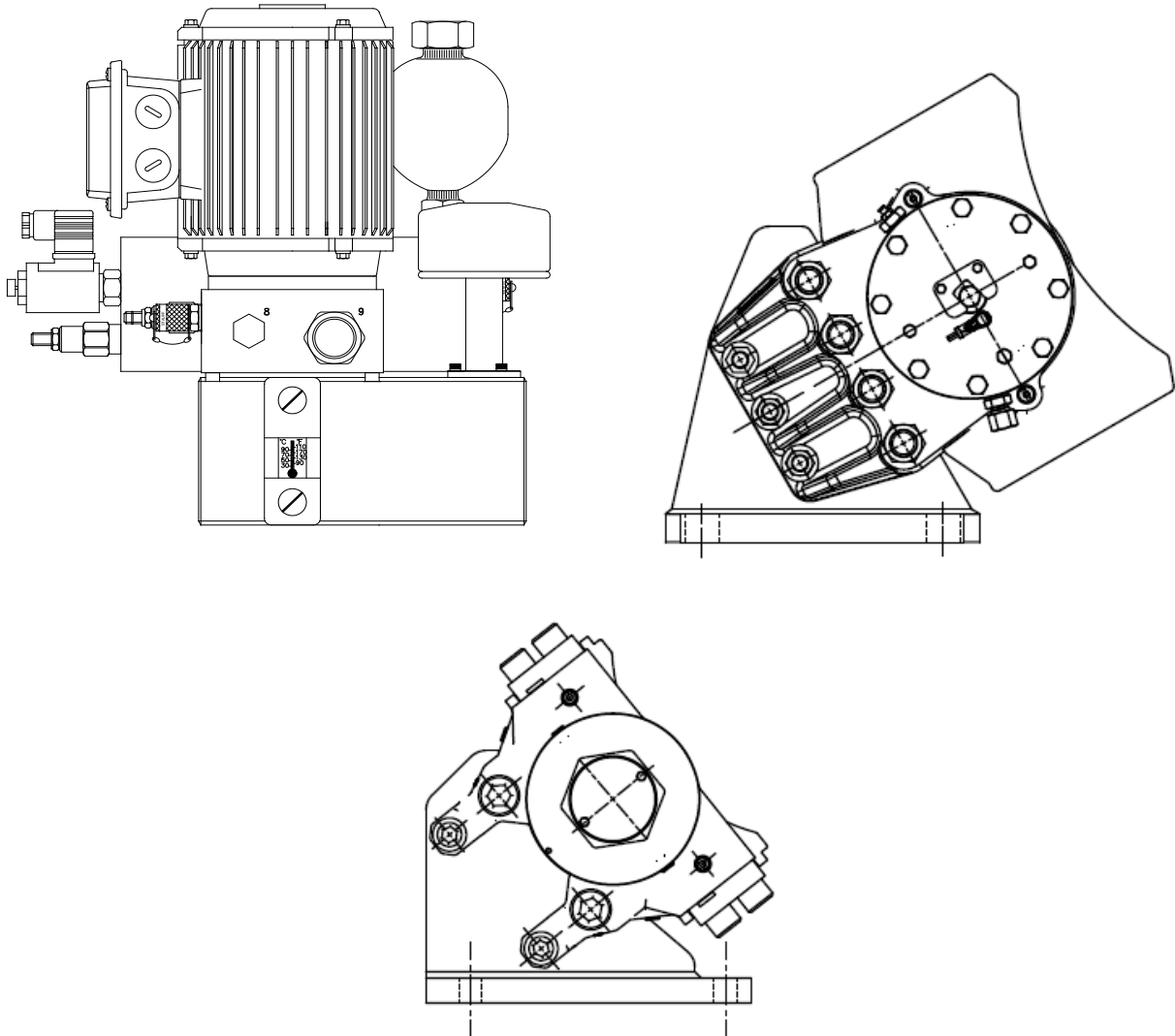


BRAKE SYSTEM MANUAL



PT Proteknik Mega Persada – Arutmin Mine
Conveyor Drives: OLC-1, OLC-1A & OLC-2

Supplied to: PT Proteknik Mega Persada

Order No: 902-12/PMP/ES & 904-12/PMP/ES

KIP Contract No K22-12-04
Written by: M. Boustead, Revision 0, 6/12/12

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2. LUBRICATION SCHEDULE
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7. COMMISIONING REPORT FORM
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9. DRAWINGS AND SETTINGS
10. NOTES

1.SYSTEM DESIGN PARAMETERS

Conveyor: OCL-1

<u>Scope of Supply</u>	
Brake caliper assemblies	1 x VKSD-VR119-KS-40
Manual:	M1596
Brake Control unit	1 x BCU-I75-02-24-567
Manual:	MAU-075-001
<u>Design Parameters (per drive)</u>	
Dynamic torque required	60,000 Nm at brake disc
Stopping time	32 seconds
Disc speed start	200 rpm (estimate)
Disc diameter	1,200 mm
Disc thickness	40 mm
Friction co-efficient	$\mu=0.40$ nominal (Mild Steel grade 350, or Bisalloy 80 disc)
Torque provided	60,095 Nm @ 2mm pad to disc air gap
Note: The above torque figures are theoretical only	

Conveyor: OLC-1A

<u>Scope of Supply</u>	
Brake caliper assemblies	1 x VKSD-VR96-KS-40
Manual:	M1596
Brake Control unit	1 x BCU-I75-02-24-567
Manual:	MAU-075-001
<u>Design Parameters (per drive)</u>	
Dynamic torque required	40,000 Nm at brake disc
Stopping time	55 seconds
Disc speed start	200 rpm (estimate)
Disc diameter	1,200 mm
Disc thickness	40 mm
Friction co-efficient	$\mu=0.40$ nominal (Mild Steel grade 350, or Bisalloy 80 disc)
Torque provided	48,480 Nm @ 2mm pad to disc air gap
Note: The above torque figures are theoretical only	

Conveyor: OLC-2

<u>Scope of Supply</u>	
Brake caliper assemblies	1 x VCSD-VR62-KS-40
Manual:	M1591
Brake Control unit	1 x BCU-I75-02-24-567
Manual:	MAU-075-001
<u>Design Parameters (per drive)</u>	
Dynamic torque required	20,000 Nm at brake disc
Stopping time	32 seconds
Disc speed start	200 rpm (estimate)
Disc diameter	1,200 mm
Disc thickness	40 mm
Friction co-efficient	$\mu=0.40$ nominal (Mild Steel grade 350, or Bisalloy 80 disc)
Torque provided	33,027 Nm @ 2mm pad to disc air gap
Note: The above torque figures are theoretical only	

2.LUBRICATION SCHEDULE

Application / Conveyor	Unit type	Oil Grade	Oil Volume Nominal litres
ALL	BCU-I75-02-24-567	ISOVG 68	5.0

Note:

- a) Oil Grades - ISOVG 68 hydraulic oil, for a temperate to hot climate
- b) Oil quantities are stated above but are to be checked for correct fill level by sight glass in the tank or reservoir.
- c) Greater oil volume will be required for bleeding.
- d) All oil must be filtered to a 10-micron nominal filter rating when filling.
- e) For more detail on oils see brake control unit manual, section 2.8.

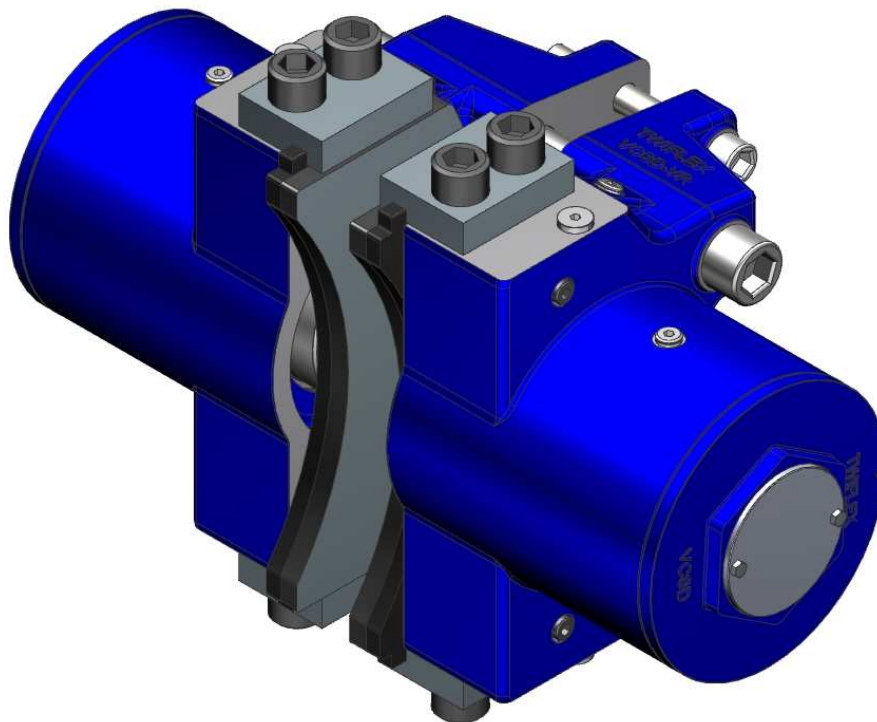
3. MANUAL FOR BRAKE CALIPER VCSD-VR

Caliper Type: VCSD-VRxxx-xx-xx

Manual: M1591

This manual covers the complete range of VCSD-VR brake calipers

**INSTALLATION, OPERATION
AND MAINTENANCE MANUAL
FOR VCSD-VR DISC
BRAKE CALIPER
M1591**



Registered Office, Amphill Road, Bedford MK42 9RD. Registered in England, Number 00404531

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Matrix International • Twiflex Limited • Bibby Transmissions • Huco Dynatork • Warner Linear

Amendment and Revision Record – M1591

Issue	Date	Amendment	Issued by
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Approved By	A. Robins
Date	Nov 2012
Prepared By	D. Roguski
Date	Nov 2012

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1. SPECIFICATION

1.1 General Description

The VCSD-VR disc brake caliper is a spring applied, hydraulically retracted unit which may be used on any disc having a diameter of at least 0.6m and a thickness of 20mm or greater. There is no upper limit on diameter or thickness. The caliper consists of 2 spring modules bolted to a mounting plate.

The VCSD-VR caliper has a special operational feature called the 'Parked-Off' position. With the brake in the "parked off" position, disc spring load and hydraulic pressure are zero, allowing pads to be removed in safety. This feature also permits brake installation without the need for hydraulic pressure.

The braking force is applied by the disc spring packs (Item 2) through a piston (Item 37) and adjusting screw (Item 39) to the friction pad (Item 28). The pads are guided between pad retaining bars (Item 44), to which the torque reaction forces are applied. For available ratings see Table 1.

VCSD Model	Air Gap [mm]	Hydraulic pressure [bar]		Clamping Force	Braking Force *	Brake pad surface pressure (N/mm ²) - VR
		Full Retraction	Brake Release	[kN]	[kN]	
60	1	138	131	80.0	64.0	2.76
50		100	97	59.1	47.3	2.04
40		88	84	51.3	41.1	1.77
30		65	62	37.9	30.3	1.31
25		56	50	30.4	24.3	1.05
20		49	41	24.8	19.8	0.85
15		35	34	20.5	16.4	0.71
10		24	23	14.1	11.3	0.49
60	2	138	124	75.7	60.6	2.61
50		100	94	57.2	45.7	1.97
40		88	80	49.0	39.2	1.69
30		65	59	36.2	28.9	1.25
25		56	52	31.7	25.4	1.09
20		49	44	26.5	21.2	0.92
15		35	32	19.6	15.7	0.68
10		24	22	13.6	10.9	0.47
60	3	138	117	71.3	57.0	2.46
50		100	90	55.1	44.1	1.90
40		88	76	46.5	37.2	1.60
30		65	56	34.4	27.5	1.19
25		56	57	34.8	27.8	1.20
20		49	46	28.2	22.6	0.97
15		35	31	18.7	15.0	0.65
10		24	21	13.1	10.5	0.45

Table 1. VCSD-VR Performance Data

* Braking force values based on 0.4 of coefficient of friction assuming fully bedded pads and frictional losses.

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Brake adjustment is manual and is effected by turning adjusting bar (Item 38).
Prior to adjustment spindle locking plate (Item 36) must be removed (see p.2.3).
The nominal Air Gap for each brake is 2mm.

Maintenance of the correct Air Gap ensures that braking force is effectively constant and minimizes the stress range for the springs leading to an extended fatigue life.

The Loss of braking force for 1mm pad wear is between 3 and 6% (depending on rate).

It is recommended to adjust the Air Gap after every 0.5 to 1mm of Pad Wear (each side).
Minimum adjustment to be 1/8 of a turn to give approx 0.375mm.
Minimum fatigue life of the springs is 2×10^6 operations.

Failure of an individual spring is rare and would result in a proportional loss of braking torque in the affected caliper half only.

Two G1/4 tapped ports are provided in the housing for a bleed screw and hydraulic feed connection (Items 26, 32).

Four G1/8 tapped ports are provided in the housing as drain ports, for piping away any hydraulic leakage from the braking surfaces.

The drain ports are supplied plugged for transit and storage (Item 33).

Seals are designed to be leak free but by using clear plastic flexible tubing connected to a small bottle and fitted to the drain ports any leakage can be observed.

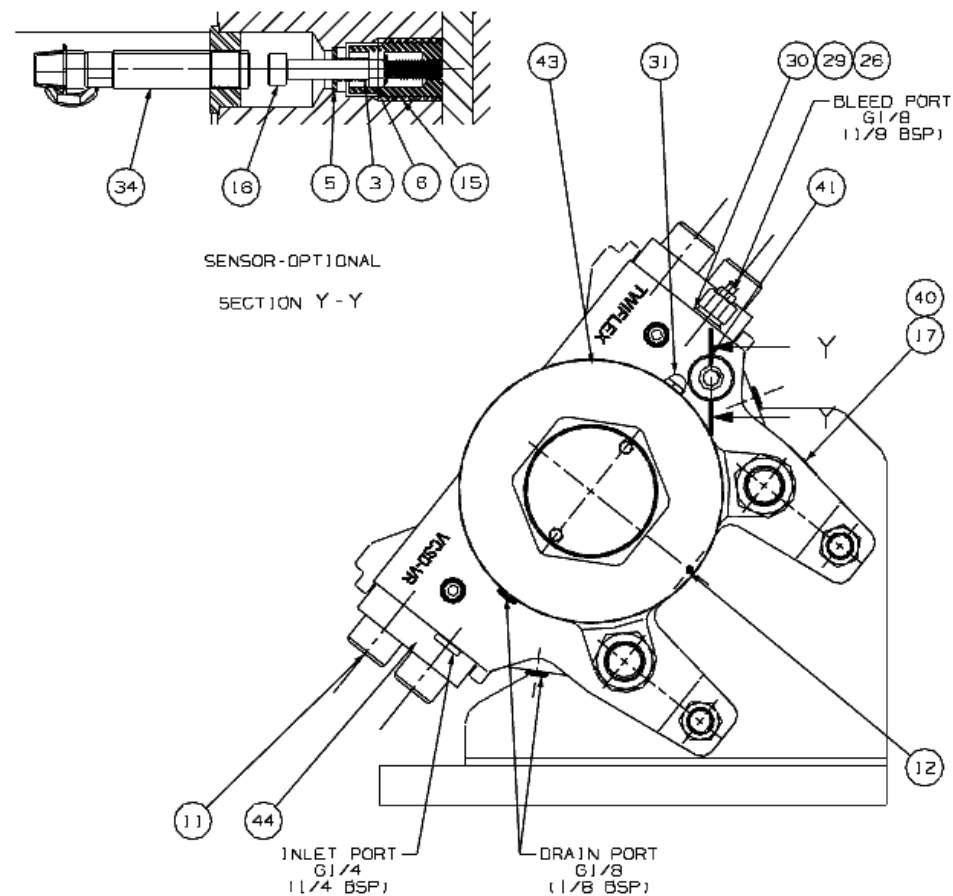


fig 1. VCSD-VR

Item	Description	Qty
1	COMPRESSION SPRING,	4
2	DISC SPRING	20
3	SPRING, COMPRESSION	2
4	SPRING PIN,	2
5	WASHER M6, FORM C	2
6	NUT, NYLOC, M6, TYPE T	2
7	NUT,FULL,M16	2
8	NUT, FULL, M24	2
9	SET SCREW, H/HD,M5	4
10	SCR, SOCKET CAP, M8	4
11	SCR, SOCKET CAP, M20	8
12	SCR, GRUB, M5x8, CUP POINT	2
13	SCR,SOCKET CAP, M16	2
14	SCR, SOCKET CAP, M24	2
15	PLUG, SPECIAL	2
16	SCR, SOCKET CAP	2
17	SCREW, No2 X 1/4" HAMMER DRIVE	4
18	ROD SEAL	2
19	ROD SEAL	2
20	SCRAPER	2
21	WIPER	2
22	SLYDRING	2
23	SLYDRING	2
24	O-RING	2

Item	Description	Qty
25	O-RING	2
26	BLEEDSCREW, 1/8"BSP	2
27	PEDESTAL (Optional)	1
28	PAD ASSEMBLY	2
29	ADAPTOR	2
30	BONDED SEAL, 1/4 BSP	2
31	SILENCER	2
32	BLANKING PLUG,1/4BSP	2
33	1/8 BSP PLUG	6
34	PROXIMITY SWITCH (Optional)	2
35	PLUG CONNECTOR (Optional)	2
36	SPINDLE LOCKING CAP	2
37	PISTON	2
38	ADJUSTING BAR	2
39	ADJUSTING SCREW	2
40	LABEL	2
41	BUSH	2
42	HOUSING	2
43	END COVER	2
44	PAD RETAINING BAR	4
45	SPRING GUIDE	2
46	SHIM/SPACER	2
47	RED CAP TAPER 1/4"	2

Table 2. Items List

*** Drawings /Item lists are particular to each part number and therefore available on request.**

1.2 Detail Specification

1.2.1 Braking Force & Torque

The braking force 'F' (kN) acting tangentially on the disc is given by:

$F = 2 \times \mu \times S$ where μ is the pad friction coefficient
S is the spring pack thrust (kN)

The braking torque 'T' (kNm) is given by:

$T = F (R - 0.055)$ where R is the outside radius of the disc (m).

Design CoF $\mu = 0.4$ has been assumed (standard pad material when used under clean, dynamic conditions with disc of correct material and finish) with pads fully bedded in.
For other conditions a service factor should be used.

1.2.2 Retraction Cylinders

Piston Outer Diameter	= 140 mm
Piston Inner Diameter	= 105mm
Effective Area	= 6734.8mm ²

For Hydraulic pressures (brake release and full retraction) see Table 1.

Hydraulic fluid: Only mineral oil based fluids must be used, e.g. Shell 'Tellus' 37'.
Water based fluids would require the caliper to have a high degree of corrosion protection.

Bleeding: G1/8 bleed screw is fitted to each cylinder.

Fluid displacement: Total oil displacement to give 1mm retraction = 6.8 ml (per/module)

1.2.3 Friction Pads

Material: Asbestos free bonded to backing plate.

Pad thickness	= 22 mm
Max Pad Wear allowance (each)	= 10 mm
Total pad area (two pads)	= 57980 mm ²

Replacement: see section 3.4

1.2.4 Materials & Quality Assurance

The housing and end cover castings are made from S.G. iron to BS2789 grade 450-10. Calipers may be supplied with non-destructive test (N.D.T.) certificate for all critical components at additional cost. The main hydraulic seals are polyurethane. O-rings are nitrile rubber. Each caliper is fully tested after assembly and issued with a test certificate.

1.2.5 Safety Features

The VCSD-VR caliper has a special operational feature called the 'Parked-Off' position. This is made possible by having a piston stroke long enough to permit the disc springs to fully expand.

No internal spring loads exist in this condition.

- a) The caliper can be installed without the need for hydraulic pressure.
- b) The brake pads can be changed with complete safety with the hydraulics at zero pressure.
- c) The spring module, including the spring pack, can be safely and easily disassembled and reassembled.

1.2.6 How to achieve the 'Parked-Off' position

- a) With the brakes on, apply hydraulic pressure to retract the piston (Item 37).
- b) Loosen and remove the spindle locking plate screws (Item 9). Remove the spindle locking plate (Item 36) and turn the adjusting spindle (Item 38) anticlockwise until the pad (Item 28) is retracted from the brake disc as far as the housing (Item 42) allows (pad should touch the housing). This should leave approximately 4mm air gap between the pad face and brake disc.
- c) Turn the adjusting spindle (Item 38) anticlockwise by approximately 1 turn to give additional 3mm clearance between pad and adjusting screw (Item 39).
- d) Release the hydraulic pressure. This results in the piston moving forward to relax the disc spring pack (Item 2).
- e) The brake is now in the 'Parked Off' position. No internal disc spring loads exist in this condition.

2. INSTALLATION AND ADJUSTMENT

VCSD-VR is supplied in "Parked Off" position-no internal spring load in the modules.

Hydraulic fluid: Only mineral oil based fluids must be used.

Hydraulic Inlet Port - 1/4 BSP (G1/4)

Bleed Port - 1/8 BSP (G1/8)

Drain Port - 1/8 BSP (G1/8)

Max Permissible Pressure - 160 bar

For tools that may be required see p. 3.9

2.1 Lifting

TWIFLEX LIMITED DO NOT SUPPLY LIFTING BOLTS

- a) When brake supplied with mounting bracket / pedestal (Item 27) use M12 lifting holes to lift the brake (fig 2). It is recommended to use the shackle when using M12 lifting eye.

Ensure M12 lifting eye is parallel to pedestal face as shown

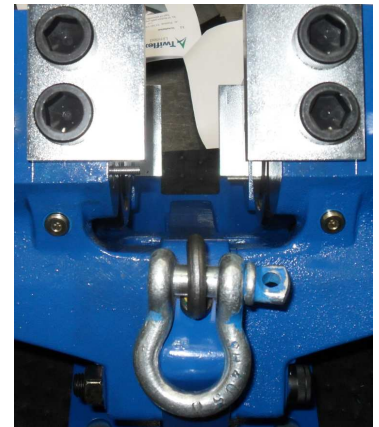


fig 2. M12 Lifting Bolt

- b) When brake supplied without mounting bracket one of M20 screws may be removed and M20 lifting bolt may be used instead (only for lifting/installation purpose).

Next the M20 screw must be replaced and tightened to 400 Nm.

ENSURE ALL M20 SCREWS ARE TIGHTENED TO 400 Nm.



fig 3. M20 Lifting Bolt

2.2 Installation

- Position the VCSD-VR caliper in place (ensure all fixing bolts have been tightened) *.
- If friction pads are supplied separately - slide the friction pads in so retaining bars are between lugs (if necessary top retaining bar should be removed to ease assembly).

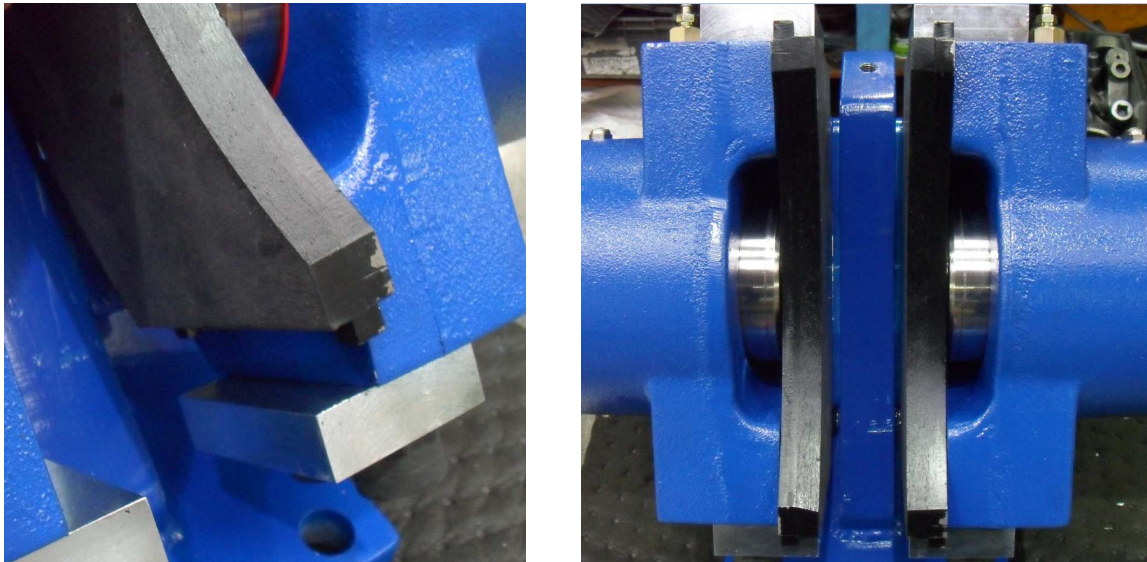


fig 4. Friction Pads-Assembly

- Insert springs and M8 bolt and screw in until it stops (Items 1,10).



fig 5. Pad Retraction Screws

- Remove 2 off 1/4 BSP Plugs from hydraulic inlet ports to connect hydraulic hoses.
- Adjust the brake as per p.2.3

* A suitable method for checking the security of the mounting bolts visually is to mark them with white paint.

* The drain connections (two each side of the spring module) must be piped away to waste, with a transparent flexible tube, any leakage past the seals will be immediately apparent and drained to the tank.

2.3 Adjustment

- a) Remove the spindle locking plate (Item 36) by unscrewing two M5 hex screws (Item 9).

2 off M5 screws can be used to assist in removing the locking plate (fig 6).

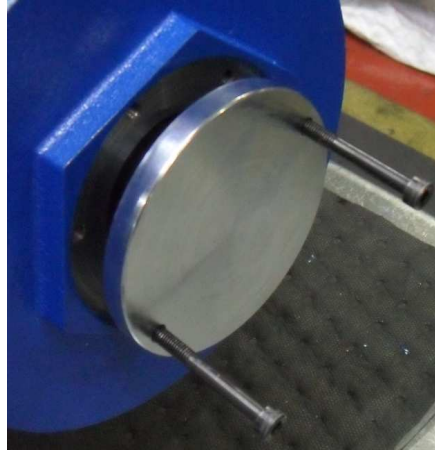


fig 6. Spindle Locking Plate

- b) When fitted with proximity sensors ensure sensors are removed.
c) Apply hydraulic pressure to fully retract the piston.
d) Turn the adjusting bar (Item 38) in a clockwise direction to move the pad forward.
Adjust the air gap accordingly (see table 1).



fig 7. Adjustment-Adjusting Spindle

- e) Replace both o-ring (item 24) and spindle locking plate (item 36).
f) Tighten 2 off M5 screws to 5 Nm.
g) Brake is adjusted for operation.

2.4 Monitoring Units (optional)

VCSD-VR brake may be equipped with different types of sensors that measure brake on/off and/or pad wear. If sensor kit supplied separately follow as specified below:

- Remove friction pad as per p.3.4.
- Screw sensor kit (fig 8) in until flush with the housing (item 42) - apply Loctite 243 on M20 thread.
- Remove M6 nut in a way shown (fig 10).
- Replace friction pads.
- Screw the sensor in and adjust it as per p.2.5.



fig 8. Sensor Kit

M6 Nut

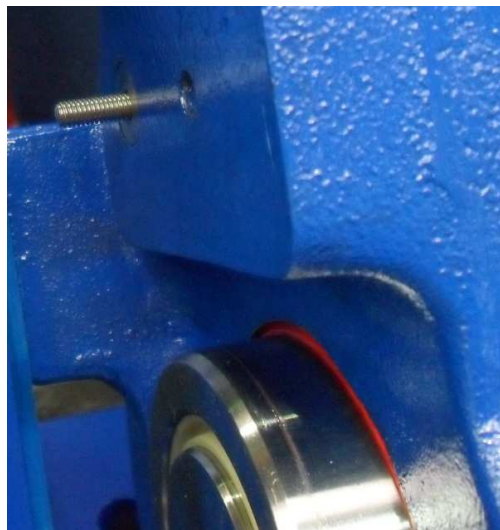


fig 9. Sensor Kit Assembly



fig 10. Sensor Kit Assembly

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2.5 Adjustment for Inductive Analogue Sensor

- Fully retract the brake by applying the full retraction pressure.
- Screw the sensor in until it touches the plunger (Item 16) then move approx 1/2 turn back.
- Adjust required air gap as per Table 1.
- Release the hydraulic pressure to apply the brake.
- When the plunger is in the sensing area the LED comes off (PLC controller to be set to measure the electric current in mA) - see p.3.10 for details.
- Operate the brake a few times to ensure the switch is working properly.

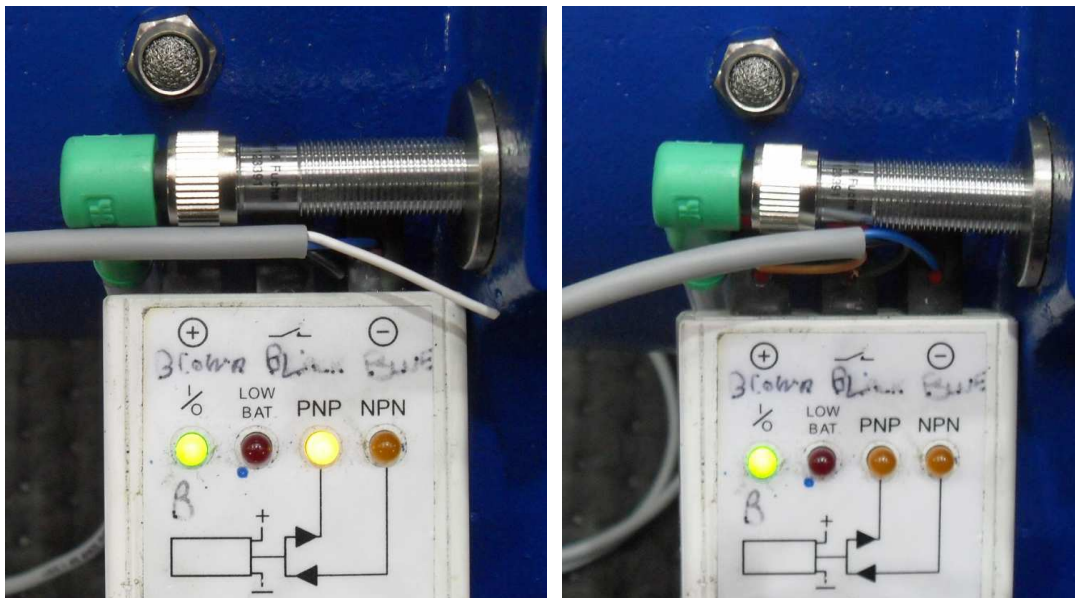


fig 11. Sensor setting- FOR REFERENCE ONLY

Notes:

*Before inserting the sensor use the air gun to blow any debris away.
This exercise should be repeated when running the routine check.
Lubricant (spray) may be applied into the sensor hole.*

3. MAINTENANCE

3.1 Running-In Period

For the first week of full operation make a DAILY CHECK of the following items:-

- a) Hydraulic Fluid Leakage; Note and report any evidence of fluid leakage. If the leakage occurs at pipe connections or at the bleed screws, tighten as appropriate, carefully wipe away any spilled fluid and check for further leaks with brakes off.
If friction pads become fouled with fluid, they must be replaced after curing the leak and the disc(s) wiped clean.
- b) Observe brake operation, including smooth retraction and application of brake pads. Note and report any unusual observations.
- c) Inspect the brake disc braking path for smoothness and freedom from contamination. Clean if necessary with a clean dry cloth. If a solvent is required to remove deposits it must be of a type which dries without residue i.e. acetone
- d) At the end of the running-in period, check the security of all bolted connections and pipe connections.

3.2 Maintenance Inspection Program

After the running-in period, the following schedule should be followed in addition to any statutory requirements.

MAINTENANCE SCHEDULE			
ITEM	WEEKLY	MONTHLY	PERIODIC
a) Check for hydraulic fluid leakage	x (note i)		
b) Observe brake operation	x (note ii)		
c) Clean calipers of accumulated deposits		x	
d) Clean brake paths on disc		x (note iii)	
e) Observe bolt security		x (note iv)	
f) Note and record condition of friction pads, pad movement (stroke). Adjust if required		x (note v)	
h) Check availability and condition of spares and tools			x

NOTES:

- i) Any appreciable flow from the drain pipes indicates that one or more of the hydraulic seals needs replacing. The most convenient way to replace seals on site is to replace the complete spring module.
- (ii) See note (b) under "Running-In Period".
- (iii) Under adverse conditions this may be required more frequently.
- (iv) Check that no movement has occurred at paint-marked bolts, and test security of other fasteners.
- (v) Check the friction pad condition

3.3 Friction Pad Wear Adjustment

See p.2.2/ p.2.3

3.4 Friction Pad Replacement

Brake pads should always be replaced in pairs

- a) Secure the disc against rotation.
- b) Remove the spindle locking plate (Item 36).
- c) When fitted with proximity sensors (Item 34) ensure sensors are removed.
- d) Apply hydraulic pressure to fully retract the piston.
- e) Retract brake pads by turning adjusting spindle (Item 38) anti-clockwise (see p.1.2.6).
- f) Remove hydraulic pressure.
- g) Remove the pad retaining bars (Item 44) and M20 screws (Item 11).
- h) Unscrew two M8 screw (Item 10) from the friction pad plate (Item 28).
- i) Remove the pad by sliding it through the spring module / reactive back plate.
- j) Remove any dust/dirt from the backing plate.
- k) Insert new pads.
- l) Replace two M8 screws and springs into the pad plate.
- m) Replace pad retaining bar and M20 screws and tighten them to 400 Nm.
- n) Adjust air gap (see p.2.3).

3.5 Loss of Braking Force

Possible causes are:

- a) Contaminants on surface of disc or pads, especially lubricant.
- b) Increased hydraulic back pressure when brakes are applied.
- c) Increased stroke (due to pad wear-necessitating adjustment), or incorrect adjustment.
- d) Damaged or broken disc spring or springs.

Spring force can be checked by the use of a calibrated oil pressure gauge suitable for the maximum back pressure employed.

Pressures required for full retraction are given in table 1.

3.6 Releasing Brake after Hydraulic Failure

The brakes can be retracted by connecting a hand pump direct to the feed port. After the brake has been released the “parked-off” position can be obtained by turning the adjusting spindle fully anti-clockwise. The hand pump can then be removed.

3.7 Replacement of seals

Leakage of cylinder seals is indicated by oil seeping from the G 1/8” cylinder drain ports.

If new seals are required, it is recommended spring modules be returned to Twiflex for seal replacement. Otherwise the following procedure should be carried out:

- a) Secure the disc against rotation.
- b) When fitted with proximity sensors ensure sensors are removed.
- c) Apply hydraulic pressure to fully retract the piston and remove locking plate (item 44).
- d) Retract brake pads fully back (Item 28) by turning adjusting spindle(Item38) anti-clockwise.
- e) Take hydraulic pressure off so the pressure is zero and spring pack is in safe position.
Ensure there is a clearance between pad and disc when pressure is zero.
- f) Disconnect the hydraulic hoses.
- g) Remove the Spring Module from the pedestal by unscrewing 2 x M24 bolts (Item 14) and 2 x M16 bolts (Item 13).

- h) Take assembly to work bench to continue seal replacing.
- i) Remove spindle locking plate (Item 36) and end cover (Item 43).
- j) Remove disc springs (Item 2).
- k) Remove friction pad (Item 28) as under 3.4.
- l) Remove the piston (Item 37) from the housing (Item 42) by tapping it ("soft" face mallet to be used).
- m) Remove the seals and/or bearings (Items 18-25) and inspect surfaces.
- n) Re-assembly is generally as dismantling procedure but in reverse order .

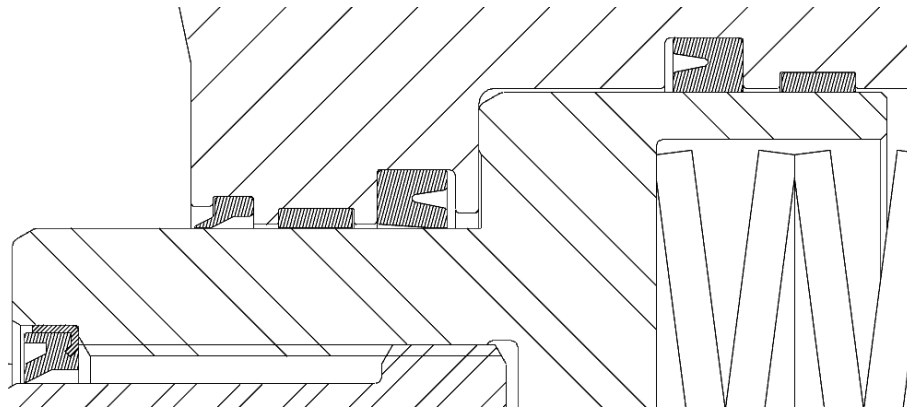


Fig. 2 Seals Layout

NOTE:

EXTRA CARE AND SPECIAL TOOLING IS REQUIRED TO PREVENT SEAL DAMAGE WHEN REPLACING PISTON INTO THE CYLINDER. IT IS RECOMMENDED BRAKES BE RETURNED TO TWIFLEX FOR SEAL REPLACEMENT .

- o) Refit the piston.
- p) Replace the disc springs, end cover and spindle locking plate (ensure springs are covered with lithium grease and replaced in the right order / arrangement).
- q) Refit the Spring Module onto pedestal and retighten 2 x M24 bolts to 730 Nm and 2x M16 bolts to 200 Nm.
- r) Reconnect the hydraulic hoses into the brake.

3.8 Recommended Spares

For each brake in use at remote sites or in marine installations, the following spares should be carried:

Item	Description	Part No.	Qty
18	Rod Seal	6000736	2
19	Rod Seal	6000737	2
20	Scraper	6000738	2
21	Wiper	6000739	2
22	Slydring	6000740	2
23	Slydring	6000741	2
24	O-ring	6000753	2
25	O-Ring	6000754	2
*28	Pad Assembly	70A0370-9	2

* These are to be stored in a dry place away from contact with oil or grease.

3.9 Recommended Tools

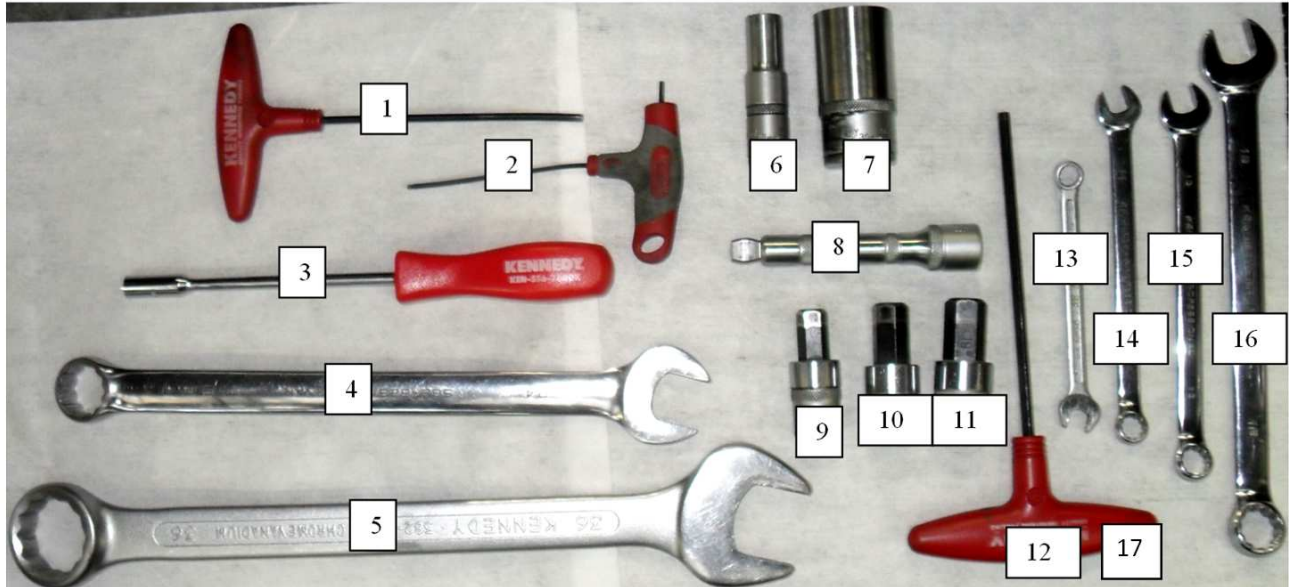


fig 12. Tools

Item	Description	Size	Used On
1	Allen Key	6 mm	M8 Socket Cap Screw
2	Allen Key	2.5 mm	M5 Grub Screw
3	Socket	8 mm	M5 Hex Screw
4	Spanner	24 mm	M16 Nut
5	Spanner	36 mm	M24 Nut
6	Socket	13 mm	Adjusting Spindle
7	Socket	36 mm	M24 Nut
8	Extension		
9	Socket Allen	14 mm	M16 Bolt
10	Socket Allen	17 mm	M20 Screw
11	Socket Allen	19 mm	M 24 Bolt
12	Allen Key	8 mm	Pad retraction Screw
13	Spanner	10 mm	Monitoring Kit
14	Spanner	11 mm	Bleed Screw
15	Spanner	13 mm	Breather
16	Spanner	19 mm	Bleed adaptor
17	Allen Key	5 mm	1/8 BSP Plug, Plunger

M20 and M12 eye bolts-required for lifting only-NOT SUPPLIED

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Matrix International • Twiflex Limited • Bibby Transmissions • Huco Dynatork • Warner Linear

3.10. Inductive analogue sensor



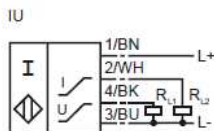
Model Number

IA6-12GM50-IU-V1

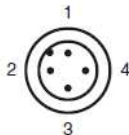
Features

- Inductive analog
- 0...6 mm quasi-embeddable
- Analog output 0 V ... 10 V / 4 mA ... 20 mA

Connection

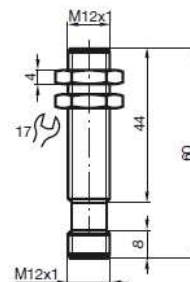


Pinout



Wire colors in accordance with EN 60947-5-2

1	BN	(brown)
2	WH	(white)
3	BU	(blue)
4	BK	(black)



Technical Data

General specifications

Switching element function	analog, current or voltage output
Installation	quasi embedd.
Output polarity	DC
Measurement range	0 ... 6 mm

Nominal ratings

Operating voltage	U_B	15 ... 30 V
Limit frequency (3dB)		1000 Hz
Reverse polarity protected		yes
Short-circuit protection		yes
Repeat accuracy		0 ... 10 μ m
No-load supply current	I_0	≤ 10 mA

Analog output

Output type	0 ... 10 V / 4 ... 20 mA
Load resistor	voltage output: $\geq 500 \Omega$ current output: $\leq 500 \Omega$
Recovery time	1 ... 10 ms, typ. 5 ms
Adjustment tolerance zero point	$\leq \pm 10$ % of full-scale value
Temperature drift	± 5 % (0 ... 70 °C) ± 10 % (-25 ... 0 °C)
Residual ripple	20 %

Ambient conditions

Ambient temperature	-25 ... 70 °C (-13 ... 158 °F)
---------------------	--------------------------------

Mechanical specifications

Connection type	Device connector M12 x 1, 4-pin
Housing material	chromium plated brass
Sensing face	PBT
Protection degree	IP67

Compliance with standards and directives

Standard conformity	
Standards	EN 60947-5-2:2007 IEC 60947-5-2:2007 EN 60947-5-7:2003 IEC 60947-5-7:2003

Approvals and certificates

UL approval	cULus Listed, General Purpose
CSA approval	cCSAus Listed, General Purpose
CCC approval	Products with a maximum operating voltage of ≤ 36 V do not bear a CCC marking because they do not require approval.

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4. MANUAL FOR BRAKE CALIPER VKSD-VR

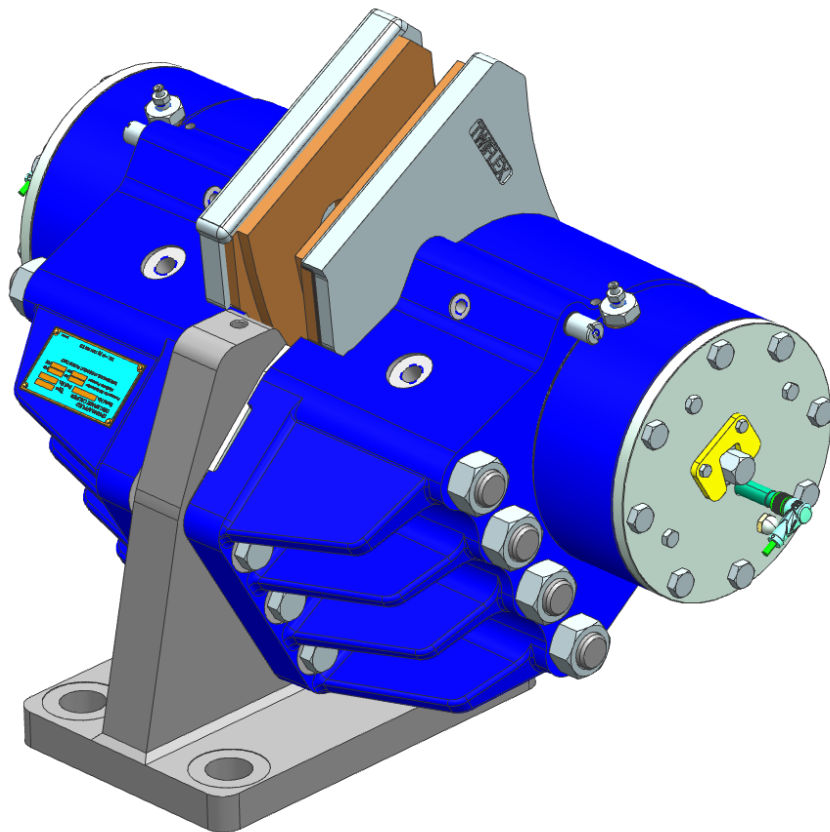
Caliper Type: VKSD-VRxxx-xx-xx

Manual: M1596

This manual covers the complete range of VKSD brake calipers.



**TWIFLEX
DISC BRAKE CALIPER
TYPE 'VKSD-VR'
INSTALLATION, OPERATION AND
MAINTENANCE INSTRUCTIONS
M1596**



AMENDMENT AND REVISION RECORD – M1596

AMENDMENT NUMBER ISSUE AND DATE		SIGNATURE AND DATE WHEN AMENDMENT ISSUED
01	November 2012	

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1. SPECIFICATION

1.1 GENERAL DESCRIPTION

THE OPERATIONAL SEQUENCE is illustrated in Appendix 3. Numbers in brackets refer to item numbers shown in Appendix 1.

The VKSD-VR disc brake caliper is a spring applied, hydraulically retracted unit which may be used on any disc having a diameter of at least 1.0m and a thickness of 20mm or greater. There is no upper limit on diameter or thickness. Braking torque may be controlled by any combination of air gap setting, varying shim thickness, or by the use of hydraulic backpressure during braking. The caliper consists of two halves which are mounted each side of a mounting plate or pedestal by means of 4 M24 tie rods and 3 M16 bolts. No special bleeding arrangements are required for mounting at any angle as the fluid is almost entirely evacuated when the hydraulic pressure is fully released with the pads fully wound back.

The braking force is applied by the disc spring packs (1) through spring guides (52) and thrust screws (54) to the pressure plates (45) and hence to the friction pads. The caliper is available in two models: the VKSD 119/62 with 5 pairs of disc springs per module, and the VKSD 64/28 with 7 single disc springs per module. Conversion of model VKSD 119/62 into VKSD 64/28 is possible and is described in Section 5. The pads are kept in position by pad retraction hooks (53) on to the pressure plates (45), which transmit the braking force to the main housing (44).

The housing contains the spring chamber. The hydraulic retraction cylinder (51) is spigoted to the housing and the bleed screw and inlet port can be positioned by releasing two set screws (15) and rotating the cylinder. The hydraulic retraction piston (46) is held onto the spring-guide by means of a piston collet (47) with its retainer (48).

Brake rating can be varied by removing or adding shims between piston (5) and end cover (43). Gap adjustment is manual and is effected by turning the hexagonal thrust screw (54). Maintaining the correct air gap ensures that the braking torque is effectively constant and minimises the stress range for the springs, leading to an extended fatigue life. Reduction in torque due to a 1mm stroke increase is about 6% to 8% for the VKSD 119/62 and 4% to 15% for the 64/28. Failure of an individual spring would result in a substantial loss of braking torque in the affected caliper (see 3.7d). Connections are provided for piping away any oil leakage.

Optional monitoring units are available which indicate:

- a) Full retraction of the brake - i.e. correct functioning of the hydraulic system. In conjunction with a suitable hydraulic pressure switch this facility can be used to give an indication of spring breakage. (See section 3.7 (d))
- b) Need for adjustment for pad wear.
- c) Need for pad replacement.

1.2 **DETAIL SPECIFICATION**

1.2.1 The VKSD-VR series has a special operational feature called 'Parked-Off' position. This is made possible by having a piston stroke long enough to permit the disc springs to fully expand. Hence no internal spring loads exist in this condition. Yet the brake pads can be retracted to be within the mounting face and 11.5mm clear of the disc. This brings the following advantages:

- a) The unit can be shipped without a retraction tool and with the cylinder end-cover and monitoring unit in position.
- b) The unit can be installed without the need for hydraulic pressure.
- c) The pads can be safely exchanged without a retraction tool and with the hydraulics at zero pressure.
- d) Hydraulics can be operated and tested, and air gap set without placing unbalanced axial load on the brake disc and shaft.
- e) The module, including spring pack, can be safely assembled and disassembled.
- f) In the case of hydraulic failure, the retraction tool is only required to release the brake. Full retraction is not necessary as upon release the pad can be wound back in the parked-off position, and the retraction tool removed.
- g) For 'ship-shafts' maintenance use, the brake when placed in the 'parked-off' position cannot be applied accidentally or due to inadvertent operation or hydraulic failure (see also 3.11.2). The 'parked-off' feature combined with a separate cylinder allows:-
- h) Cylinder to be rotated to position the bleed, at any time, even after installation.
- j) Cylinder can be dismantled, permitting replacement of seals and cylinder wearing surface 'in-situ'.

1.2.2. **BRAKING FORCE & TORQUE (SEE CURVES PUBLICATION G1457 and G1458)**

The braking force 'F' (kN) acting tangentially on the disc is given by $F = 2 \times \mu \times S$ where μ is the friction coefficient of the brake pad and S is the spring pack thrust (kN) (lbf). Nominal values for braking force are given on the drawings together with the appropriate air gap settings; please note that these values are based on a coefficient of friction $\mu = 0.4$.

This value can only be achieved under ideal conditions. Factors which influence μ are: - material and surface finish of disc, amount of surface contact (after bedding in), cleanliness of pad and disc, humidity, temperature and grade of pad material, and finally, rubbing speed. In a near static application for example μ can be as low as 0.15.

Braking torque T (kNm) is given by $T = F \times (R - 0.095m)$ where R is the outside radius of the disc in metres.

(T lbf.ft = F lbf x (R - 0.312)ft).

The braking force obtained is also dependent on the gap setting and the amount of shims fitted see performance curve publication G1457 and G1458. Please note that these curves are also based on $\mu = 0.4$ and that a service factor has to be applied for a particular application. Also shown on these curves is the fatigue life for (99%) of the springs for a particular gap setting. See Publication S1025 for an explanation of how to read the curves.

1.2.3 **RETRACTION CYLINDERS STANDARD.**

Diameter = 150mm

Spring guide diameter = 70mm

Effective area = 13823mm²

Hydraulic pressures for brake release and full retraction are given on the drawing for the nominal gap setting. For other gap settings or shim settings these pressures can be read from the curves. (See Publication S1025 - Performance Curves - Explanation)

In order to avoid overstressing the 8 M10 set screws on the end cover it is recommended that the maximum hydraulic pressure shall not exceed the retraction pressure by more than 25 bar.

Recommended hydraulic fluid is a good quality mineral oil such as Shell 'Tellus 37' or an approved fire resistant type.

Fluid inlet connections are 3/8" BSP (G3/8) and fittings are normally provided to connect the cylinders by means of 12mm o/d. x 1.5mm seamless steel tubing to DIN 2391/C.

1.2.4 **FRICTION PADS**

Friction pads in non-asbestos material are supplied; some types may have a lower coefficient of friction. Initial thickness of the pads (including back plate) = 22mm.

Total pad area (two pads) = 119952mm²

The permissible wear on these brakes is 10 mm.

1.2.5 **MATERIALS AND QUALITY ASSURANCE**

The housings and pressure plates on the VKSD-VR are castings of SG iron, to BS 1563, grade 450-10. Other metal components are steel. Calipers can be supplied with non-destructive test (NDT) certificates for all critical components, if required. The main hydraulic seals are PTFE; 'O' rings are nitrile rubber.

Each caliper is fully tested after assembly and a test certificate issued.

1.3 **MONITORING UNITS** **(See Appendix 4)**

- a) Fully retract the brake by applying the full retraction pressure.
- b) Screw the sensor in until it touches the spring guide (Item 52) then move approx 1/2 turn back.
- c) Adjust required air gap as required.
- d) Release the hydraulic pressure to apply the brake.
- e) When the piston is in the sensing area the LED comes off (PLC controller to be set to measure the electric current in mA) - see appendix 3 for details.
- f) Operate the brake a few times to ensure the switch is working properly.

2. INSTALLATION AND ADJUSTMENT

2.1 INSTALLATION

Refer to Drawing Appendix 1.

Ideally, two calipers should be used on each disc, mounted diametrically opposite each other, in order to neutralise the braking torque reaction forces on the shaft bearings. Any other position can also be catered for, as the bleed screw on the cylinder can be indexed to any position, after releasing the two set screws (15); the set screws should be retightened to 6Nm.

Brake mounting plates or pedestals should be of adequate strength and rigidity, and must be flat, and parallel to the disc surface. Each caliper is secured by four M24 tie bolts and three M16 bolts, (can be supplied by Twiflex) which should be tightened to a minimum torque of 675Nm and 200 Nm respectively.

NOTE: VKSD-VR 119/62 calipers subject to REVERSING BRAKING LOADS over 70kN must be provided with one M24 'fitted' tie rod at either end of the row of 4 tie rods. One end hole in the mounting plate must also be machined to $\varnothing 25(+0.025/+0.000)$.

The mounting plate should be positioned with respect to the disc as shown on the drawing with the mounting surfaces lined up with, but 6 mm proud of, disc rubbing surface.

Sufficient space should be allowed for access to the calipers for routine inspection and maintenance and for pad renewal. Calipers are supplied in the "parked off" position in which the springs are fully relaxed and the pad is retracted within the mounting face. When the caliper halves have been bolted into position, the hydraulic supply should be connected and all air bled from the system using the bleed screws (29).

2.2 INITIAL ADJUSTMENT

- a) With the bleed screws closed the cylinders may be pressurised to a maximum of 25 bar above the retraction pressure as given on the drawing. In the fully pressurised condition the back of the spring guide (52) should be 2.0 mm from the inner face of the cylinder end cover (43) or shims where fitted.

- b) The Air Gap should be set as follows:

Remove the M5 screws in the spindle locking plate (50) and disengage its slot from the adjusting thrust screw (54). Using a 19mm AF spanner turn one of the adjusting thrust screws clockwise until the pad is in firm contact with the disc, then back off by 2/3rds of a turn for 2mm air gap, 1 turn for 3mm air gap or 1 1/3 turns for 4mm air gap. (The pitch of the thrust screw is 3mm).

The actual retraction dimensions will be slightly greater when the caliper deflects under load. Repeat the setting procedure on the other side of the caliper. Operate the caliper and check that the air gap or stroke is correct; adjust if required. The minimum adjustment which can be made is 1/6th of a turn = 0.5mm by leaving the locking plate in the same position or 1/12th of a turn = 0.25mm by removing the M5 screws and inverting the locking plate (12).

- c) Please note that the correct shim stack height will have been fitted, commensurate with the brake performance required. The braking force can however be reduced by adding shims or increased by removing shims.

Notes i) Maximum shim stack height is 7mm for VKSD-VR 119/62 and 6mm for VKSD-VR 64/28, both with the step on the end cover (43) inwards. Exceeding these amounts will under stress the disc springs and shorten their life.

WARNING The position of the step on the end cover (43) of the VKSD-VR 119/62 **MUST ALWAYS BE INWARD (See Fig 4) - NEVER OUTWARD (See Fig 3), as this would severely overstress the disc springs.**

- ii) Always remove or add the same amount of shims on each side of the caliper.
- iii) Always, after refitting end cover (43), retighten set screws (11) evenly in turn to a torque of 50Nm.
- d) The drain connections (two each side of the caliper) can now be piped away to waste. It may be convenient to use flexible tubing, and if transparent tubing is used any leakage will be immediately apparent.

3. MAINTENANCE

3.1 RUNNING-IN PERIOD

For the first full week of operation make a DAILY CHECK of the following items:

- a) Hydraulic Fluid Leakage. Note and report any evidence of fluid leakage. If the leakage occurs at the pipe connections or at the bleed screws, tighten as appropriate, carefully wipe away spilled fluid and check for further leaks with brakes off.

Note: The air gap is measured between pad and disc at full retraction. The stroke is the air gap plus caliper deflection and is the movement of the adjusting spindle from brake 'on' to full retraction, measured with respect to the end cover.

If friction pads become contaminated with fluid, they must be replaced after curing the leak and the disc(s) wiped clean. A very slight seepage of oil past seals (22 & 24) is normal, but should not be such as to cause a flow in the drain pipes.

- b) Observe brake operation including smooth retraction and application of brake pads. Note and report any unusual observations.
- c) Check condition of the brake paths. They should remain smooth and free from contaminants, for example rope lubricant. Remove any deposits.
- d) At the end of the running-in period, check the security of all bolted connections and pipe connections.

The caliper mounting bolts should be paint marked to provide a visual security check.

3.2 MAINTENANCE INSPECTION PROGRAMME

After the running-in period the following schedule should be followed in addition to any statutory requirements.

MAINTENANCE SCHEDULE			
ITEM	WEEKLY	MONTHLY	PERIODIC
a) Check for hydraulic fluid leakage	x (note i)		
b) Observe brake operation	x (note ii)		
c) Clean calipers of accumulated deposits		x	
d) Clean brake paths on disc		x (note iii)	
e) Observe bolt security		x (note iv)	
f) Note and record brake pad thickness		x (note v)	
g) Note and record brake pad movement, (stroke). Adjust if required.		x (note vi)	
h) Check availability and condition of spares and tools			x
j) Remove and inspect brake pads (see 3.4)			x

NOTES

- (i) See note (a) under "Running-in period" (3.1). Any appreciable flow from the drain pipes indicates that one or more of the hydraulic seals needs replacing. This may be conveniently done by replacing the complete half as in Section 3.8. However, if servicing on site is essential, consult Section 3.9.
- (ii) See note (b) under "Running-in period".
- (iii) Under adverse conditions this may be required more frequently.
- (iv) Check that no movement has occurred at paint-marked bolts and test security of other fasteners.

- (v) Note brake pad wear by measuring proximity of pad backing plate edges to disc, with brakes on. A dimension of 2mm at any position indicates that the pad needs to be replaced. Replacement thickness is 12mm measured over backing plate. New thickness is 22 mm over backing plate.
- (vi) Air gap is nominally as shown on the drawing or on the curves but is affected by disc axial float and run-out and will vary by that amount. Actual stroke measured at the pad will be greater, due to deflections of the caliper, movement of the adjusting spindle is the most convenient measurement.

Where no monitoring units are fitted and when pad wear is very low (as is generally the case), comparative checks of pad movement may be made monthly with accurate measurements at less frequent intervals.

3.3 **ADJUSTMENT FOR FRICTION PAD WEAR**

The adjustment is made by turning the thrust screw clockwise using a suitable spanner. Adjustment can be made in 0.5mm increments (1/6th of a turn) by leaving the locking plate in the same position or in 0.25mm by inverting the locking plate.

- a) Determine adjustment required to bring piston stroke into line with original gap setting (see note (vi), section 3.2 above).
- b) Remove the M5 screws from spindle locking plate and slide it back to disengage spindle.
- c) Turn spindle clockwise by one 'flat' (60°) for 0.5mm adjustment, two 'flats' (120°) for 1mm etc. Should adjustment required be 0.25mm then the locking plate must be inverted for 1/12 turn.
- d) Replace locking plate and tighten M5 screws to 6Nm so locking the adjusting spindle.
- e) Repeat for opposite pad; pads should always be adjusted in pairs.

3.4 FRICTION PAD REPLACEMENT

Brake pad replacement is indicated when:-

- a) The monitor switch (where fitted) has signalled that pad is fully worn.
- b) Top of the plunger (41) is level with housing face.
- c) The backing plate edges are 2mm or less from disc.
- d) If the pad thickness has more than 2mm taper, as measured at the four corners, it should be replaced, even if not fully worn.
- e) Minimum pad thickness is 12mm.

Before removing brake pads, secure the disc against rotation. Retract brakes fully, hydraulically, and as a SAFETY measure place the brake in the 'parked-off' position by turning the adjusting spindle fully anti-clockwise. The hydraulic pressure may now be released. Unhook the pad retraction hook (53) by inserting a large screwdriver in the pad retraction plunger (41), depressing spring (2) and turning the hook through 180°. The pad can now be removed, first by pushing it away from the pressure plate through the holes provided, then by sliding it along the disc-rubbing surface.

Remove dirt using a clean dry cloth and note condition of pad friction surface, which should be polished, uniformly marked and free of serious pitting. Replacement of the pad is carried out by reversing the removal procedure, but first clean the pressure plate. Ensure that pad is properly seated by turning adjusting spindle such that pad is forced firmly against disc. Refit pad retraction hooks.

After fitting the pads the piston stroke must be reset as in Section 2.2 (Initial Adjustment) paragraph (b).

3.5 HYDRAULIC SUPPLY FAILURE

The brake may be released by means of a hand pump connected direct to the feed port. After the brake has been released the 'parked-off' position can be obtained by turning the adjusting spindle fully anti-clockwise. The hand pump can then be removed.

3.6 **MECHANICAL RETRACTION**

The retraction tool kit can be purchased separately.

The retraction tool should only be required to release the brake when a hand pump is not available or there is a hydraulic failure, such as when the seals leak to such an extent that the brake cannot be released by hand pump. The retraction tool should then be used as follows:

- a) Remove end covers (43) complete with monitoring units, if fitted. Take care not to lose any shims.
- b) With nuts (M12) already threaded onto the retraction screws, attach the retractor flange to the hydraulic piston (46).
- c) Release brake by tightening the nuts evenly in turn.
- d) Turn the adjusting spindle fully anti-clockwise thus placing the brake in the 'Parked-Off' position.
- e) The retractor flange may now be removed, by loosening the nuts evenly in turn.
- f) Always remove retractor flange before resuming normal operation.

3.7 **LOSS OF BRAKING FORCE**

Possible causes are:-

- a) Contaminants on surface of pads or disc, especially lubricants,
- b) Higher than normal hydraulic back pressure when brakes are applied.
- c) Increased stroke (due to pad wear, necessitating adjustment; or incorrect adjustment).

d) **Damaged or broken disc spring or springs**

Where the monitoring switch assembly is fitted any significant reduction in the force developed by a spring can be detected by the addition of a pressure switch to the hydraulic fluid supply. If the above system is not used, spring force can be checked by use of a calibrated oil pressure gauge suitable for the maximum back pressure employed. Pressures required for full retraction are given on the drawing or can be taken from the curves. Springs should be changed if the reduction in pressure is 12.5% or more. Normally the easiest procedure would be to remove the complete caliper half as in 3.8 below and return it to Twiflex for service or replacement. However, for cases where this is not possible the procedure for seal replacement is described in section 3.9 and replacement springs is described in section 3.10.

3.8 REMOVALS AND REPLACEMENT OF CALIPER HALF

- a) Balance load and secure the installation from rotation.
- b) Remove monitoring units if fitted.
- c) Release the brakes hydraulically or mechanically and place the brake in the 'Parked-Off' position by turning the adjusting spindle fully anti-clockwise; hydraulic pressure may then be released.
- d) If the brake pads are re-usable, remove them and put them in a dry clean place; otherwise ensure that spares are available.
- e) Disconnect the hydraulic unions, controlling the fluid loss.
- f) Support the weight of the caliper half to be removed.
- g) Unscrew and remove the four M24 tie rod nuts and remove the three M16 mounting bolts.
- h) Carefully slide the caliper half from the tie rods.
- j) Replacement of the caliper half is the reverse of removal. It is essential to tighten the M24 bolts to 675Nm. The M16 bolts should be tightened to 200Nm See Note under 2.1 on calipers subject to reversing loads.

3.9 REPLACEMENT OF SEALS 'IN SITU'

Leakage of large cylinder seal (26) is indicated by oil seeping from the cylinder cover drain. Leakage of rod seal (28) is indicated by oil seeping from the 1/8" BSP holes at the sides of the housing fitted with plugs (32) on delivery. At least one plug, normally the lower one, must be removed.

- a) Follow instructions as under 3.8 a, b, c, and e.
- b) Remove end cover (43) complete with bolts (11) and shims if fitted.
- c) Slacken off set screws (15).
- d) Remove retaining ring (5) from piston.
- e) Retract and remove collet retainer (48) by inserting M4 x 40 screws in the tapped holes in this retainer.
- f) Remove split piston collet (47). It may be necessary to free the segments by tapping them towards the pad end of the caliper. Note that collet segments are marked with a batch number, e.g. B/20.
- g) Remove cylinder (51) and piston (46) together, by means of bar bolted to M12 retraction holes in piston. 'O' ring (18) may be reused unless severely damaged.
- h) Take assembly to work bench and carefully remove piston (46).
- j) Inspect wear surfaces, seals, wear rings and 'O'-rings for damage and refurbish or replace as necessary.
- k) Re-assembly is generally as dismantling procedure but in reverse order; note the following:-
 - i) Exercise great care not to damage seals and wearing surfaces.
 - ii) Clean wearing surfaces on spring guide (52) and cylinder (51).
 - iii) First assemble piston into cylinder.
 - iv) Note that cylinder seal (25) has to be 'sized' by means of the cylinder, and rod seal (24) by means of a tool, with a diameter of 69.970/69.940 (Ø 70f7) with a chamfer of 8mm long @ 15° blended into this diameter; surface finish to be 0.4 microns.

- v) Slide cylinder-piston assembly onto spring guide, ensuring that cylinder spigot is fully home into housing. Move piston into cylinder so that split collet segments (47) can be entered into groove in spring sleeve. Refit 'O'-ring (19) to hold segments in place. Insert collet retainer (48) and apply some air pressure to the cylinder port to 'blow' piston into place. Collet segments should be replaced in their original positions, referring to batch number and identifying marks.
- vi) Refit retaining ring (5).
- vii) Turn cylinder so that bleed screw (29) is uppermost. Note that thread at hydraulic feed port and at bleed screw port are the same size; hence adaptor (33) and bonded seal (36) can be removed to port which is nearest top and thus save turning cylinder an unnecessary amount.
- viii) Retighten two set screws (15) to 6Nm.
- ix) Refit end cover (43), (note correct position - see note 1), with monitoring unit if fitted, taking care to enter the 3 tension pins (3) into the holes in the spring guide. Tighten set screws (11) evenly in turn to a torque of 50Nm. Refit external 'O' Ring (26). Finally reconnect hydraulic pipes, bleed and, readjust brakes as under 2.1 and 2.2

Note 1 Note position of end cover (43) and refit as shown on drawing, i.e. flat side is **always** on outside.

Note 2 Tension pins (3) have to be tapped into inside of lid.

3.10 REPLACEMENT OF SPRINGS

3.10.1 DISMANTLING

- a) Proceed as under 3.8 (a) to (h) and transfer caliper half to workshop.
- b) Proceed as under 3.9 steps (b) to (f).
- c) The pressure plate (45), thrust screw (54) and spring guide (52) with springs (1) can now be removed from the housing.
- d) Alternatively the pressure plate (45) and thrust screw (54) can be removed first by rotating the latter clockwise by means of the thrust screw (54).

The spring guide (52) with springs (1) can then be removed for inspection.

Note: Take great care not to damage the rubbing surface for the rod seals (29). Keep springs in their order as assembled if possible.

3.10.2 **NON-DESTRUCTIVE TESTING**

If the springs are visually sound an N.D.T. method may be employed. It is essential that the surface finish of the disc springs is not damaged by electrical contacts, sparking or otherwise, so an N.D.T. should be adopted, which does not give rise to any such danger.

Inspect other parts as may be required.

3.10.3 **RE-ASSEMBLY**

Follow generally the reverse of the dismantling procedure, but note:

- i) Exercise great care not to damage seals and bearing surfaces.
- ii) Re-assemble spring in the reverse order to that in which they were originally assembled (to extend their fatigue life).
- iii) Observe assembly notes under 3.9.
- iv) Clean wearing surfaces of cylinder (51), thrust screw (54) and spring guide (52) before re-assembly.

Retighten set screws (15) to 6 Nm.

- v) It is essential when reinstalling caliper half to preload M24 tie rods and M16 bolts as under 3.8(j).
- vi) Re-fit brake pads and adjust for correct air gap.

NOTE: The disc spring bearing on the spring guide abutment must do so with its smallest diameter, as shown on the drawing.

3.10.4 **TESTING**

Carry out functional testing and operational testing as may be required. Check settings of monitoring units (if used) after refitting the caliper. Adjust if necessary.

3.11 **OPERATION AS 'SHIP SHAFTS' HOLDING BRAKE**

It is assumed that the brakes are in the "Parked Off" Position - 1.2.1g.

3.11.1 **HYDRAULIC PRESSURE AVAILABLE**

Apply hydraulic pressure and turn the thrust screw (54) on both sides, until the pads are firmly against the disc. Releasing the hydraulic pressure will apply the brake, and Braking Force will be as for zero gap setting as indicated for the shim height as fitted.

3.11.2 **NO HYDRAULIC PRESSURE AVAILABLE**

Remove end cover and fit mechanical retraction tools - see section 3.6 a, b, and c.

Continue tightening the nuts until piston is fully retracted, i.e. when spring guide (52) is within 8 mm of retraction tool face. Turn adjusting spindles until the pads are firmly against the disc. Releasing the nuts on retraction bolts will apply the brake. Braking Force will be as for caliper at zero gap setting at no shim height. Braking force will, however, be affected by the factors as under 1.2.2.

4. **ROTABOLTS**

4.1 Special Tie bolts are available, optionally, with a 'Rotabolt' tell-tale at one end.

This tell-tale can be rotated by hand when the pre-load is below specification and cannot be rotated when the pre-load is correct.

5. CONVERSION OF VKSD 119/62 INTO VKSD 64/28 OR VICE VERSA

5.1 To convert VKSD 119/62 into VKSD 64/28

See Section 3.10.1 and remove the ten disc springs (1). Discard three disc springs and re-assemble springs facing alternate directions, and with the disc spring bearing on the spring guide abutment with its smallest diameter as shown in Figure 1. Add spacer, part no. 7952008, at the other end of the spring stack if required.

5.2 Follow further re-assembly instructions as under 3.9 and 3.10.3 and fit shims as required up to a maximum of 4mm with the end cover (43) step outward (See Fig 3) or 6mm with the cover step inward (See Fig 4).

5.3 To convert VKSD-VR 64/28 into VKSD-VR 119/62

See section 3.10.1 and remove 7 disc springs and discard spacer. Add 3 disc springs, part no. 2500154, and re-assemble springs into 5 pairs, facing alternate directions, and with the disc spring bearing on the spring guide abutment with its smallest diameter.

5.4 Follow further re-assembly instructions as under 3.9 and 3.10.3 and add or discard shims as required up to a maximum of 7mm.

NOTE: Position of end cover (43) on VKSD-VR 119/62 the flat side is on outside; on VKSD-VR 64/28 the stepped side is on the outside for the higher capacity settings and to the insider for the lower capacity settings.

5.5 Follow installation and adjustment instructions as under 2.1 and 2.2.

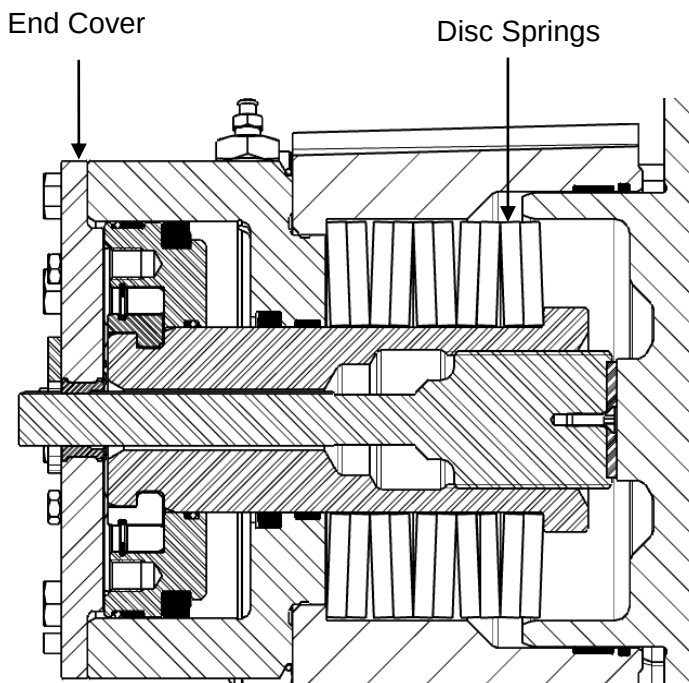


Fig 1. – VKSD119/62

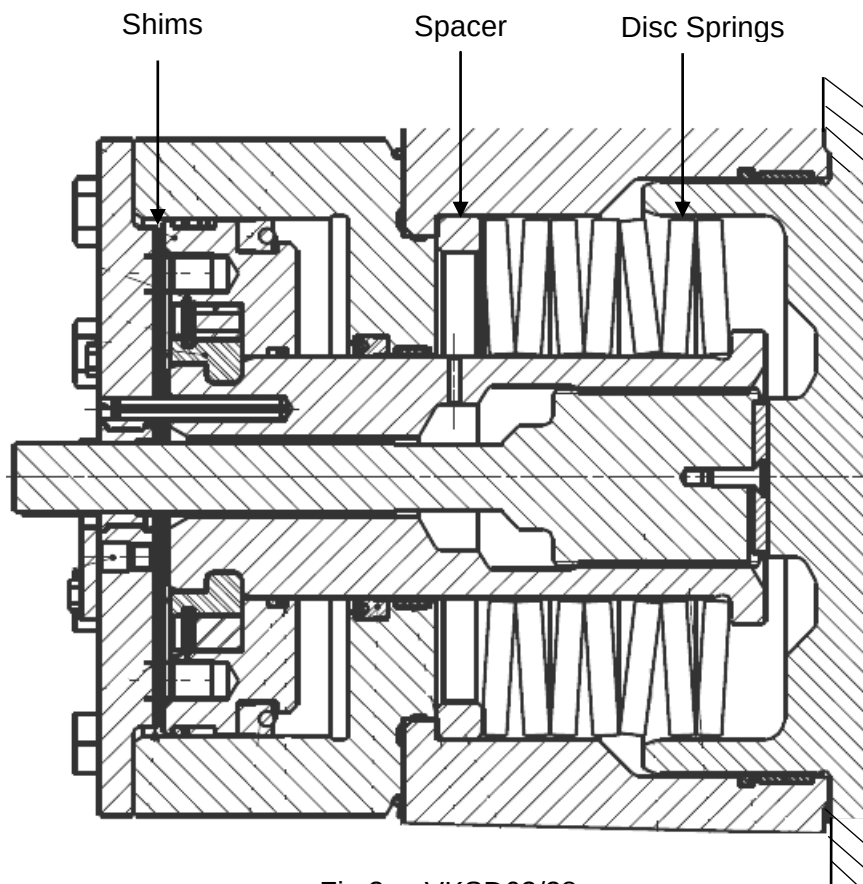


Fig 2. – VKSD62/28

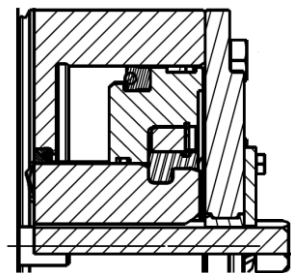


Fig 3.
Part section showing End
Cover positioned outwards

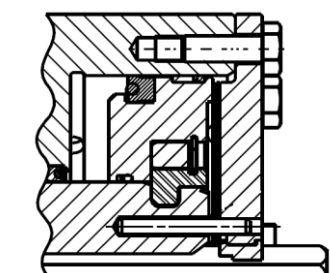


Fig 4.
Part section showing End
Cover positioned inwards

VKSD-VR PART IDENTIFICATION



APPENDIX 2

RECOMMENDED SPARES FOR VKSD-VR SERIES

For each brake in use at remote sites or in marine installation, the following spares should be carried:-

ONE SET OF BRAKE PADS

These are to be stored in a dry place away from contact with oil or grease.

Please order correct grade of friction material.

Component	Part No.	Qty.
Brake Pad: Ferodo 3904F (non asbestos) - High Temperature	7000358-9	2
One set of seals	6000543	2
2 Part cylinder seal	6000428	2
2 Part rod seal	6000427	2
Bearing ring (piston) 150 I/D	6000402	2
Bearing ring (rod) 70 I/D	6000403	2
Bearing ring 172 I/D	6000404	2
'O' ring 150 x 145 x 3	6000399	2
'O' ring 74 x 70 x 2.4	6000398	2
'O' ring 190 x 185 x 3	6000167	2
'O' ring 65 x 75 x 3	6000445	2
Back-up ring 70 I/D	6000490	2
Quad ring seal 171.05 I/D	6000450	2

APPENDIX 3 TO PUBLICATION M1438

OPERATIONAL SEQUENCE (REF; FIGS 5 TO 9).

Brake 'Parked Off' (Fig.5)

The brake is supplied in the 'Parked-Off' position, so that it can be mounted with disc springs fully relaxed and yet with brake pads 11.5 mm clear of disc.

Brake cannot come on accidentally and can be dismantled safely. Connect hydraulic feed and bleed brake in this position.

To Operate Brake

- Apply hydraulic pressure as for full retraction. This compresses disc springs. Thrust screw and pressure plate will separate at 'A' (Fig. 6).
- Check for oil leaks.
- Turn adjusting spindle clockwise:- up to 5 turns max. On the VKSD-VR 119/62; up to 7½ turns on the VKSD-VR 64/28, thus closing the gap at 'A' (Fig. 7). The number of turns depends on the number of shims fitted.

The brake is now operational with about 11.5 mm retraction.

To Set Correct Air Gap

- Maintaining full retraction pressure, continue turning spindle clockwise until brake pad is firmly up against disc (Fig. 8).
 - Slacken off brake by turning spindle anti-clockwise to obtain required air gap.
 - Thread pitch on thrust screw is 3 mm
- e.g. 1/6 turn for 0.5mm Air Gap
 1/3 turn for 1mm Air Gap
 1/2 turn for 1.5mm Air Gap
 1 turn for 3mm Air Gap etc.
 1/12 turn for 0.25mm Air Gap can be obtained by inverting locking plate.

Releasing hydraulic pressure will now apply the brake.

APPENDIX 3 CONTINUED

Replacing Brake Pads

Return brake to 'Parked-Off' position as follows:

- Apply just sufficient hydraulic pressure to release brake
- Turn spindle fully anticlockwise.
- Release hydraulic pressure.

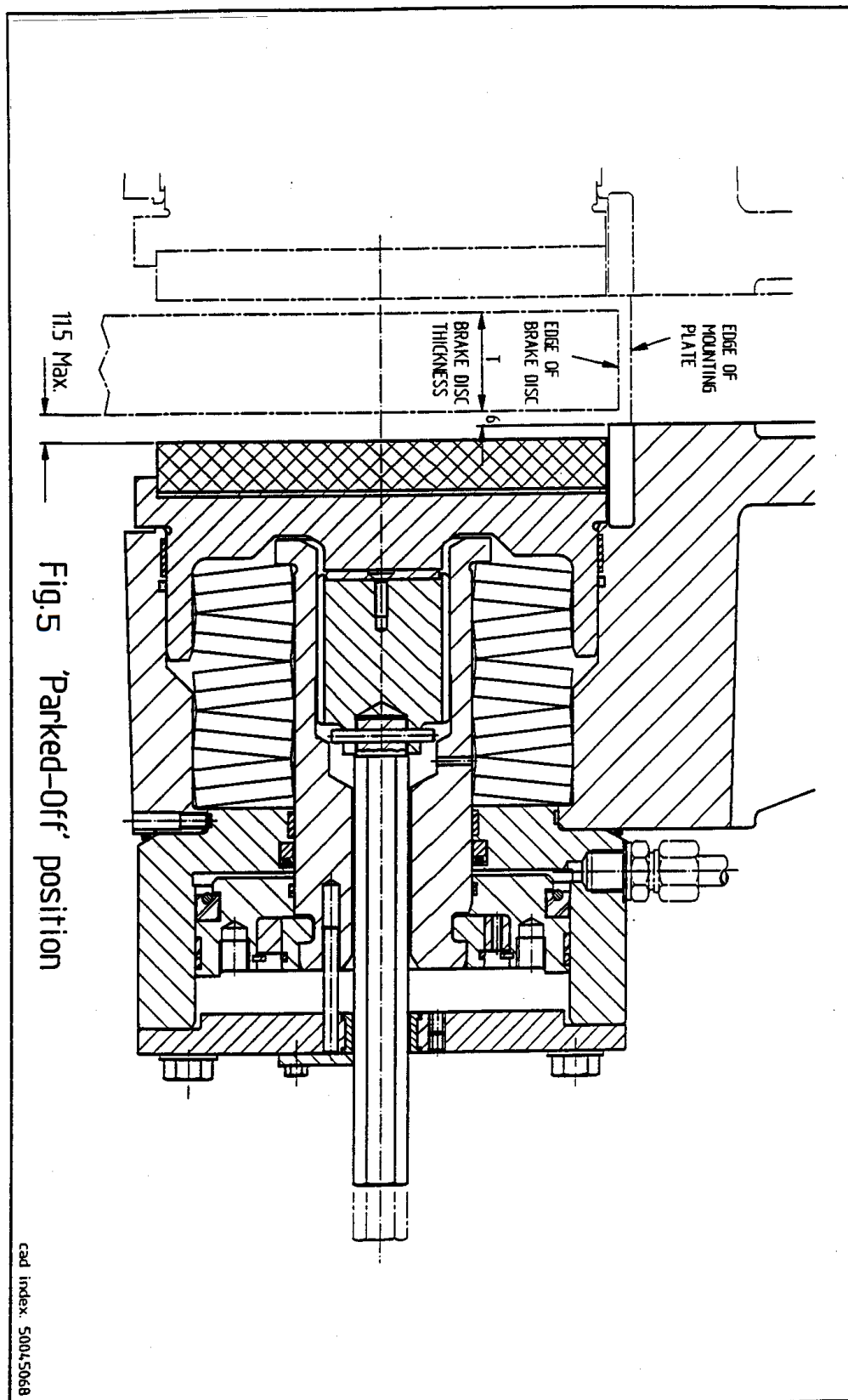
Sufficient space is now available to remove old pad and insert new pad.

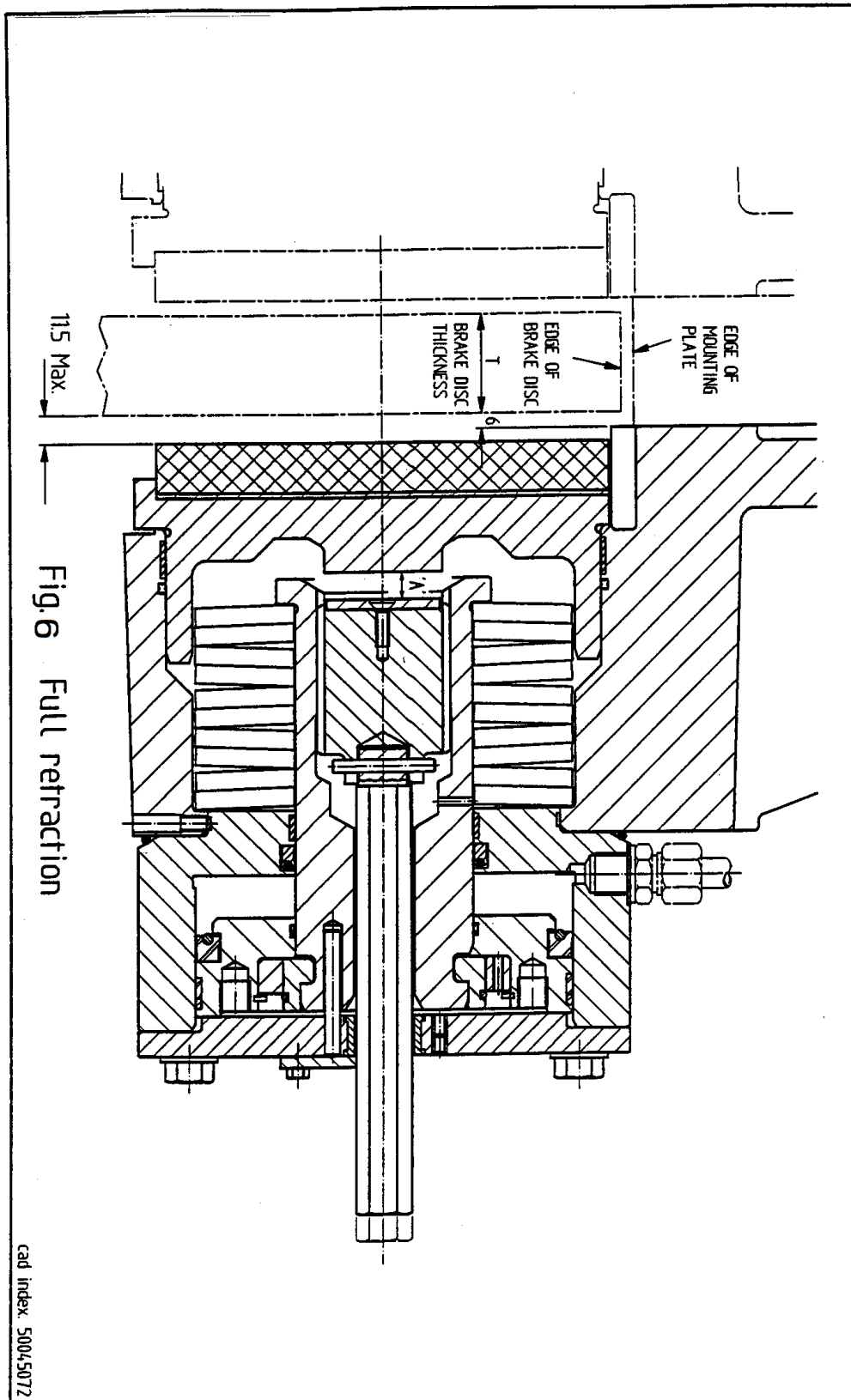
Hydraulic Supply Failure

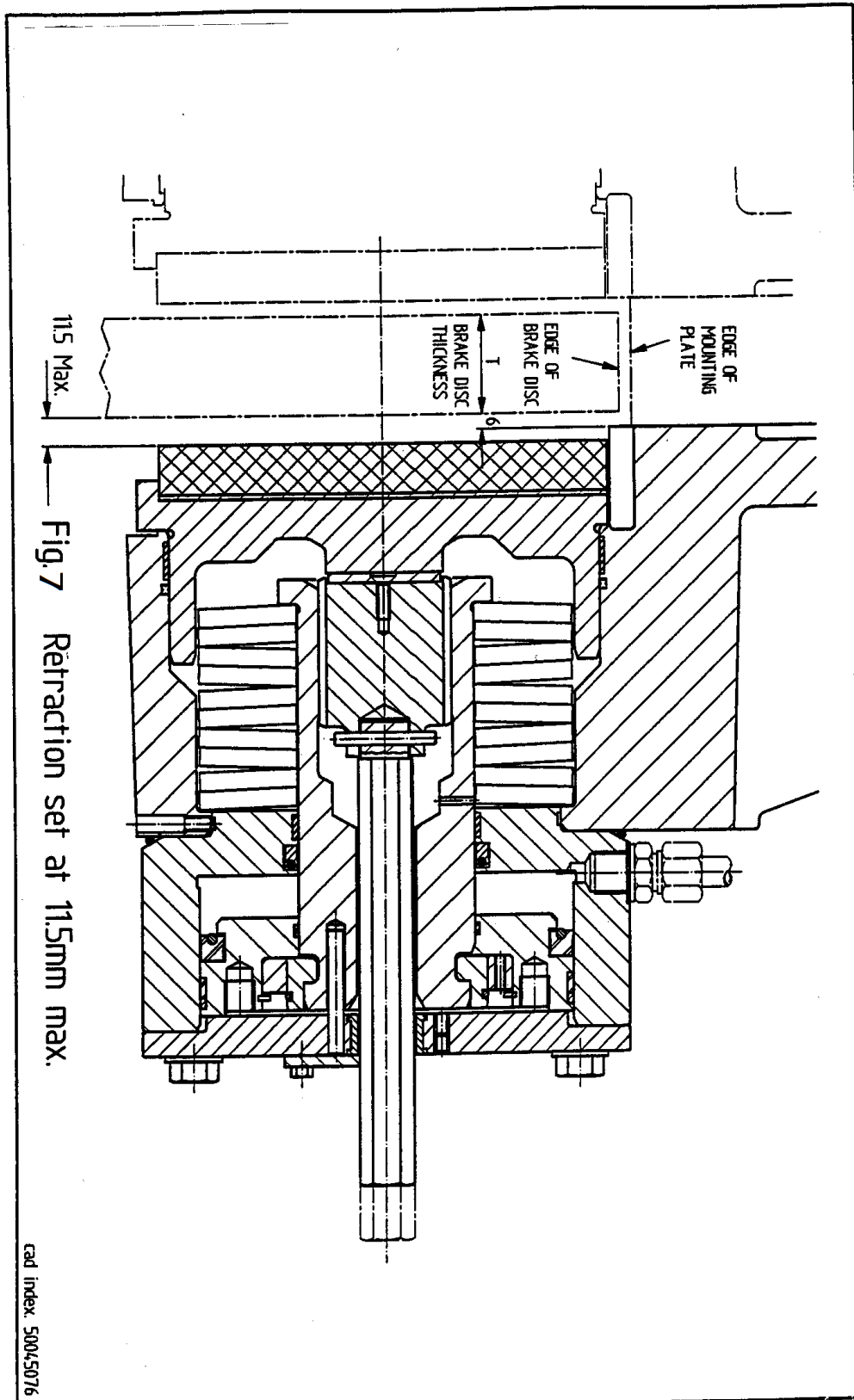
- Release brake by means of hand-pump connected direct to feed port.
- Turn spindle anti-clockwise to obtain 'Parked-Off' position.

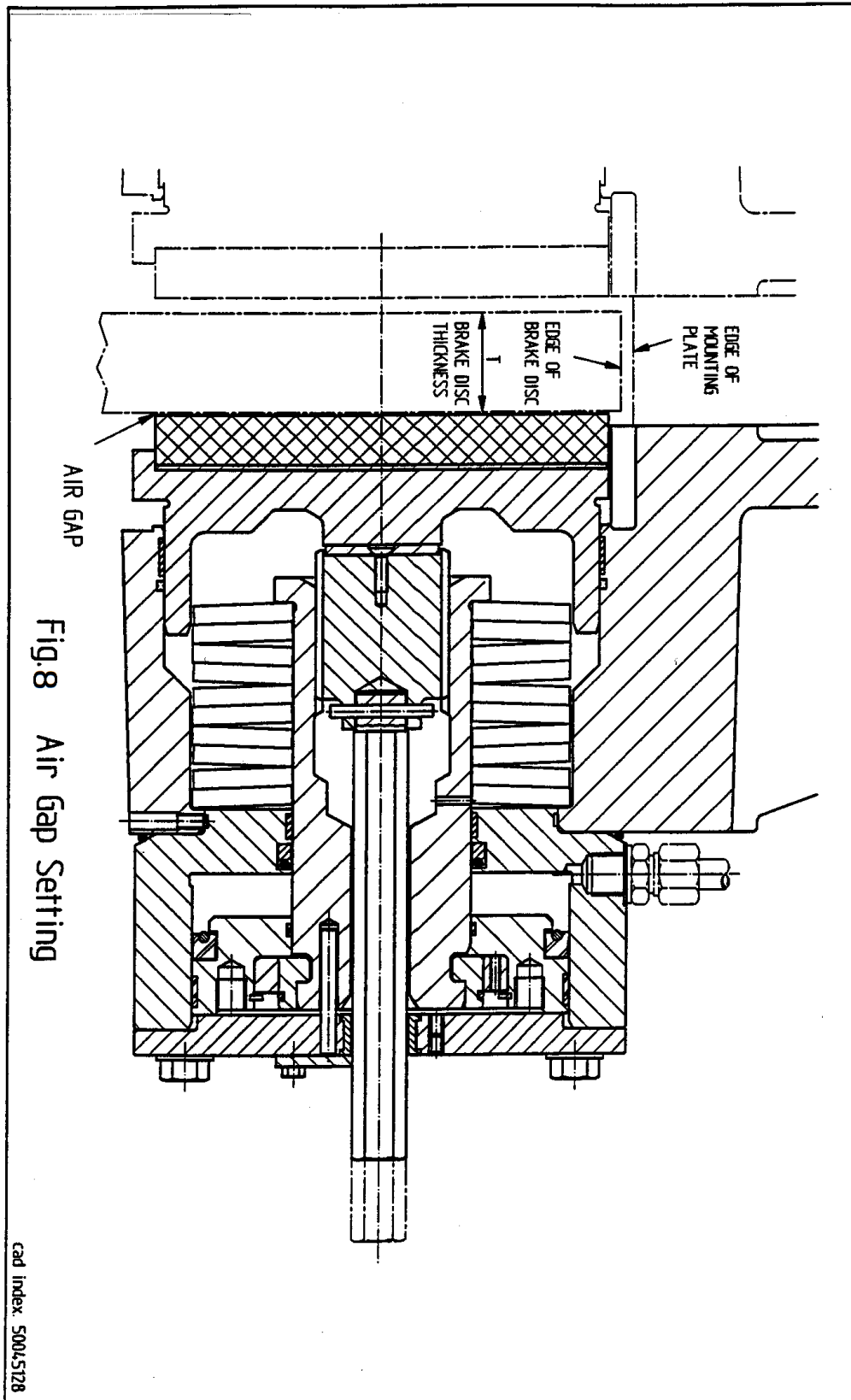
Mechanical Retraction

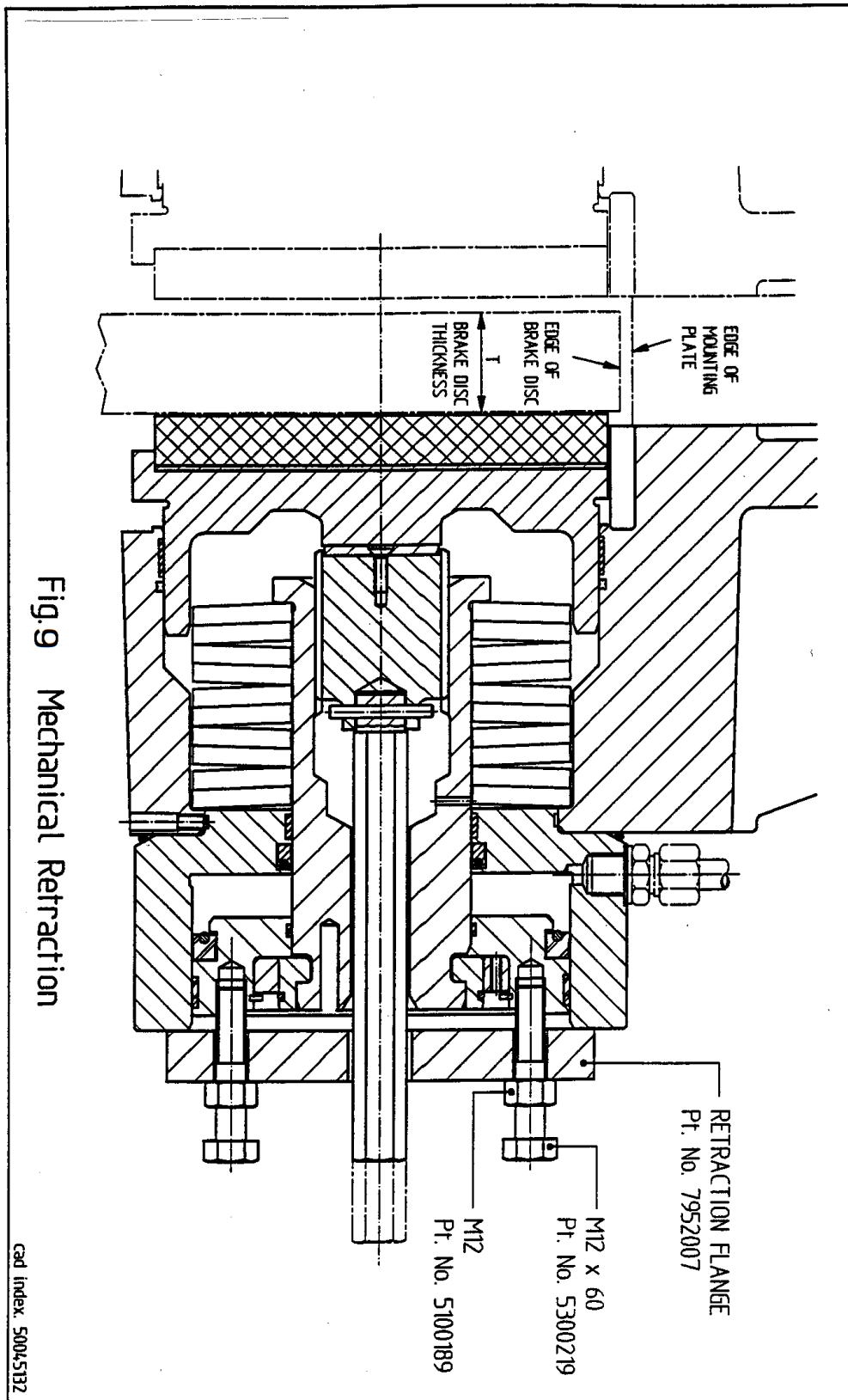
- Fit retraction tool (Fig. 9).
- Release brake by turning nuts the same amount in turn.
- Turn spindle anti-clockwise to obtain 'Parked-off' position.











Appendix 4 - Inductive Analogue Sensor



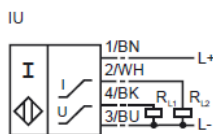
Model Number

IA6-12GM50-IU-V1

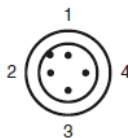
Features

- Inductive analog
- 0...6 mm quasi-embeddable
- Analog output 0 V ... 10 V/4 mA ... 20 mA

Connection

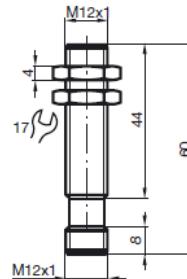


Pinout



Wire colors in accordance with EN 60947-5-2

1	BN	(brown)
2	WH	(white)
3	BU	(blue)
4	BK	(black)



Technical Data

General specifications

Switching element function	analog, current or voltage output
Installation	quasi embedd.
Output polarity	DC
Measurement range	0 ... 6 mm

Nominal ratings

Operating voltage	U_B	15 ... 30 V
Limit frequency (3dB)		1000 Hz
Reverse polarity protected		yes
Short-circuit protection		yes
Repeat accuracy		0 ... 10 μ m
No-load supply current	I_0	≤ 10 mA

Analog output

Output type	0 ... 10 V / 4 ... 20 mA
Load resistor	voltage output: $\geq 500 \Omega$ current output: $\leq 500 \Omega$

Recovery time	1 ... 10 ms, typ. 5 ms
Adjustment tolerance zero point	$\leq \pm 10$ % of full-scale value
Temperature drift	± 5 % (0 ... 70 °C) ± 10 % (-25 ... 0 °C)
Residual ripple	20 %

Ambient conditions

Ambient temperature	-25 ... 70 °C (-13 ... 158 °F)
---------------------	--------------------------------

Mechanical specifications

Connection type	Device connector M12 x 1, 4-pin
Housing material	chromium plated brass
Sensing face	PBT
Protection degree	IP67

Compliance with standards and directives

Standard conformity	
Standards	EN 60947-5-2:2007 IEC 60947-5-2:2007 EN 60947-5-7:2003 IEC 60947-5-7:2003

Approvals and certificates

UL approval	cULus Listed, General Purpose
CSA approval	cCSAus Listed, General Purpose
CCC approval	Products with a maximum operating voltage of ≤ 36 V do not bear a CCC marking because they do not require approval.

5. MANUAL FOR CONTROL UNIT MAU-075-001

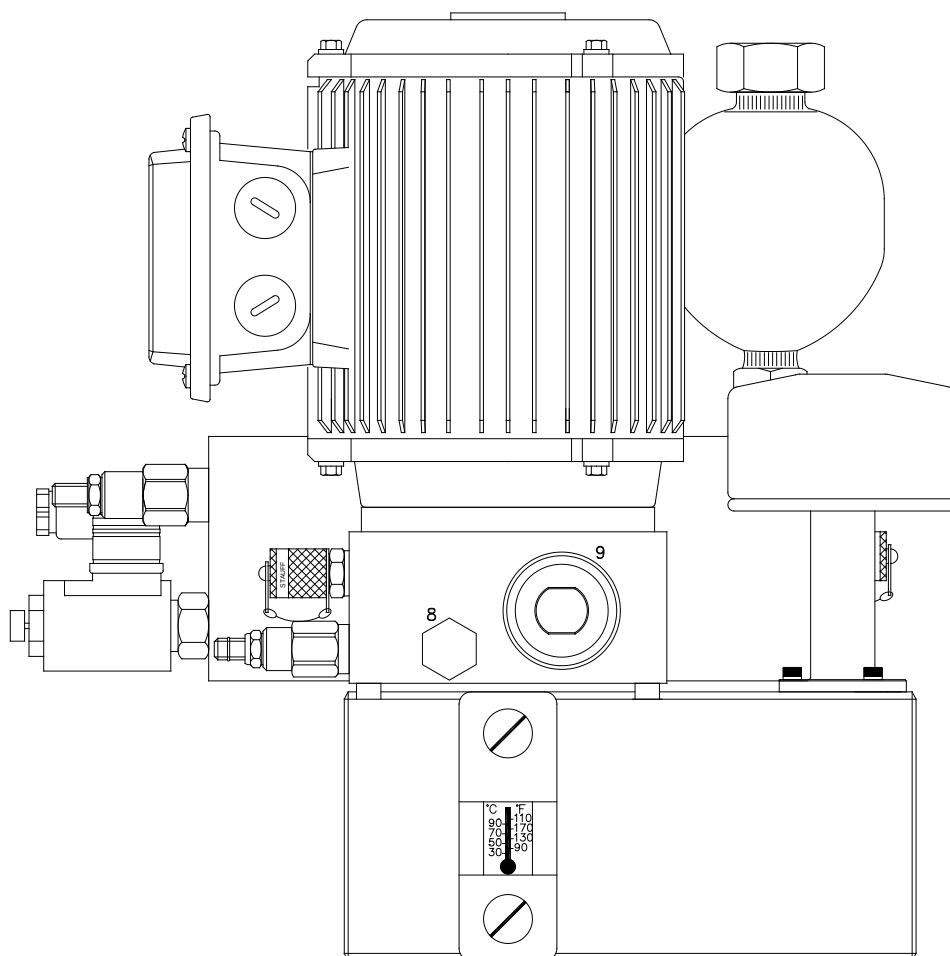
Brake Control Unit Type: BCU-A/I 75-02-2X-XX

This manual covers the complete range of Brake Control Units with an 02 control function, The A/I indicates A as the pre 2008 style manifolds and the I indicates the post 2007 manifolds (both styles available as some installations like commonality across the site). The “X” signifies the control voltage i.e. 4 = 24 VDC & 5 – 48VDC. The last two digits of the code represent a design parameter for internal use only.

OPERATION, INSTALLATION and MAINTENANCE MANUAL

MAU-075-001

BCU-A/I 75-02-2X-XX Brake Control Unit



Written by: M. Boustead, Revision 0 06/12/12

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1. Operation

The hydraulic brake control unit BCU-A/I75-02-2X-XX has been developed to be either a stand-alone or integral with brake control unit known as an 'OMEGA' unit for the operation of Svendborg failsafe spring applied hydraulic release brakes. This version '02' option provides easy site adjustable braking torque and site adjustable braking time. The system provides constant torque braking under all conditions. The brake control unit can be connected to one or more brakes. (qty subject to approval)

This manual deals with the type mentioned only. The control unit can be supplied with motors and directional control valves of different voltages.

1.1. Lubrication Schedule

Oil grade	ISOVG 46 hydraulic oil, for a temperate to cool climate ISOVG 68 hydraulic oil, for a temperate to hot climate
Oil volume	5.0 Litres nominal

Note:

- a) Oil quantities are stated above but are to be checked for correct fill level by sight glass in the tank or reservoir.
- b) Greater oil volume will be required for bleeding.
- c) All oil must be filtered to a 10-micron nominal filter rating when filling.
- d) For more detail on oils see section 2.8.

1.2. Function

(See schematic, sect 9 3.1)

When the brake control unit is connected to the brake, the unit is to be operated with the electric motor (3) powered via the pressure switch (13), via the MOTOR contact [PS-1].

- 1.2.1. Brake off: Power is connected to the electric motor (3) and solenoid coil (17) [SV-1], so that the directional control valve (16) is energized in the closed position. The pump increases the pressure, the brake releases, and the accumulator (12) is filled with oil. On reaching the set rising pressure the pressure switch (13) via the MOTOR contact [PS-1] will disconnect the supply to the motor, and pressure will be retained in the system. If the pressure should drop to the falling setting, the pressure switch (13), via the MOTOR contact [PS-1] will reconnect the supply to the motor and thereby boost the pressure. Pressure switch (13), via the Analogue output [PT-1] provides an indication of caliper pressure and can be monitored to indicate the "brake off" condition.
- 1.2.2. Brake on: Power is disconnected to both motor (3) and solenoid (17) [SV-1]. Oil pressure is dumped to tank through the orifice (19) and back-pressure relief valve (18), the brake is applied with set pressure (constant torque) as described below.

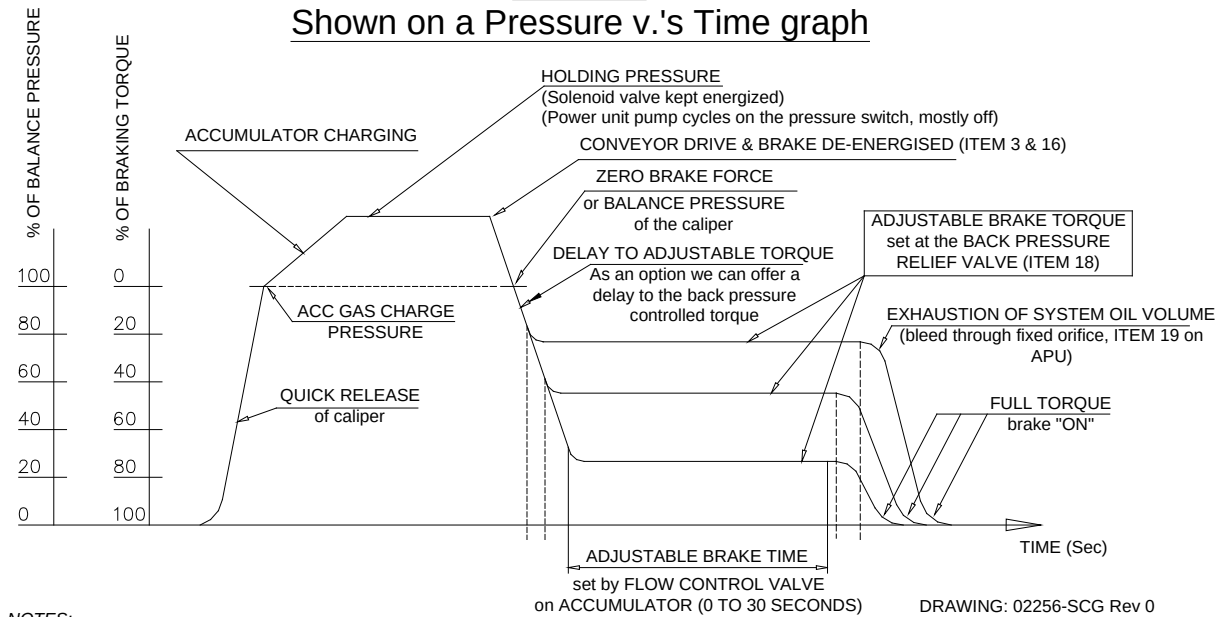
Controlled braking torque: The oil pressure is bled to the tank through the orifice (19) and back-pressure relief valve (18), resulting in a ramp to a reduced constant braking torque. Controlled time is set on the flow control valve (11). When the accumulator (12) is fully discharged, the orifice (19) releases the residual pressure, and full torque is applied.

The constant brake torque and braking time is pre-set from factory as per the table in Sect 9-1, this can be adjusted on site.

Please see the graph on the following page detailing the control philosophy.

CONVEYOR BRAKE SET UP - OPTION 02
(Typical)

Shown on a Pressure v.'s Time graph



2. Installation

2.1. Transport

When leaving the factory the brake control unit is always packed in such a way to guarantee maximum security during transport.

2.2. Lifting and handling

The BCU-A75 brake control unit weighs approx. 31.5 kg. If it is necessary to use a lifting harness, under no circumstance must it be placed around valves, gauges or the accumulator etc.

2.3. Mounting

For mounting purposes the brake control unit has 4 x 8 mm holes in the side of the manifold (23). If the brake control unit is supplied as part of an integral 'Omega' unit the brake control unit will be mounted on the brake, or can be supplied mounted in an enclosure.

2.4. Hydraulic connections

The brake control has 3 x 3/8" BSP pressure ports, in the manifold (23). If the brake control unit is supplied as part of an Omega unit, the brake control unit and brake caliper will be connected.

When installing connecting hydraulic lines, ensure that BSPP (**parallel**) fittings are used in both the brake control unit and brake ports.

We only accept for system integrity if rigid lines are chosen of the JIC flared tube type. We do **NOT** accept the cutting or compression ring fitting type rigid line termination.

All tubing shall be seamless annealed tube, rated to the operating pressure. If stainless steel tube is considered it shall be seamless annealed grade 304 or 316 material. Flexible lines rated to operating pressure and conditions can also be used. All lines shall be suitably supported at a maximum interval of 600mm

Steel lines should be externally protected by grit blast to class 2 ½ and epoxy coated.

Lines must be cleaned and flushed with turbulent oil velocity.

All lines must be pressure tested to 1.5 times the operating pressure of the system.

For further details on hydraulic lines see Kempe Industrial Products Specification SPAU-004.

2.5. Options

If required a valve cover kit can be supplied to protect and make tamperproof all adjustable valves. Please enquire.

2.6. Electrical connections

Refer to the electrical schematic Sect 9-4.

It is the customers responsibility to ensure that all electrical connections are made in accordance with AS/NZS 3000-2000 and are sealed from contaminate and moisture ingress. No warranty will be considered if this is not completed accordingly. If required we can supply IP 67 Fly lead valve coil and pressure switch connectors, standard cable length 5m, other on request.

2.7. Set up and commissioning

Following the installation manual with a pressure gauge and stopwatch easily sets of the timed constant torque control.

However if required, Kempe Industrial Products can offer full site commissioning with data recorders to more accurately set provide printed documentation of the timed constant torque control.

If you are in any doubt regarding installation of this unit ASK, contact the local branch quoting model and serial number.

For further details on installation and commissioning see Kempe Industrial Products specification SPAU-005.

Kempe Industrial Products
Kempe Engineering Services Pty Ltd
50 Gheringhap Street Geelong VIC 3220 AUSTRALIA
Ph 03 5225 2900 Fax 03 5225 2950

2.8. Hydraulic fluid

2.8.1. General

The following oil types are recommended by Kempe Industrial Products – other oil types that are compatible to this can be used.

The minimum and maximum temperatures for standard components are -20°C to +60°C or (-14°F to +140°F). Extreme temperature applications may require special options, please contact Kempe Industrial Products.

2.8.2. Type of fluid

A selection can be made with the table below (see section 4. on viscosity for selection)

TYPE	SUPPLIER	ISO Viscosity class	
		VG 46	VG68
Mineral Oils	BP	Energol HLP 46	Energol HLP 68
	CASTROL	HYSPIN AWS 46	HYSPIN AWS 68
	ESSO	NUTO H 46	NUTO H 68
	FUCHS	RENOLIN MR15 VG 46	RENOLIN MR 20 VG 68
	MOBIL	Mobil DTE 25 Mobil DTE Excel 46	DTE 26 Mobil DTE Excel 68
	SHELL	Shell Tellus Oil 46 Shell Tellus Oil DO 46 Shell Tellus Oil S 46	Shell Tellus Oil 68 Shell Tellus DO 68 Shell Tellus S 68
	TOTAL	Azolla ZS 46	Azolla ZS 68
	SAE Motor Oils		SAE 20 W 20
Bio Oils (1)	Shell	Naturelle HF-E46	Naturelle HF-E68

Note: (1) There must be less than 200 PPM water in oil.

2.8.3. Filtration

The oil in a hydraulic system and the oil added to the hydraulic system must always be filtered. The level of cleanliness in a hydraulic system is an important factor to the lifetime of the system. When refilling the hydraulic tank, it is recommended that the oil added to the system is filtered through an off-line filter unit with minimum 10µm nominal filtration.

The hydraulic brake control unit, supplied by Kempe Industrial Products, has a build-in oil pressure filter with a standard minimum 10-micron nominal filter rating. To maintain a reliable system it is recommend that only hydraulic oil of the following classes of purity is used: NAS 1638, Class 8, or ISO 4406, Class 19/17/14.

It is recommended that the filter be changed at a minimum, once every six months or more often depending on the level of exposure to contamination.

2.8.4. Service life of hydraulic fluids (oil)

Mineral oil: 8.000 Hours or at least once a year.

Other fluids: 2.000 Hours or at least once a year.

2.8.5. Viscosity

Be sure that the oil viscosity is within the recommended range of 20 cSt / 98 SSU or 20-200 cSt at working temperature. The chart below will assist in determining this

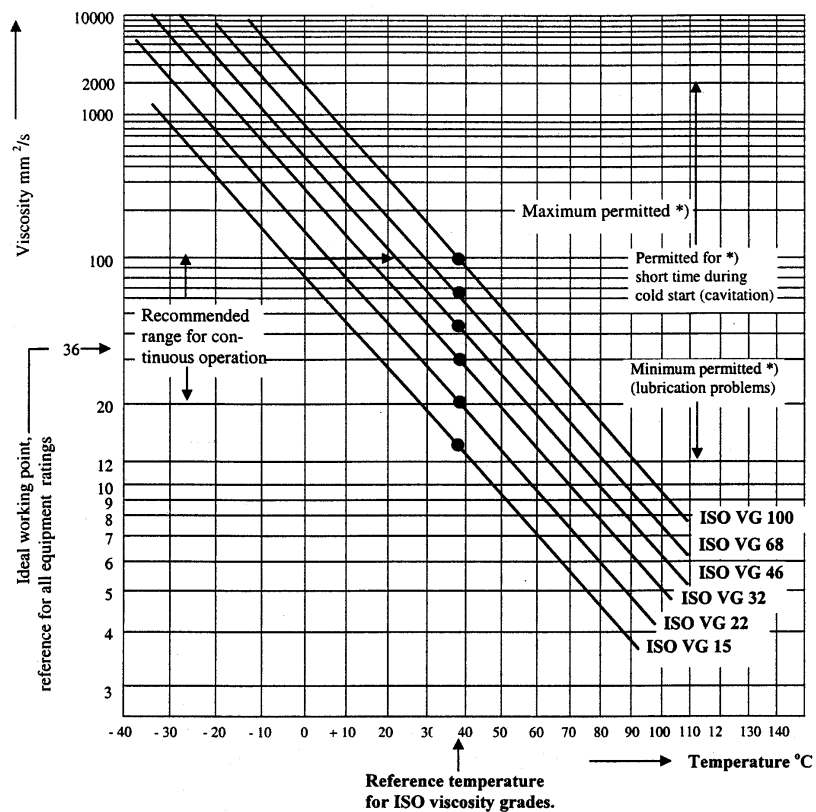
In Australia, the general recommendation is as follows:

VG 46 – Cool to Temperate climates (+10°C to +60°C)

VG 68 – Temperate to Hot climates (+30°C to +70°C)

Installation and Maintenance Manual MAU-075-001

When selecting an oil, keep in mind the temperature variance between summer, winter, day and night and its effect of oil bleed down rated through orifice (19 / 33)



2.8.6. Change of pressure fluid

Note: The mixing or blending of different brands or types of pressure fluid can cause unintended chemical reactions, such as sludging, gumming etc. The complete hydraulic system should be thoroughly flushed prior to changing from one oil type to another.

The respective manufactures should be contacted prior to changing from one pressure fluid to another.

2.9. Before starting the brake control unit

- 2.9.1. Fill up the tank on the brake control unit with oil of the recommended quality (see section 2) filtered to 10 micron. The oil volume is approximately 5.0 ltr. Fill up the tank until the oil level can be seen in the sight glass to maximum level.
- 2.9.2. Connect the electric motor (3), pressure switch (13), and solenoid coil (17) to recommended voltage (see section 9-5 for details).
- 2.9.3. The relief valve (7) is pre-set at the factory to the correct pressure, (see section 9-1). If adjustment is needed, loosen the lock nut and, using an Allen key, screw out (anti-clockwise) to decrease the maximum pressure. Screw in (clock-wise) to increase maximum pressure. Remember to re-tighten the lock nut.
- 2.9.4. The pressure switch (13) is mounted on the brake control unit. The pressure switch is set at the factory, if adjustment is needed (see 3.5 and section 9-1 for settings).
- 2.9.5. Start the electric motor: Check that the rotation of the electric motor (3) is anti-clockwise viewed from the rear end, with the NENV motor the direction cannot be viewed, therefore ensure that hydraulic pressure increases and that the disc brake pistons are lifted within a maximum of 15 seconds.
- 2.9.6. **Note:** If the brake needs to be released without the brake control unit running, it is necessary to use a hand pump. The solenoid valve (16) is fitted with a manual override button. The hand pump can be connected at the port (A4) on the brake control unit or any other test point in the system, see 3.6 for more detail.

2.10. Adjusting torque & brake time

The setting of items 07, 11, 13, 18 and 19 should not be touched as they are preset (See section 9-1 for settings). If however they are required to be adjusted follow as noted.

2.10.1. Setting Controlled Braking Torque:

- ⇒ Loosen lock nut on valve (18).
- ⇒ Use allen-key: Turn out (counter-clockwise) to increase braking torque.
Turn in (clockwise) to decrease braking torque.

(Unless stated otherwise in section 9-1, the pressure is factory set at 50% of the balance pressure)

- ⇒ Check torque (pressure) with a pressure gauge in test point (15). This is a "Minimess" coupling.
- ⇒ Start electric motor (3) and energise solenoid coil (17) [SV-1] to close the directional control valve (16), opening the brake, and allow the pressure switch (13) via the MOTOR contact to stop the motor, and turn off manually when maximum pressure is reached.
- ⇒ De-energize the solenoid coil (17) [SV-1] to open directional control valve (16). Now you can read counter pressure. Full braking torque is 1-2 bar counter pressure.

Zero braking torque is equal to balance pressure - see table Sect 9-1.

- ⇒ Tighten lock nut on valve (18).

Installation and Maintenance Manual MAU-075-001

2.10.2. Setting Controlled Braking Time:

- ⇒ Loosen the lock nut on valve (11).
 - ⇒ Use allen-key: Turn out (counter-clockwise) to decrease braking time.
Turn in (clockwise) to increase braking time.
- (Unless stated otherwise in section 9-1, the valve is factory set ½ turn out from fully closed ~ 12 secs.)
- ⇒ After adjusting the flow rate, re-check the torque setting. The braking time is a balance between the 2 valves.
 - ⇒ Check timing with a pressure gauge in test point (15).
 - ⇒ Start electric motor (3) and energise solenoid coil (17) [SV-1] to close the directional control valve (16), opening the brake, and allow the pressure switch (13) via the MOTOR contact to stop the motor, and turn off manually when maximum pressure is reached.
 - ⇒ De-energize the solenoid coil (17) [SV-1] to open directional control valve (16), and count the seconds until the pressure drops to approx. 3-4 bar.
 - ⇒ Tighten the lock nut on valve (11).

3. Maintenance

All pressure control valves, flow control valves, pump regulators, and signalling devices such as pressure switches are adjusted during the initial manufacture and subsequent commissioning. The settings must be monitored continuously during the initial stages of operation and then subsequently at reasonable intervals.

By remaining alert and paying attention to detail it is possible to detect faults in their very early stages and so prevent them from developing into more serious malfunctions. This is particularly true during the early stages but also remains true throughout the service life of the equipment. A constant look out should be kept for: -

- External leaks
- Dirt
- Damage, especially to hoses and pipes
- Unusual noises from pumps, motors, couplings etc
- Proper function of equipment including by way of instruments

A maintenance regime and log should be established to ensure the equipment is kept in peak operating condition.

3.1. Maintenance schedule.

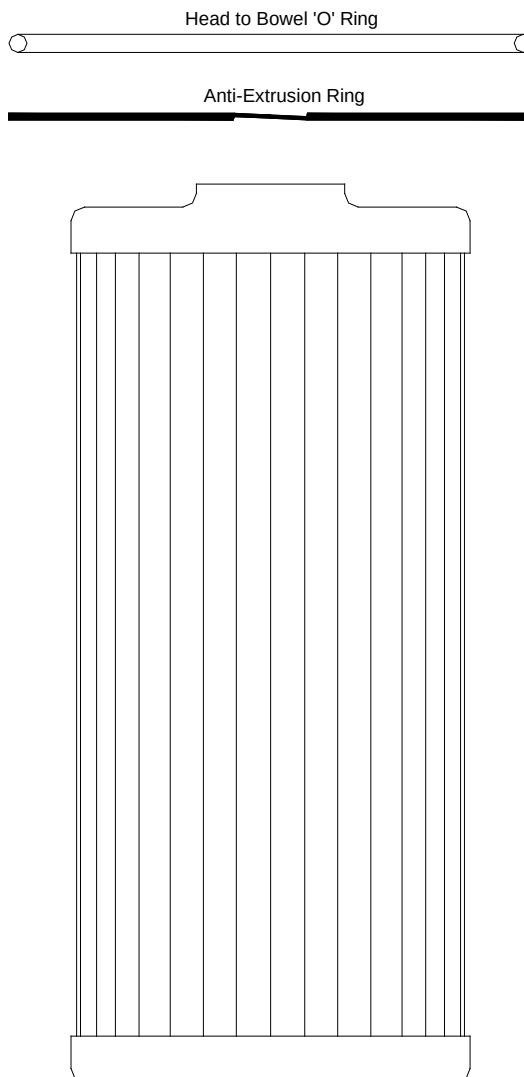
The recommended schedule is as per the following table, however in arduous condition or when conditions dictate the interval should be increased

Checks	Continuously / Daily	Weekly	Monthly	Six Monthly	Yearly
Fluid					
Level		√			
Temperature		√			
Condition (change)					√
Filters					
Changing element					√
Changing breathers				√	
Accumulators					
Check gas pressure				√	
Check mounting				√	
Tests / adjustments					
Pressure & flow valves					√
Signalling devices (pressure switches & sensors)					√
Miscellaneous checks					
External leaks	√				
Dirt	√				
Damage	√				
Noises	√				
Instruments	√				

3.2. Replacing the hydraulic filter

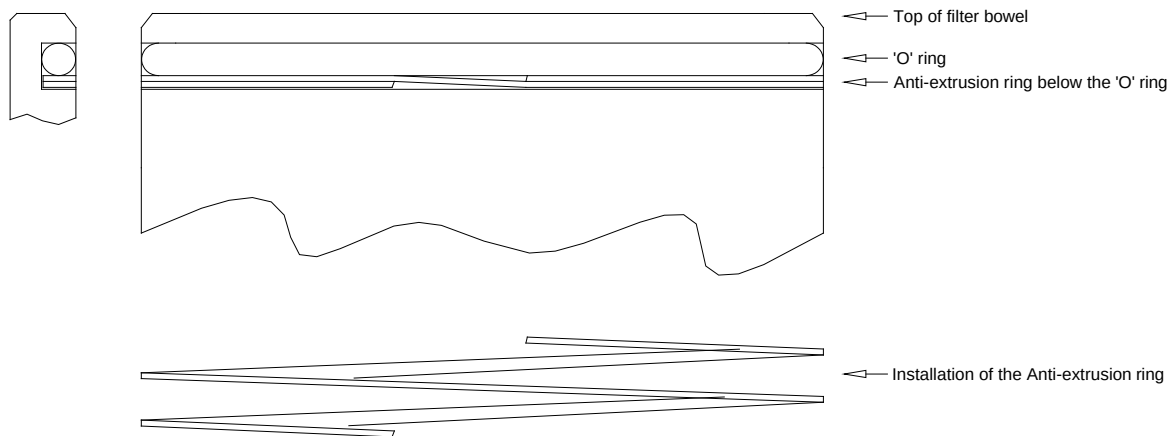
The brake control unit is supplied with a high-pressure filter (9) mounted inside the manifold (2). The filter must be replaced at a minimum once a year.

- 3.2.1. Ensure a new clean element is at hand of the correct type including micron rating.
- 3.2.2. Make sure that the electric motor is not running, that no voltage is connected to the solenoid valve, (system isolated) and that the system oil pressure is drained (0 Bar).
- 3.2.3. Clean dust and dirt from the outside surface of the hydraulic unit before removing the bowl.
- 3.2.4. Remove the high-pressure filter (9) by screwing out the filter bowl and remove the filter element and bowl. Fit the new high-pressure filter and re-fit the cap, tighten to 100Nm.
- 3.2.5. Discard the filter element, clean the bowl internals with a clean lint free rag.
- 3.2.6. Check the condition of the filter bowl 'O' ring and Anti-extrusion ring and replace as necessary. Use Hydac seal kit P/No "Repair Kit –E MDF30FG"



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- 3.2.7. When fitting a new bowl 'O' ring and Anti-extrusion ring ensure it is in the correct order with the 'O' ring to the top of the bowl.



- 3.2.8. Fit the new high-pressure filter and re-fit the cap, tighten to 100Nm.
- 3.2.9. Check the oil level and re-fill if necessary.
- 3.2.10. Start the brake control unit and make sure that there is no leakage from the filter bowl.

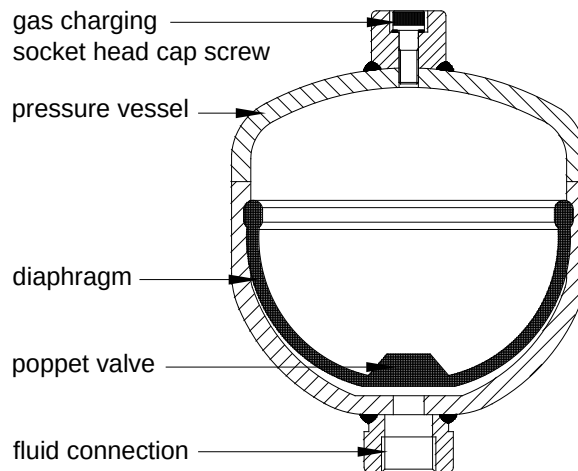
3.3. Replacing the hydraulic fluid

The brake control unit is supplied with a hydraulic oil tank that must be drained, cleaned and fluid replaced at a minimum once a year.

- 3.3.1. Make sure that the electric motor is not running, that no voltage is connected to the solenoid valve, (system isolated) and that the system oil pressure is drained (0 Bar).
- 3.3.2. Clean dust and dirt from the outside surface of the hydraulic unit before removing plugs or fittings.
- 3.3.3. Remove the four bellhousing / tank screws and the tank will fall away below, take care to ensure it does not drop.
- 3.3.4. Clean the tank out with dry clean rag, ensuring that any contaminates are removed and recorded in a log.
- 3.3.5. Refill with hydraulic fluid as per 2.8, ensuring the tank square section seal is in place mounted on the tank, top entry hole.
- 3.3.6. Start the brake control unit and make sure that there is no leakage.
- 3.3.7. Check the oil level and re-fill if necessary.

3.4. Accumulator

3.4.1. General



On no account must any welding, soldering or mechanical work be carried out on the accumulator shell. After the hydraulic line has been connected it must be completely vented. Work on systems with accumulators (repairs e.g. connecting pressure gauge etc.) must only be carried out once the pressure and fluid has been released, (system isolated). All care must be taken in charging accumulators with reference to industry standard, Workcover and manufacturer work procedures. Any supplier recommendations take precedence over the below text and must be adhered to.

3.4.2. Function

Fluids are practically incompressible and cannot, therefore, store pressure energy. In hydro-pneumatic accumulators the compressibility of a gas is utilized for storing fluids. The diaphragm accumulators are based on this principle using nitrogen as the compressible medium. The diaphragm accumulator consists of a fluid section and a gas section with the diaphragm acting as a gas-proof screen. The fluid section is connected to the hydraulic circuit, so that the diaphragm accumulator draws in the fluid when the pressure increases and the gas is compressed. When the pressure drops, the compressed gas expands and forces the stored fluid into the circuit. At the base of the diaphragm is a pre-vulcanized poppet valve. This shuts off the hydraulic outlet when the accumulator is completely empty and thus prevents damage to the diaphragm.

3.4.3. Charging with Gas

Use only dry nitrogen for charging accumulators. Never use oxygen! If the gas pressure in the nitrogen bottle is higher than the maximum operating pressure of the accumulator, a gas pressure-reducing valve has to be fitted.

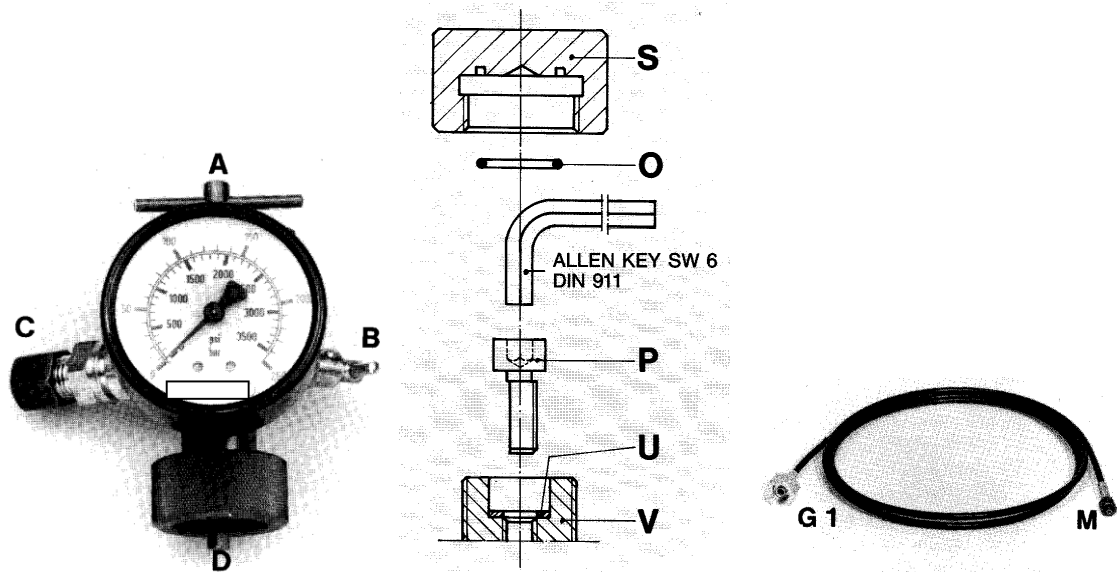
Equipment required

- Charging head c/w hose
- Bottle of nitrogen.
- 6mm Allen key
- Leak detector
- Grease

The flexible charging hose is connected to the shut-off valve of a commercially available nitrogen bottle by means of the cap nut G1. Cap nut M of the charging hose is for the connection with nipple C, of the charging and testing unit, tighten.

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- 3.4.3.1. Using an allen key loosen the screw P in the accumulator only enough without loosening gas.
- 3.4.3.2. Install the charging head onto the accumulator.
- 3.4.3.3. Close valve B
- 3.4.3.4. Opening the spindle valve A thus opening the screw P in the accumulator
- 3.4.3.5. Open the shut-off valve on the nitrogen bottle. Slowly release nitrogen into accumulator, to ensure better pressure compensation and to prevent damage to the gas valve insert.
- 3.4.3.6. Close shut-off valve of the nitrogen bottle and check charging pressure on the gauge.
- 3.4.3.7. Repeat this process until the required charging pressure is reached. After a period of approx. 5 minutes, during which time temperature compensation takes place, re-check pressure and adjust if necessary.
- 3.4.3.8. When the required charging pressure is reached, tighten unscrew spindle A on the accumulator. This is so that the screw P in the accumulator closes the gas valve V. Now the charging and testing unit is unscrewed and the cap-screw P is re-tightened with a 6mm allen key to a torque of 20 Nm.
- 3.4.3.9. Check the gas valve on the accumulator for leakage's by means of leak detector spray or soapy water.
- 3.4.3.10. Smear the top of the accumulator around the gas valve and the cap thread with a little light grease to stop corrosion and screw the protective cap onto the gas valve of the accumulator and tighten.



3.5. Pressure switch

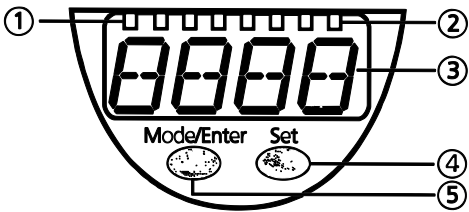
Note: Any instructions in grey are not applicable to the switch fitted to the brake control unit this manual represents.



3.5.1. Safety Instructions

Please read the following prior to installing / operating the unit. If the operating instructions or the technical data are not adhered to, personal injury and / or damage to property may occur. Please check in all applications that the product materials (see Technical data) are compatible with the media to be measured.

For gaseous media the application is limited to max. 25Bar.



Controls and indicating indication:

①	3 (4) x LED green	Lighting LED = set display unit
②	2 x LED yellow	Switching status; lights if the respective output has switched
③	4-digit Alphanumerical display	Display of the system pressure, display of parameters and parameter values
④	Set button	Setting of the parameter values (scrolling by holding pressed; incremental by pressing briefly)
⑤	Mode / Enter button	Selection of the parameters and acknowledgement of the parameter values.

3.5.2. Function and features

- The pressure sensor **detects the system pressure**,
- shows the current system pressure on its **display**,
- and generates **2 output signals** according to the set output configuration.

	Output 1	Output 2
Switching function (output 1 and output 2; function can be selected for each output separately)	Hysteresis function / N.O. (Hno)	
	Hysteresis function / N.C. (Hnc)	
	Hysteresis function / N.O. (Fno)	
	Hysteresis function / N.C. (Fnc)	
Output polarity (applies to both switching outputs)	p-switching (PnP)	
	n-switching (nPn)	

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Applications:

Part Number	Measuring range	Permissible over pressure	Bursting pressure
800 1013-XXX	0 ... 250 bar	400 bar	850 bar

Warning:

! Avoid static and dynamic overpressure exceeding the given overload pressure.

Even if the bursting pressure is exceeded for a short time the unit can be destroyed (danger of injuries)!

For gaseous media the application is limited to max. 25 bar.

! Indication of the current system pressure as from 1 % of the value of the measuring range. Display "0" does not mean that the system is free of pressure !

3.5.3. Operating modes

3.5.3.1. Run mode:

(Normal operating mode)

At power on, the unit is in the Run mode. It carries out its measurement and evaluation functions and provides output signals according to the set pressure.

The display shows the current system pressure. The yellow LEDs indicate the switching state of the outputs.

3.5.3.2. Display mode:

(Indication of parameters and the set parameter values)

When the "Mode/Enter" button is pressed briefly, the unit passes to the Display mode which allows parameter values to be read. The internal sensing, processing and output functions of the unit continue as if in run mode.

- The parameter names are scrolled with each pressing of the "Mode/Enter" button.
- When the "Set" button is pressed briefly, the corresponding parameter value is displayed for 15s. After another 15s the unit returns to the Run mode.

3.5.3.3. Programming mode:

(Setting of the parameter values)

While viewing a parameter value pressing the "Set" button for more than 5s causes the unit to enter the programming mode. You can alter the parameter value by pressing the "Set" button and confirm the new value by pressing the "Mode/Enter" button. The internal sensing, processing and output functions of the unit continue as if in Run mode with the original parameter values unless a new value is confirmed (except parameter **COF**).

The unit returns to the Run mode when no button has been pressed for 15 s.

3.5.4. Installation

! Before mounting and removing the sensor, make sure that no pressure is applied to the system.

3.5.5. Electrical connection

! The unit must only be connected by an electrician.

The national and international regulations for the installation of electrical equipment must be observed.

Voltage supply to AS/NZS 3000:2000

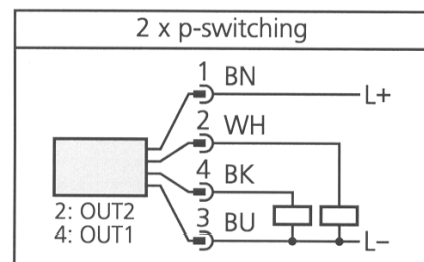
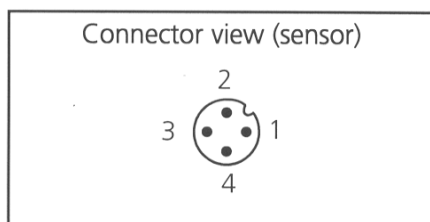
The device shall be supplied from an isolating source and protected by an over current device such that the limited voltage circuit requirements in accordance with UL 508 are met.

Disconnect power before connecting the unit as follows.

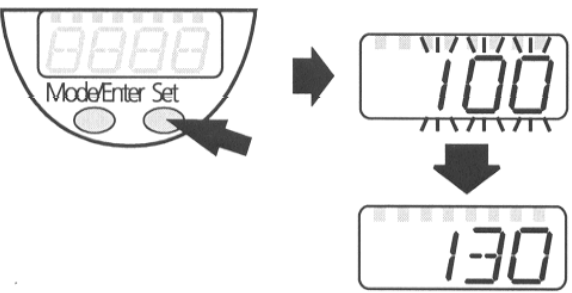
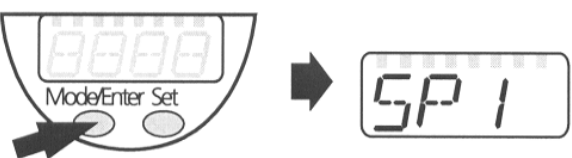
Core colours of IFM sockets:

1 = BN (brown), 2 = WH (white),

3 = BU (blue), 4 = BK (black).



3.5.6. Programming

1		Press the Mode/Enter button several times until the respective parameter is displayed.
2		Press the Set button and keep it pressed. The current parameter value flashes for 5s, then the value is increased* (incremental by pressing briefly or scrolling by holding pressed).
3		Press the Mode/Enter button briefly (= acknowledgement). The parameter is displayed again, the set parameter value becomes effective .
4	Change more parameters: Start again with step 1.	Finish programming: Wait for 15s or press the Mode/Enter button until the current measured value is indicated again.

*Decrease the value: Let the display of the parameter value move to the maximum setting value. Then the cycle starts again at the minimum setting value.

Select the display unit (**Uni**) **before** setting values for the parameters **SPx** and **rPx**. This avoids rounding errors generated internally during the conversion of the units and enables exact setting of the values. Setting at the factory: **bAr / mbAr**.

If no button is pressed for 15s during the setting procedure, the unit returns to the Run mode with unchanged values.

The unit can be electronically locked to prevent unwanted adjustment of the set parameters: Press both pushbuttons until **Loc** is displayed.

To unlock: Press both pushbuttons until **uLo** is displayed. Units are delivered from the factory in the unlocked state.

With the unit in the locked state **Loc** is indicated briefly when you try to change parameter values

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3.5.7. Installation and set-up / Operation

After mounting, wiring and setting check whether the unit operates correctly.

Fault indication:

OL	Overload (above measuring range of the sensor)
UL	Underload (below measuring range of the sensor)
SC 1	Flashing: short circuit in the switching output 1*
SC 2	Flashing: short circuit in the switching output 2*
SC	Flashing: short circuit in both the switching outputs*

* The output concerned is switched off as long as the short circuit continues.
These faults are indicated even if the display is deactivated.

3.5.8. Technical information / Functioning / Parameters

Adjustable parameters

SP1 SP2	Switch-on point $\frac{1}{2}$: Upper limit value at which the output changes its switching status. setting range 2 to 250 bar indicated in bar. For motor switch this is the motor switch OFF point. For the release switch this is the release switch ON point.
rP1 rP2	Switch-off point $\frac{1}{2}$: Lower limit value at which the output changes its switching status. rPx is always lower than SPx. The unit only accepts values which are lower than SPx. Changing the switch-on point also changes the switch-off point (the distance between SPx and rPx remains constant). If the distance is higher than the new switch point, it is automatically reduced (rPx is set to the minimum setting value). For the motor switch this is the motor switch ON point. For the release switch this is the release switch OFF point.
OU1 OU2	Switching functions of the switching outputs 4 settings can be selected: Hno = hysteresis / normally open Hnc = hysteresis / normally closed Fno = window function / normally open - Not used in this application Fnc = window function / normally closed - Not used in this application
EF	Enhanced functions This menu item contains a submenu with additional parameters. You can access these parameters by pressing the SET button briefly

Enhanced functions

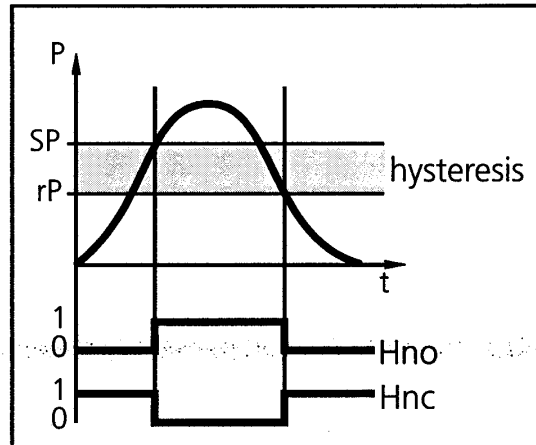
HI LO	Min-Max memory for the system pressure • Hi: displays the highest measured pressure • LO: displays the lowest measured pressure Erase the memory: - Press the "Mode/Enter" button until HI or LO is displayed. - Press the "Set" button and keep it pressed until "- - -" is displayed. - Then press the "Mode/Enter" button briefly.
------------------------	---

COF	Calibration offset The internal measured value (operating value of the sensor) is offset against the real measured value. • Setting range: -5 ... +5% of the span. The new value of the parameter COF becomes effective immediately. Press the Mode/Enter" button briefly to store the new value permanently.									
CAR	Calibration reset Resets the calibration set by COF to the value set at factory. - Press the "Mode/Enter" button until Car is displayed. - Press the "Set" button and keep it pressed until "- - -" is displayed. - Then press the "Mode/Enter" button briefly.									
dS1 dS2 dr1 dr2	Delay time for the switching outputs - Not Used In This Application dSx = switch-on delay; drx = switch-off delay The output does not immediately change its switching status when the switching condition is met but when the delay time has elapsed. If the switching condition is no longer met when the delay time has elapsed, the switching state of the output does not change. setting range: 0 -0.2 ... 50s in steps of 0.2 s									
P-n	Output polarity 2 options can be selected: - PnP = positive switching - nPn = negative switching - Not Used In This Application This setting applies to both switching outputs									
dAP	Damping for the switching outputs (Standard setting 6 or 80 Hz) Pressure peaks of short duration or high frequency can be filtered out. dAP-value = response time between pressure change and change of the switching status in ms. • the value for dAP defines the switching frequency (f) of the output:									
	dAP	3	6	10	17	30	60	125	250	500
	F [Hz]	170	80	50	30	16	8	4	2	1
Di S	Setting the display 7 options can be displayed d1 = update of the measured value every 50ms d2 = update of the measured value every 200ms d3 = update of the measured value every 600ms The update interval only refers to the display. It has no effect on the output. rd1, rd2, rd3, rPh = display as d1, d2, d3, Ph; but rotated 180°. Off = In the Run mode the display of the measured value is deactivated. If one of the buttons is pressed for 15 s. Another press of the Mode/Enter button opens the Display mode. The LEDs remain active even if the display is deactivated									
Uni	Display unit The measured value and the values for SPx, rPx can be displayed in the following units • Bar / mbar, • PSI, • MPa / kPa, Select the display unit before setting the values for the parameters SPx, rPx. This avoids rounding errors generated internally during the conversion of the units and enables exact setting of values. Setting at factory: Uni - bAr									

3.5.9. Hysteresis

The hysteresis keeps the switching stable if the system pressure varies about the preset value. When the system pressure is rising, the output switches when the switch-on point has been reached (SPx); when the system pressure is falling again, the output switches back when the switch-off point (rPx) has been reached.

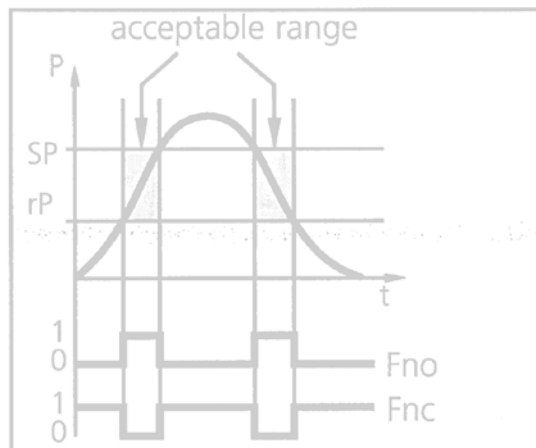
The hysteresis can be set: First the switch-on point is set, then the switch-off point with the requested difference.



Window Function - Not Used In This Application

The window function enables the monitoring of a defined acceptable range. When the system pressure varies between the switch-on point (Sx1) and the switch-off point (rPx), the output is switched (window function / NO) or not switched (window function / NC).

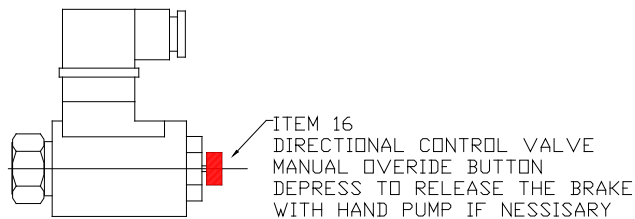
The width of the window can be set by means of the difference between Spx and rPx = upper value, rPx = lower value.



3.6. Manual release of the caliper

Manual release of the caliper is by use of a hand pump fitted with a quick release coupling matching that fitted to the brake control unit.

- 3.6.1. The hand pump must be filled with 10um filtered fluid of the same type as the braking system / brake control unit.
- 3.6.2. The hand pump must also have the internal relief set to a maximum of 250 Bar.
- 3.6.3. Connect the minimess hose to the quick release coupling, item 15 on the brake control unit.
- 3.6.4. Depress the override button on the directional control valve, item 16.
- 3.6.5. Operate the hand pump and build pressure to a minimum of the fully lifted pressure as listed in the table, section 9-1.
- 3.6.6. Pressure should increase and the caliper pistons release.
- 3.6.7. Care must be taken at pinch point between the brake pads and disc.
- 3.6.8. Always release the oil back through the hand pump otherwise the brake control unit tank will be overfilled.



3.7. Dismantling the Brake control unit

This should only be carried out in an authorized hydraulic workshop, please contact your local Svendborg Brakes supplier.

If any of the hydraulic valves are removed / replaced in the brake control unit, the valve seals are to be lubricated with hydraulic oil and tightened to the following torques:-

Valve / Filter Item No	Torque
7	35 Nm
8	35 Nm
9	100 Nm
11	40 Nm
16	40 Nm
18	35 Nm
19	35 Nm
30	40 Nm
44	40 Nm

3.8. Storage

The units must be stored indoors in a clean, dry area. The unit must be stored in such a way that passing or falling objects or equipment cannot damage the components.

The units do not require further surface treatment for indoor storage for up to 3 months. For storage up to 1 year, cover completely with plastic to protect from dust.

3.9. Trouble shooting table

Sheet 1 of 2

Source Effect	1	Mechanical Drive	2	Suction Conditions	3	Pump	4	Delivery Line	5	Return Line	6	Delivery Valves
A Excessive noise	1	Coupling out of alignment		Excessive resistance in the suction line due to:	1	Pump speed too high	1	Pipe mounting loose or absent	1	As per 4A	1	Valve chatter due to dirty or damaged seat
	2	Coupling Loose	1	Suction line blocked or leaking	2	Pump maximum pressure exceeded	2	Incorrect installation	2	Return line not submerged	2	Insufficient damping (unsuitable type)
	3	Coupling defective	2	Fluid Level too low	3	Pump defective	3	Insufficient size	3	Return Filter Blocked	3	Flow noises when operating
	4	Pump or motor mounting loose	3	Suction line too small or to many bends	4	Shaft seal or suction seals defective	4	As per 4C			4	Unsuitable characteristic
	5	Electric motor bearings defective			5	Delivery and return connections transposed						
	6	Pump or motor defective			6	Control system hunting						
	7	Wrong direction of rotation			7	As per 1A, 4						
B Insufficient output forces & torque's (Insufficient pressure)	1	Power transmission defective (lost power phase)		As per 2A	1	Control pressure set too low or defective control device	1	Leaks	1	Excessive line resistance	1	Operating pressure set to low
	2	Wrong direction of rotation			2	Internal leakage due to wear	2	Excessive Line resistance	2	Filter blocked	2	Internal leaks due to wear
	3	Electric Motor defective			3	Output port o'ring defective or missing	3	Delivery filter blocked			3	Valve seat dirty or damaged
	4	Coupling defective			4	Pump defective.	4	System not bleed at highest point			4	Valve spring broken
	5	Key of pump or motor shaft sheared			5	Unsuitable type					5	Insufficient valve adjustment range
C Jerky brake piston movement (pressure & flow fluctuations)		As per 1A, 1-7		As per 2A	1	Defective regulator on variable delivery pumps		System not properly bleed		As per 5B		As per 6A, 1-2
					2	Defective pump					3	Excessive un-damped remote control line
					3	Effects of pump regulators or flow controls					4	Faulty remote control valve
					4	Pilot valves unsuitable						
D Brake not operating or operating too slowly (insufficient flow or no flow)		As per 1A, 1-7		As per 2A	1	Internal leaks due to wear		As per 4B		As per 5B	1	As per 6B, 1-5
					2	Pump Defective					6	With sequence controls
					3	Supply and return connections transposed						Start valve set too high or defective
E Excessive operating temperature					1	On cycle system, pump running too long, max 3 minutes	1	Insufficient line size giving excessive frictional resistance		As per 4E	1	Excessive continuous delivery
					2	As per 1, main relief set higher than pressure switch	2	Delivery filter blocked			2	Pressure setting too high
					3	Reduced efficiency due to wear					3	Excessive cycle time
					4	Regulator defective on variable delivery pumps						
					5	Excessive speed or delivery						
F Foaming of oil			1	Suction line leaking	1	Shaft seal or suction side seals defective			1	Return line not submerged		
			2	Fluid level too low	2	Suction line not submerged						
			3	Insufficient oil level								
G Brake control pressure (torque incorrect)							1	Hoses too elastic				
							2	Lines not properly bleed				
H Line hammer during control operation								As per 4A, 1-4	1	Pipe work loose	1	Cycle rate too rapid
							5	Excessive pipe work volume			2	Throttle valves or orifices damaged
I Pump starts and stops too frequently					1	Leak in the hydraulics system, either , brake, power unit or lines					1	Start or stop valve not controlled correctly or defective.
					2	In accumulator systems, pump flow to low						

3.9 Trouble shooting table (Con't)

Sheet 2 of 2

Source Effect	7	Flow valves	8	Control Valves	9	Fluid	10	Brake caliper	11	Miscellaneous
A Excessive noise	1	Valve vibrating and effecting other control devices	1	Valves chatter due to defective solenoid or low voltage	1	Suction difficulties due to	1	Surface wear		
	2	Flow noises	2	Valve defective due to wear or dirt		a) Low fluid level	2	As per 3A, 6		
	3	As per 3A, 6	3	Excessive flow		b) High viscosity (low temperature)				
			4	Control pressure fluctuations	2	Dirty fluid damaging and blocking equipment				
			5	Incorrect adjustment of valves with adjustable dampening	3	Fluid foaming				
			6	Check electrical control system						
B Insufficient output forces & torque's (Insufficient pressure)	1	Excessive pressure drop	1	Incorrect switching point (e.g does not switch off on zero pressure)	1	Viscosity too low therefore leakage excessive	1	Internal leakage (eg piston seals)	1	Fault in control circuit with pressure control systems
	2	Incorrect adjustment	2	Solenoid defective	2	Viscosity too high therefore excessive flow resistance	2	As per 10A, 1	2	Instruments defective
	3	Valve defective	3	Internal leakage due to wear	3	Fluid foaming	3	Excessive internal friction (poor efficiency)		
			4	Excessive flow resistance						
			5	Valve jammed						
C Jerky brake piston movement (pressure & flow fluctuations)	1	Valve Dirty		As per 8A	1	Fluid contaminated	1	Stick-slip effect due to excessive friction at piston seals	1	On new units, corrosion protection not removed
	2	As per 7A, 1			2	Fluid foaming			2	Back pressure too low when compared to seal resistance
D Brake not operating or operating too slowly (insufficient flow or no flow)	1	Flow setting too low		As per 8B		As per 9B		As per 10B		Starting requirements not satisfied (defective regulator). Open circuit in electrical control wires (connectors). Signaling devices incorrectly set or defective (e.g. pressure switches)
	2	Insufficient adjustment range	5	Valve jammed			4	Caliper sized (e.g. seized piston)		
	3	Valve blocked (dirt)	6	Manual valves not open / closed						
E Excessive operating temperature	1	Flow setting too low (excessive pump delivery through pressure relief valve)	1	Excessive leakage		As per 9B	1	Reduced efficiency due to wear	1	Insufficient fluid in the system
	2	Valve defective	2	Zero pressure circulation inoperative			2	Excessive internal friction (poor efficiency)	2	Cycle rate in excess to that designed for
			3	Valve jammed			3	Internal leakage	3	Ambient temperature in excess to that designed If cooling fitted a) Cooling water inoperative, b) Thermostat set to high c) Fan defective, d) Cooling water temperature too high e) Deposits in the cooler Reduced cooling due to enclosure or such like
									4	
F Foaming of oil					1	Unsuitable fluid			1	Excessive cycle rate to that designed
G Brake control pressure (torque incorrect)			1	Switching time set too slow			1	Internal leakage	1	Remote controlled check valve not closing immediately due to: a) Dirty or defective seat b) Control fault
			2	Solenoid valve defective			2	Insufficient bleeding of oil in system		
			3	Leakage						
			4	Valve dirty						
H Line hammer during control operation			1	Switching time setting too fast	1	Fluid foaming	1	Excessive load	1	No throttling before switching valves in accumulator systems
			2	Unsuitable type			2	No damping		
I Pump starts and stops too frequently									1	Accumulator gas pressure too low
									2	Accumulator bladder (diaphragm) defective
									3	Pressure switches set incorrectly

3.10. Recommended spare parts

FOR 5 YEARS OF NORMAL OPERATION (PER UNIT):

1 off Solenoid coil as per required voltage	800-1017-008 24VDC
5 off Oil filter cartridges	800-1009-009 (Long) - 10 micron
5 off Air filter cartridges	800-1021-008 – 5 micron
1 off Seal Kit	800-2001-200

For purchasing and further information, please contact your local Kempe Branch.

Kempe Industrial Products
Kempe Engineering Services Pty Ltd
50 Gheringhap Street Geelong VIC 3220 AUSTRALIA
Ph 03 5225 2900 Fax 03 5225 2950

3.11. Optional Extras

3.11.1. Programmable digital or analogue pressure switches

If not already supplied, we can supply your unit with the IFM brand of intelligent digital programmable pressure switches. Features include:-

- Dual digital or digital and analogue models available.
- Programmable simply via two buttons with a simple menu, on board programmer.
- Digital display of the pressure and digital output contact status.
- All Stainless Steel Construction and IP65 rating
- Duty life of 1 Million operations.
- Setting and checking status without the need hand pump, pressure gauge or long down time.

3.11.2. Hand pump

If not already supplied, we can supply your unit with a stand along or integrated hand pump for manual release of the brake calipers.

3.11.3. Enclosure

Optional for all BCU-A75 brake control units is an enclosure to mount the unit, not that is mandatory as the unit without the enclosure is suitable for outside mounting and is IP65 rated. However an enclosure extends the life of the systems and keeps foreign material from building up on the unit effecting servicing and maintenance.

The basic enclosure offers:-

- Compact size 600mm wide, 450mm deep and 500mm high.
- Material Stainless Steel ASTM grade 316 B2 brushed finish.
- Rating of IP55 for the main enclosure as the brake control unit is IP65 rated.
- Ease with fixed mounting points that enable mounting on a vertical surface.
- Standard brass gland plate for electrical cable entry.
- Unique quick release mounting bracket for the brake control unit for easy install and removal.
- Access to all serviceable components of the brake control unit without removal.
- Swing away single hinge point door that is removable.

Installation and Maintenance Manual MAU-075-001

Options in addition to the basic unit:-

- Laminated drawings of the unit and caliper mounted behind polycarbonate inside the door.
- Removable drip tray inside under the brake control unit to hold 140% of the reservoir capacity.
- Internal switches neon light activated when the door is opened

6. SPECIFICATIONS

Spec No.	REV	Description
SPAU-004	0	Installation of hydraulic line specification for brake system
SPAU-005	0	Commissioning specification for brake system

HYDRAULIC LINE INSTALLATION SPECIFICATION

**For the design, installation and testing of hydraulic
lines for Svendborg Brakes systems**

TITLE				Hydraulic Line Installation Specification		NUMBER	SPAU-004
REVISION RECORD							
Rev	Date	Revision Description		Author	Checked	Approved	
0	19/11/06	First Issue		D.Tonitto	D.Tonitto	D.Tonitto	
1	06/12/12	Formatting Update		MB	MB	MB	

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1. Introduction

This specification will outline the requirements for installing and testing of hydraulic lines as a part of a brake system comprising Twiflex Brake Calipers and Kempe Brake Control Units supplied by Kempe Industrial Products.

This specification provides a general outline of the scope of works, with detailed instructions to be found in the specific installation and maintenance manual provided for each application and reference documents.

At all times the client, Twiflex Brakes and Kempe Industrial Products safety requirements must be complied with.

1.1. General

- 1.1.1. The purpose of this specification is a guide to ensure all brake systems supplied by Kempe Industrial Products are operating at peak performance and to a consistent standard.
- 1.1.2. This specification is the Kempe Industrial Products standard and customer specifications and requirements may take precedence.
- 1.1.3. This specification covers the minimum requirements for the installation of hydraulic lines for a brake system supplied by Kempe Industrial Products.
- 1.1.4. Brake systems are to be installed in strict accordance with the installation and maintenance manual for the unit concerned.

1.2. Applicable reference documents

- 1.2.1. Kempe Industrial Products Installation and Maintenance Manual for particular application
- 1.2.2. SAE hose standard
- 1.2.3. ASTM A179 Cold draw seamless annealed Hydraulic tube (current version of AS1835)
- 1.2.4. ASTM A106 Black seamless carbon steel pipe (current version of AS/NZS 1835)
- 1.2.5. Parker Catalog 4300, June 2003, Industrial Tube Fittings, Adapters and Equipment.
- 1.2.6. AS/NZS 1627.5, Pickling, descaling and oxide removal

- 1.2.7. AS/NZS 3750 Paints for steel structures
- 1.2.8. AS/NZS 2312 Guide to protection of iron and steel against exterior atmospheric corrosion.
- 1.2.9. AS/NZS 2700S-X53 Golden Tan, Color standards for general purposes swatch.

2. Design

2.1. General

- 2.1.1. Design hydraulic lines with best engineering practice in mind.
- 2.1.2. Design hydraulic lines to the shortest practicable route, without inhibiting through passage of access ways.
- 2.1.3. Never design rigid lines between two-fixed point straight with no bends or allowance for thermal expansion or movement. This can be achieved with a bend in the line
- 2.1.4. Ensure that allowance be made for any moving or vibrating point in reference to another that rigid hydraulic lines are attached or connected. This can be achieved by hoses or clamping such that line flexibility is retained.
- 2.1.5. Allow for thermal expansion of rigid lines. This can be achieved by hoses or clamping such that line movement is permissible.
- 2.1.6. All calipers are supplied as standard with minimum two (2) pressure ports. The bottom most port shall always be used for incoming or supply oil from the brake control unit. The top most port shall always be used for a bleed point, which is supplied as standard with a hydraulic test point.
- 2.1.7. All calipers it is recommended have the lowest drain port connected to either a sealed bottle or back to tank above the oil level. The purpose of this is to isolate the internal cavity of the caliper from the ingress on contaminates and moisture as well as to allow any seal leakage to drain and not cause hydraulic "lock-up". If the drain bottle option is installed then this can be used to identify any excessive seal leakage to a particular caliper. Note that some minor seal leakage is normal, and not a mandatory requirement to repair the caliper. All calipers are supplied as a minimum with standard drain bottle kits, which may be effected by UV light over time. If the environment requires, we can supply offshore or heavy duty bottles UV resistant bottles.

2.1.8. Two methods of connecting hydraulic line between caliper and brake control unit exist.

a) Piping single line from brake control unit to caliper.

b) Piping a supply and return line from brake control unit to caliper, thus a flushing circuit is achieved. This method offers the advantage of flushing any trapped air and contaminants continually in operation. This method will require the brake control unit to be design to enable this, please enquire at the time of order.

3. Material

Generally all equipment should be selected from high quality materials designed to operate under the conditions and pressures required by the installation.

Recommended Standards:

Hoses	-	SAE or equivalent
Tubes	-	ASTM A179 Cold draw seamless <u>annealed</u> Hydraulic tube (current version of AS1835)
Tube	-	ASTM A269 Stainless Steel Cold seamless <u>annealed</u> Hydraulic tube
Pipe	-	ASTM A106 Black seamless carbon steel pipe (current version of AS 1835)

3.1. Tube.

- 3.1.1. Mild steel tube shall not be used when the installation is in contact with corrosive environments, i.e. coal, iron ore, marine or offshore conditions. All of these installations require the use of stainless steel tubing.
- 3.1.2. All mild steel tube shall be either zinc plated externally or external protected prior to installation.
- 3.1.3. If the tube is externally zinc plated, the internal surface must be moisture and rust free. Any internal rust renders the tube un-usable and must be discarded.

3.2. Fittings

- 3.2.1. In selecting hydraulic fitting material, environmental considerations should be considered in respect to atmospheric corrosion and also consideration to Galvanic corrosion. For example if stainless steel tubing were chosen (a Cathodic '-' metal) with zinc plated free machining steel fittings (a Anodic '+' metal) galvanic corrosion has a greater tendency to occur than if stainless steel fitting were chosen.

4. Connection joints

4.1. Connection type

- 4.1.1. All connection joints are recommended be of the JIC 37 Deg type. Cutting or compression ring fitting types are NOT accepted due to the fact that if they are not installed or maintained correctly instantaneous loss of oil pressure can occur. This will result in instantaneous application of brake torque and possible damage to equipment as well as possible oil coverage of the disc, contaminating the brake pads.

It is highly recommended that all JIC 37Deg tube flares be produced on hydraulic tubing by a power flaring tool specifically designed of the purpose. It is not recommended that a hand operated or copper tube flaring tool be used.

- 4.1.2. When installing connecting hydraulic lines, ensure that BSPP (parallel) fittings are used in both the brake control unit and brake ports. This should be sealed either with o'ring and pressure ring of "Dowty" seal. BSPT fittings are NOT accepted.

4.2. Stainless steel

- 4.2.1. If stainless steel is the chosen tube material is recommended that "Flaretite" seals of the stainless steel grade 316 material are used in the connection joint.

5. Line sizes

It is important for the lines between the connections of the brake and brake control unit to be as short as possible.

This gives the following advantages:

- 5.1.1. A minimum safety factor of 1:4 shall be used for all tube and hose from burst pressure to the working pressure rating.
- 5.1.2. Small compression volume in the hydraulic system. This is particularly important in hydraulic systems with large and rapid load variations, and in systems where high speed accuracy is essential ensuring brake caliper reaction times are as responsive as possible. For this reason it is also important to limit the use of hydraulic hose which will act as an accumulator.
- 5.1.3. Small power losses in the hydraulic system.
- 5.1.4. Lower installation costs in that similar pipes sizes and shorter pipe lengths can be used.
- 5.1.5. For a normal hydraulic system, the total pressure drop in the pressure and return line must be not more than 5 percent of the normal working pressure.
- 5.1.6. The following flow rates are recommended values and should not be exceeded:
 - Main lines 4.5 m/s
 - Drain lines 1.0 m/s

In certain systems, where very small compression volumes are required, higher flow rates are acceptable. Contact Kempe Industrial Products in these specific cases.

6. Pipe / tube clamps

6.1. Type

- 6.1.1. It is recommended that clamp of the Polypropylene material be used.

6.2. Spacing

- 6.2.1. Pipes / tubes should be clamped with uniformly spaced clamps. The distance between the clamps may not exceed the following. Pipes must be clamped immediately before and after a bend and immediately before transition to a hose.

Tube OD (mm)	Spacing between clamps (mm)	Spacing from joint to clamp (mm)	Spacing from bends to clamp (mm)
6.0 – 12.7	1,000	50	100
12.7 - 22.0	1,200	100	200
22.0 – 32.0	1,500	125	250
32.0 – 38.0	2,000	150	300
38.0 – 57.0	2,700	175	350

- 6.2.2. The pipe clamps shall be secured to a stable and non-vibrating surface.

- 6.2.3. The pipe clamps must be of a type permitting a certain amount of axial and radial motion in the clamped joint and have good vibration-absorbing property.

7. Hoses

7.1. General

- 7.1.1. To avoid damage caused by vibrations or movement it maybe appropriate for the brake control unit and brake caliper to be connected to pipes by means of hoses.
- 7.1.2. All hose shall be of the permanent crimp type, the use of re-usable type fittings is NOT permitted. All fittings shall be of steel or stainless steel, suitably protected from corrosion and shall have a factor of safety of 4:1 based on yield strength calculations.
- 7.1.3. The material of the hose bore shall be compatible with the hydraulic oil specified.

- 7.1.4. The hoses must be short and have the smallest possible diameter in order to obtain a small compression volume. However the hose must be of sufficient internal diameter not to allow pressure drop must also be made for the flow velocities listed under 'Pipe sizes' and or minimum recommended size on the hydraulics schematic of manual.
- 7.1.5. The hose bend radii must be greater than the minimum permissible bending radius according to the data provided by the supplier of the hose.
- 7.1.6. Hydraulic hoses always to be designed with at least one straight female (swivel) end connection fitting. No hoses shall be fitted with angled connection fittings on both ends.
- 7.1.7. Supports shall be located on the rigid piping as close as practical to the hose connection and shall be size to accommodate the pressure forces developed at the hose connections
- 7.1.8. In manufacture, cut hose ends shall be de-burned and any hose dust or particles must be cleaned out prior to the fixing of hose ends.

7.2. Monospring calipers

- 7.2.1. Monospring type calipers must be connected by way of flexible hoses to allow for the caliper to move on axles.
- 7.2.2. Noting that hose must be installed such that the natural straightening of the hose under pressure must not act to influence the movement of the caliper on the axles in either direction.

7.3. Hose protection

- 7.3.1. The installation of hoses across the radial path of a brake disc shall be avoided, this is due to the risk of damage from pad material coming free of the pad and also any possible sparking especially with sintered metal pads. If it is not possible to avoid such, the hose shall be protected accordingly with fire sleeve and or mechanical protection sleeve as required.
- 7.3.2. Hydraulic hoses must be protected from chafing on foreign objects and other hoses as well as damaged from rubbing, chafing on any sharp object, kinking or twisting and hoses shall only flex in one plane. The recommended method is the use of Polypropylene clamps as appropriate positions to prevent this taking into account any vibration of machine movement.

7.4. Hose burst protection

- 7.4.1. A recommended method of protection of oil spread in the scenario of hose burst to protect the disc or other from oil, is to cover the hose with fire sleeve or similar. This should be installed such as to allow the oil to be contained within the sleeve and drain away from the brake caliper and disc direction to a container.

8. Weld procedures

If welded pipe is to be used then the following applies

- 8.1.1. Root weld to be made with T.I.G process with Argon gas purge to minimize slag build up and reduce chemical cleaning time.
- 8.1.2. Full penetration butt weld joints must be used on all high pressure applications.
- 8.1.3. Do Not use M.I.G weld, because of difficulty in obtaining a 'leak free' joint.
- 8.1.4. All welding to comply with the current revision of Australian Standards CB15.

9. Cleaning

9.1. Tube

- 9.1.1. All tubing shall be wad cleaned under air pressure with abrasive scouring pads and then with plain pads until clean (no contaminate observed).

9.2. Pipe

- 9.2.1. Pipe bore must be chemically pickled and neutralised after welding to remove all scale and slag. Procedure to be in accordance with AS1627 Part 5.

9.3. General

- 9.3.1. After manufacture pipes (after chemical cleaning), tubes and hoses must be flushed with mineral based hydraulic oil. This is to be done by an external pump system, the brake control unit pump must not be used. Flushing flow should ideally be at least equal to the main pumps or accumulator design flow, and the recommended oil flow at such a rate that the oil is turbulent, @ greater than 5m/s

- 9.3.2. A suitably sized filter should be installed during flushing to remove all contamination. The flushing filter should be sized to take the pumps maximum flow without excessive pressure drop. The filter should be checked regularly to ensure contamination build up does not block filter element. Replace filter element as necessary.
- 9.3.3. The main lines at the Twiflex Brakes Caliper should be connected together and the entire system flushed at maximum flow for at least 30 minutes, longer if system is large. An oil sample taken for future reference to ensure the oil cleanliness level is equal to or better than ISO 4406 - 16/13 level
- 9.3.4. This flushing process may need to be repeated with clean oil, if the flushing oil appears to be badly contaminated with chemical cleaning acid. All cleaning acid must be thoroughly removed before the main system is connected.
- 9.3.5. After the flushing process, pipes should be oiled with clean oil and capped to prevent corrosion and contamination.

10. Pressure testing

All lines should be pressure tested to at least 1.5 times the maximum working pressure and checked for leaks. The main brake control unit should not be used for this test.

11. External protection of tube and pipe

11.1. Mild steel tube or pipe

Any mild steel tubing that is not external zinc plated or otherwise protected shall be protected as follows prior to installation.

- 11.1.1. All painting to be carried out by qualified personnel in accordance with a latest revision of AS/NZS 3750
- 11.1.2. All threaded connections shall be temporarily protected from blast and paint with metal plugs fixed with thread tape or equivalent. After painting all traces of thread tape must be removed from reservoir threads.
- 11.1.3. Tube to be lightly abrasive garnet blast externally. Pipe to be grit blasted externally to the latest revision of AS/NZS 2312 class 2 ½.
- 11.1.4. The exterior is to be epoxy primed to a dry film thickness of 50µm, then given two coats of epoxy top coat to a dry film thickness of 75 to 100 µm.
- 11.1.5. The top coat colour is to be to the latest revision of AS/NZS 2700, X53 Golden tan.

INSTALLATION & COMMISSIONING SPECIFICATION

For the installation and commissioning of Svendborg Brakes systems

TITLE				Hydraulic Line Installation Specification		NUMBER	SPAU-005
REVISION RECORD							
Rev	Date	Revision Description		Author	Checked	Approved	
0	19/11/06	First Issue		D.Tonitto	D.Tonitto	D.Tonitto	
1	06/12/12	Formatting Update		MB	MB	MB	

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1. Introduction

This specification will outline the requirements for installing and commissioning of a brake system comprising a Twiflex Brake Caliper and Kempe Brake Control Unit supplied by Kempe Industrial Products.

This specification provides a general outline of the scope of works, with detailed instructions to be found in the specific installation and maintenance manual provided for each application and reference documents.

At all times the client and Kempe Industrial Products safety requirements must be complied with.

1.1. General

- 1.1.1. The purpose of this specification is a guide to ensure all brake systems supplied by Kempe Industrial Products to are operating at peak performance and to a consistent standard.
- 1.1.2. This specification is the Kempe Industrial Products standard and customer specifications and requirements may take precedence.
- 1.1.3. This specification covers the minimum requirements for the installation of a brake system supplied by Kempe Industrial Products.
- 1.1.4. Brake systems are to be installed in strict accordance with the specific installation and maintenance manual for the unit concerned.

1.2. Applicable reference documents

- 1.2.1. AS/NZS 3709 - Balancing rigid objects
- 1.2.2. SPAU-004 - KEMPE INDUSTRIAL PRODUCTS Hydraulic Line Installation Specification
- 1.2.3. Brake disc data sheet, DAU-010 for Australian market
- 1.2.4. FAU-009 – System Commissioning Report

2. Safety

2.1. Safety with high pressure hydraulics

- 2.1.1. Isolate the hydraulic system you are to be working, possibly both electrically and mechanically.
- 2.1.2. Drain storage accumulators of oil pressure.
- 2.1.3. Check oil pressure of the system before commencing work.
- 2.1.4. Wear the right PPE, before commencing work, always wear safety glasses and gloves when working with hydraulics.
- 2.1.5. Do not allow high pressure hydraulic lines near to skin surface such that impregnation is possible.
- 2.1.6. Always stand clear of hydraulic system when first starting after working on.

2.2. Safety with brake calipers

- 2.2.1. Check with machine operators and advise other where you are working.
- 2.2.2. Isolate the machine you are to be working, possibly both electrically and mechanically.
- 2.2.3. Be aware of pinch points on the caliper, pads and brake disc, wear the correct PPE, i.e. gloves.
- 2.2.4. Ensure the active caliper is always retained with the air gap bolt as required when changing brake pads.
- 2.2.5. Never place fingers between the brake pad / piston and disc.

2.3. Information

- 2.3.1. Always read the installation and maintenance manual before commencing work on the brake system.

The Twiflex Brakes calipers are design only for the purpose described here, any other use is strictly forbidden and not condoned by Twiflex Brakes or its agents. For that reason, you should carefully read and understand the operating instructions of the hydraulic unit, the brake and the machine, where the hydraulic unit and brake is placed, before you operate it.

2.4. Do not:

- 2.4.1. Attempt to service the caliper or brake control unit unless you are qualified and suitably trained to do so.
- 2.4.2. Operate the brake before bleeding, after filling with oil.
- 2.4.3. Use higher operating pressures than specified in the manual.
- 2.4.4. Change the size of the caliper spring pack without consulting Twiflex Brakes or its agents.
- 2.4.5. Use brake pads, when they are worn greater than the minimum thickness as noted in the manual.
- 2.4.6. Use brake pads, which are not recommended by Twiflex Brakes or its agents.
- 2.4.7. Try to adjust the brake pad, or remove the brake caliper before securing the brake disc from rotating.
- 2.4.8. Operate the hydraulic unit before correct filling with oil.
- 2.4.9. Adjust valves or pressure switches to higher operating pressure than specified in this manual.
- 2.4.10. Change the size or type of components without consulting Twiflex Brakes or its agents.
- 2.4.11. Use tools or any other equipment, which is not recommended by Twiflex Brakes or its agents.
- 2.4.12. Use hydraulic fluids, which are not recommended in this manual or by Twiflex Brakes or its agents.

2.5. Note:

- 2.5.1. The customer is responsible for the monitoring of caliper control devices provided to ensure the pad to disc air gap as well as brake pad wear is within allowable limits.
- 2.5.2. The customer is responsible for the monitoring of brake control unit control devices provided to ensure it operates as described in the manual and no damage results.
- 2.5.3. Always use a torque wrench where specified, to make sure that the torque obtained is the torque described in the manual.

- 2.5.4. It is the responsibility of the customer that the brake disc is always clean, and free from dirt, grease or oil.
- 2.5.5. It is the responsibility of the customer that the caliper and brake control unit is always clean, and free from dirt, grease or oil.
- 2.5.6. Do not use spare parts, other than the original spare parts from Twiflex Brakes or its agents.

3. Unpacking

3.1. Goods received

- 3.1.1. When the system first arrives on site and unpacked, all goods are to be checked against the manual, order and delivery documents to ensure all goods have been supplied.

3.2. Condition

- 3.2.1. All goods are to be inspected to ensure that there is no damage. Any damage is to be reported and rectified.

4. Installation

4.1. Brake disc

All brake discs shall be manufactured in accordance with brake disc data sheet, DAU-010 for Australian market and or specific design drawings.

4.1.1. Balancing

- 4.1.1.1. All brake disc assemblies operating at greater than 500rpm shall be balanced to the latest version of AS 3709, balance quality grade 6.3 to the maximum operating speed. A brake disc assembly will include the shaft coupling if mounted on such.

4.1.2. Run out

- 4.1.2.1. Disc run out must be within the specification on the caliper / mounting drawing.

4.1.3. Axial float

- 4.1.3.1. Axial float must be within the specification on the caliper / mounting drawing, the same as for run out.

4.1.4. Alignment

- 4.1.4.1. Disc alignment must be within recommended specifications in the installation and maintenance manual and or caliper drawing.

4.1.5. Cleaning

- 4.1.5.1. The brake disc must be washed clean with white spirit and there-after with thinners before the installation of the brake caliper's. Any residual oil or anti-corrosion preparation will reduce the friction coefficient markedly.

4.2. Calipers

4.2.1. Caliper brackets

- 4.2.1.1. All caliper mounting brackets are to be designed with good engineering practice and suitable to the duty. Brackets should not allow vibration or axial movement greater than +/-0.2mm.

4.2.2. Caliper mounting surface

- 4.2.2.1. The mounting surface of the standard bracket or customer post between the caliper halves must comply with the relevant arrangement and mounting data drawings. Classified as a friction joint, as such NO lubricant or other compound shall be applied between the caliper and mounting surfaces. The mounting of the caliper does NOT rely on the shear capacity of the mounting bolts but the friction joint created by the clean, dry, lubricant free surface to the correct surface finish and the tension created by the mounting bolts.

4.2.3. Caliper to bracket mounting bolts

- 4.2.3.1. For caliper half mounting bolts, genuine Twiflex Brakes high tensile mounting bolts (or equivalent) are recommended to mount the caliper to standard bracket or customer post. The mounting bolts should be tightened to the recommended torque's found in the relevant Installation and Maintenance Manual or arrangement drawing. Unless noted otherwise, the torque figures given are for a dry thread, if anti-seize compound or other lubricants are used on threads then torque de-rating factors should be used to achieve the same tension rating.

If the customer supplies the mounting bolts, the customer must take responsibility for achieving the correct friction joint torque. Please note that the tightening torque's in the Installation and Maintenance Manual may NOT apply.

4.2.4. Caliper mounting bracket to machine bolts

- 4.2.4.1. Ensure mounting bolts are the minimum grade as required in the installation and maintenance manual and or caliper drawing.
- 4.2.4.2. Ensure mounting bolts are tightened to fastener specifications

4.2.5. Caliper alignment

- 4.2.5.1. Refer to the specifications in the installation and maintenance manual for each system. All dual spring calipers should be aligned such that:-
- The caliper position is the correct height and distance from the disc centre, refer the caliper drawing.
 - For dual spring calipers, the pad keeper plates are no greater than 1mm difference between each side from the disc.
 - For Monospring calipers, the correct set out distance for the base plate is set, refer the caliper drawing. Noting that if the set out distance is greater than that specified in the caliper drawing / manual the axles are

under greater torsional load, such is to be confirmed with engineering if within acceptable limits

- The caliper mounting bracket is square to the disc.
- Thermal expansion allowance shall be made between the disc OD and caliper mounting bracket, this is to allow for expansion due to temperature increase by the factor of 1/1000 of the disc diameter. i.e. for 600mm diameter disc 0.6mm
- For Monospring calipers, after the caliper is mounted and mounting bolts tightened it must be checked that the caliper assembly (passive caliper) slides freely on the base plate axles. If this is not the case it may be a sign of an uneven mounting surface or that the mounting bolt torque is incorrect and must be re-checked.

4.2.6. Electrical connections

- 4.2.6.1. Ensure all electrical connections are as per the electrical schematic.

4.2.7. Caliper drain port

- 4.2.7.1. Ensure the caliper drain ports are either fitted with a drain bottle, piped back to the brake control unit tank below oil level or fitted with a long hose.

4.2.8. Caliper air gap

- 4.2.8.1. Normally the air gap to be set on commissioning for a system within specifications for calipers is 1mm.

If however the run out or axial end float of the brake disc exceeds 0.3 mm for disc diameter up to 1,500 mm or 0.6mm for disc diameter greater than 1,500 mm. The air gap must be set to ensure they do not rub the disc at any point through a 360 degree rotation. Further if caliper indicators are fitted then the air gap must be adjusted accordingly to take into account for disc run out and axial end float of the brake disc. For standard non automatic wear adjust calipers the caliper shall always have a minimum of 1mm air gap when the disc is closest to the brake pad throughout a complete 360 deg rotation. The reason of this is the requirement of the caliper to have a minimum 1mm travel to operate the on/off of release caliper indicator switch.

WARNING: With the air gap adjusted greater than allowable to take into account disc axial end float or run out the clamping force and braking torque may be reduced.

- 4.2.8.2. For standard non automatic wear adjust calipers the air gap can operate normally out to 3mm on most systems. However this is a general rule with some systems limited to varying air gaps for design reasons. If pad wear indicators are used then they will indicate at the maximum allowable air gap. If no pad wear indicators are fitted, the air gap should be maintained within the limits that are noted in the system installation and maintenance manual.

4.2.9. Brake pads general

- 4.2.9.1. The pads for the brakes are packed in a separate plastic bag to protect them from contamination. Do not assemble the pads until after the disc has been cleaned. Avoid contaminating the pads with oil. Even a small amount of oil can reduce the friction coefficient considerably.
- 4.2.9.2. In case of contamination;
 - Organic pads cannot be cleaned and have to be replaced.
 - Sintered metal pads can be cleaned with white spirit and then thinners.
- 4.2.9.3. The disc must be thoroughly cleaned, see disc section.
- 4.2.9.4. Ensure pad retraction kits are fitted to maximize pad life
- 4.2.9.5. All pads are asbestos free and certified. If a certificate is required such can be supplied.

4.2.10. Brake pad allowable wear

- 4.2.10.1. Pads have the following allowable wear limits per pad.

Note that all Monospring calipers only allow half this wear. The reason for this is that if the limits are exceeded the active pad will move outside the pad ear guide and could possibly fall out of the caliper.

VCSD-VR Organic	10 mm
VKSD-VR Organic	10 mm

4.2.11. Monospring caliper positioning systems

- 4.2.11.1. All Monospring caliper shall have installed a positioning system unless noted otherwise.

- 4.2.11.2. All positioning system mounting screw shall be fitted with anti vibration washers and Loctite.

4.3. Brake Control unit

4.3.1. Mounting

- 4.3.1.1. Position the brake control unit to enable access to all points for service, i.e. to remove the tank, charge the accumulator with nitrogen, manually override directional control valves. It is preferable that brake control units are not mounted under guarding.
- 4.3.1.2. Position the brake control unit such that it is not covered with excessive product or water.
- 4.3.1.3. Ensure that the brake control unit is rigidly mounted such that no damage can result to pipes, hoses and electrical cable from movement

4.3.2. Electric motor.

- 4.3.2.1. Brake control units fitted inside enclosures are fitted with TEFC (total enclosed fan cooled) motors. Brake control units not fitted in enclosures are fitted with TENV (totally enclosed non ventilated) motors that do not have fans. The reason for this is to protect the motor from moisture ingress from shaft seal failure with a fan motor particularly in extreme temperature environments.

4.3.3. Electrical connections

- 4.3.3.1. Ensure all electrical connections are as per the electrical schematic.
- 4.3.3.2. Ensure all electrical control device connections are sealed water and dust tight.

4.4. Hydraulic lines

- 4.4.1. Refer to the Kempe Industrial Products Specification SPAU-004 for hydraulic lines.
- 4.4.2. Before first starting the brake control unit, ensure all hydraulic connections are tight

5. Commissioning, testing and sign off

On final commissioning the following is required.

5.1. Calipers

5.1.1. Bleeding calipers

- 5.1.1.1. Ensure calipers are bleed with the bleed point coupling fitted at the top most port of each caliper half. Bleeding shall be done by a bleed hole into a sealed container fitted with a bleed coupling such that no oil is splashed onto the disc and or pads.

5.1.2. Bedding in of brake pads

- 5.1.2.1. To obtain the rated coefficient of friction between the pad and the disc brake, and to ensure the pad surface is aligned to the disc, it is essential that the pads are bedded to the disc surface.
- 5.1.2.2. The coefficient of friction may drop by approx. 20% when not bedded in.
- 5.1.2.3. Dynamic system pad bedding.

This can be done by making a series of load-tests with increasing load or speed until sufficient coefficient of friction is achieved and the stopping time or distance is achieved. If the brake disc is new, the time of bedding in will be longer because the disc also has to be bedded in. Always use new pads with a new disc never use an old used pad with a new disc.

5.1.2.4. Static system pad bedding

Static pad bedding is sufficient when the load can be held / stopped at maximum conditions of load and speed

- Organic pads

These can normally be bedded in by driving the disc through the pads with a minimum of 25% of the rated clamping force applied, or if that is not possible by making a series of activation's with increased load or increased speed.

- Sintered metal pad

Like the organic but with a slightly higher clamping force when bedding in – approx. 40% of the rated clamping force applied.

- 5.1.2.5. During the process of “bedding in”, the disc temperature should be monitored to ensure that the disc does not overheat beyond pad maximum peak temperature limits.

Organic material D 2031.....	350 Deg C
Organic material IP 160.....	350 Deg C
Organic material BE 3521.....	400 Deg C
Sintered metal material MD 550.....	900 Deg C

5.2. Brake control unit

5.2.1. Electric motor rotation

- 5.2.1.1. All Svendborg Brake control units are fitted with left hand or anti-clockwise rotation motors. With TENV motors with no fan, direction check can be done by ensuring oil pressure builds within 10-20 seconds. Noting that the failsafe normally open DCV must also be energized, closed.

5.2.2. Hydraulic valves and control devices

- 5.2.2.1. Confirm all pressure settings are correct for pressure and flow valves, pressure switches, accumulators, valve driver cards etc as per the hydraulic schematic / installation and maintenance manual.

5.2.3. Open loop brake system set up

- 5.2.3.1. The braking setting of the fail to apply (failsafe) circuit in all system should be set with the system at worst case load, for example a fully loaded conveyor. Ensure the correct braking time is achieved and the back pressure setting is maintained a minimum of 3 seconds greater than the longest stopping time. Adjust the settings as required even if different from the hydraulic schematic / installation and maintenance manual settings.

5.2.4. Closed loop brake system set up.

- 5.2.4.1. Ensure that the maximum proportional valve pressure is set no greater than the caliper balance pressure. This is to minimize any delay on applying the brake and avoiding overshoot and ‘hunting’.

5.3. Test report

- 5.3.1. FAU-009 or equivalent must be completed on final commissioning and be signed off that the system is installed and commissioned correctly in accordance with the installation and maintenance manual.
- 5.3.2. All drawings and manuals shall be marked with any ‘As Commissioned’ changes and returned to Engineering.

7. COMMISSIONING REPORT FORM

Spec No.	REV	Description
FAU-009	0	Installation & Commissioning report

BRAKE SYSTEM COMMISSIONING REPORT

JOB No:		CUSTOMER REF No:	
CUSTOMER:			
CONTACT:		CUST ORDER No:	
DATE OF BRAKE TEST:		BY WHOM:	
INSPECT BY CUST:		REPORT REQUIRED:	
CALIPER MODEL 1 :		SERIAL No:	
CALIPER MODEL 2 :		SERIAL No:	
CALIPER MODEL 3 :		SERIAL No:	
CALIPER MODEL 4 :		SERIAL No:	
BRAKE CONTROL UNIT MODEL 1:		SERIAL No:	
BRAKE CONTROL UNIT MODEL 2:		SERIAL No:	
OIL TYPE:		OIL TEMP:	
PROCEDURE No:		PROCEDURE DATE:	

For installation and commissioning, please refer to specification SPAU-005 and the system manual for the specific project.

#	INSTALLATION ACTIVITY	CHECKED OK	SIGNED OFF
1	Unpack and check brake system components are all included and free from damage.		
2	Confirm the brake disc is manufactured to specifications also check for run out, axial float and alignment.		
3	Disc and coupling mounting, alignment and coupling installation as per installation and maintenance manual		
4	Clean the brake disc.		
5	Confirm the brake mounting support is as per specifications, including dimensionally and for rigidity / vibration.		
6	Mount caliper on pedestal / mounting plate and check disc to caliper alignment and clearance as per installation manual. Dual spring calipers with maximum 1mm difference between each side and Monospring calipers with correct set out distance.		
7	Check caliper mounting fasteners as per Installation Manual or other. Check mounting surface is dry and grease or oil free.		

JOB NO:

#	INSTALLATION ACTIVITY	CHECKED OK	SIGNED OFF
8	Check that Monospring caliper assemblies (passive caliper) slides freely on the base plate axles.		
9	Check brake pads are installed as per Installation Manual, oil /grease free for organic pads, pad retraction kits installed.		
10	Ensure all caliper <u>on/off & pad wear (not adjust for AWA) switch</u> electrical connections are correct and working		
11	Ensure all caliper <u>pad pre-worn & worn (AWA only)</u> switch electrical connections are correct and working		
12	Adjust pad air gap as per Installation Manual to 1mm, or other as required.		
13	For Monospring calipers, Install and adjust positioning kits as required. Mounting bolts must be fitted with anti-vibration washers and Loctite on the bolt threads		
14	Ensure drain kits or equivalent are fitted.		
15	Ensure power unit is mounted as per manual, fixed rigidly and filled with the correct oil grade through a 10µm filter.		
16	Ensure all power unit <u>directional control valve coil</u> electrical connections are correct and working, also the hirschmann connectors are sealed from moisture ingress		
17	Ensure all power unit <u>pressure switch</u> electrical connections are correct and working, also the hirschmann connectors are sealed from moisture ingress		
18	Ensure all power unit <u>pressure transducer</u> electrical connections are correct and working		
19	Ensure all power unit <u>level and temperature</u> electrical connections are correct and working, also the hirschmann connectors are sealed from moisture ingress		

FORM FAU-009 REVISION 19-11-06

JOB NO:

#	COMMISSIONING ACTIVITY	CHECKED OK	SIGNED OFF
20	Ensure hydraulic lines are connected and tight.		
21	Check that power unit electric motor rotates in required direction, LH by removing from power unit or pressure developed within 10-20 seconds with failsafe normally open valves closed.		
22	Check that the motor cycles on power unit motor pressure switch as per settings in Hydraulic Schematic or Installation Manual.		
23	Check that the power unit brake release pressure switch and or pressure transducer operates as per settings in Installation Manual.		
24	On running power unit, bleed air from highest point, then ensure brake caliper is lifted within required lift time & check for leaks.		
25	Check the application counter pressure setting with pressure gauge or chart recorder, adjust as necessary.		
26	Operate the drive and apply the brake under normal loaded conditions, noting all functions as previously mentioned correct, also check that the disc temperature is within normal operating limits for the pad type used.		
27	Bed in brake pad as per pad instructions		
28	Check the stopping time is within acceptable customer limits, adjust as necessary. Attached here any chart recording as a record of setting characteristics.		
29	Check to ensure all function test prove the braking system to operate correctly and all interlocks are in place		
30	Note any changes in the Installation & Maintenance Manual or drawings to be included in "AS COMMISSIONED" revisions		

JOB NO:

COMMENTS:

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JOB NO:

BRAKE CONTROL UNIT SETTINGS CHECK LIST

Item No on circuit	Item	Setting	Units
	Oil brand, type and grade used for testing		
12	Accumulator Gas Charge – Main		Bar
36	Accumulator Gas Charge – Aux 1		Bar
7	Relief valve setting – Main		Bar
56	Relief valve setting – Hand pump		Bar
18	Relief valve setting – Counter pressure 1		Bar
19	Needle valve - 1		Turns from closed
33	Relief valve setting – Counter pressure 2		Bar
32	Needle valve - 2		Turns from closed
11	Flow control valve – Main accumulator		Turns from closed
35	Flow control valve – Aux accumulator 1		Turns from closed
42	Needle valve - supply		Turns from closed
43	Needle valve - proportional cct		Turns from closed
22	Hand pump working		
	Hand pump leak rate acceptable, min 5 Bar drop in 30 min		
54	Level switch - oil reservoir working	setting 1	mm from top
		setting 2	mm from top
	Temperature switch – oil reservoir	Temp	Deg C
		Contact	Deg C
16	Directional Control Valve 1		
	Coil voltage check		Voltage AC/DC
	Powered operation working		
	Manual override working		
30	Directional Control Valve 2		
	Coil voltage check		Voltage AC/DC
	Powered operation working		
	Manual override working		
48	Directional Control Valve 3		
	Coil voltage check		Voltage AC/DC
	Powered operation working		
	Manual override working		

FORM FAU-009 REVISION 19-11-06

JOB NO:

Item No on circuit	Item	Setting	Units		
45	Proportional relief valve				
	Pressure @ min command	mA / V	Bar		
	Pressure @ max command	mA / V	Bar		
	Response time - min to max command		Seconds		
	Response time - max to min command		Seconds		
	Proportional relief valve response chart				
		Input mA	Driver output	Pressure rising	Pressure falling
	46	Proportional driver set up			
Maximum current setting (HI)			25 – 98%		
Minimum current setting (LO)			02% to HI		
Ramp up time (UP)			00% to 99%		
Ramp down time (DN)			00% to 99%		
Dither frequency (DF)			08 (80Hz) to 30 (300Hz)		
Display (01)			(00) = input, (01) = output		
1.1.	Pressure switch - Motor stop or motor contact				
	Setting – rising (SP1)		Bar		
	Setting – falling (rP1)		Bar		
	[Digital only] - Output configuration (OU1)		(Hno) = open, (HNc) = closed		
	[Digital only] - Delay for switching on (dS1)		0-50 sec		
	[Digital only] - Delay for switching off (dr1)		0-50 sec		
1.2.	Pressure switch - Release or release contact				
	Setting – rising (SP2)		Bar		
	Setting – falling (rp2)		Bar		
	[Digital only] - Output configuration (OU2)		(Hno) = open, (HNc) = closed		
	Analogue start point (ASP)		Bar		
	Analogue end point (AEP)		Bar		
	[Digital only] - Delay for switching on (dS1)		0-50 sec		
	[Digital only] - Delay for switching off (dr1)		0-50 sec		

FORM FAU-009 REVISION 19-11-06

JOB NO:

Item No on circuit	Item	Setting	Units
	Output polarity (P-n)		s
	Damping for the switching outputs (dAP)		
	Damping for the analogue outputs (dAA)		s
	Setting of the display (diS)		
	Display units (Uni)		
	[Digital only] – Damping outputs		3-500 or 170-1Hz (6 std setting)
	Pressure transducer response chart		
		Output mA	Pressure
55	2. Pressure gauge working		

Oil temperature at time of graph recording		Deg C
Brake release time with discharged system till release switch made		Seconds
Brake application time @ counter pressure setting 1		Seconds
Brake application time @ counter pressure setting 2		Seconds
Decay time between on motor pressure switch from rising setting down to falling setting, (pressure hold time)		Minutes (Minimum – 15)
Recharge time on motor pressure switch from falling setting up to rising setting, (pump run time)		Seconds

Complete brake control unit leak free & clean. The unit shall maintain pressure at the pressure switch limit for a minimum 1 hour without leaks.		
Oil reservoir filled to the correct level		

COMMENTS:

.....
.....
.....
.....
.....
.....

Signed:

Date:

8. RECOMMENDED SYSTEM SPARES & TOOLS

Tools:

Application	Part Number	Description
N/A	N/A	Only standard workshop tools required

Maintenance Spares:

Application	Part Number	Description
Caliper	70A0370-9	Brake pad set VCSD-VR
Caliper	7000358-9	Brake pad set VKSD-VR
Control Unit	800-1021-008	Spin on breather BCU-A/I-75
Control Unit	800-1009-009	Oil filter element BCU-A/I-75 (long)

Critical Spares:

Application	Part Number	Description
Control Unit	800-1003-048	Electric motor 0.75kW 380/3/50
Control Unit	800-1006-100	Pump BCU-A/I-75
Control Unit	800-1012-015	Accumulator BCU-A/I-75 0.75L
Control Unit	800-1013-012	Pressure switch, programmable digital + analogue 24VDC
Control Unit	800-1017-006	Solenoid coil 24VDC

9. DRAWINGS AND SETTINGS

Installation and Maintenance Manual MAU-075-001

1. Brake Control Unit Settings

Settings incl torque and brake time

The BCU-I75-02 settings are as follows to suit the applications:

Maximum Torque: 100% spring force - not adjustable

Caliper qty: One (1) per disc

Friction co-efficient: 0.4 nominally

DRIVE	CALIPER	BRAKING FORCE	CALIPER BALANCE PRESSURE 1mm AIRGAP	CALIPER PRESSURE FULLY LIFTED PISTON	ACCUMULATOR PRECHARGE PRESSURE	RELIEF VALVE PRESSURE	PRESSURE SWITCHES				DECELERATION TORQUE / COUNTER PRESSURE	STOP TIME / COUNTER PRESSURE (18) ITEM 11 [SECONDS]/ [BAR]	BLEED VALVE
							ITEM 13 PS-1 MOTOR OU1-Hnc		ITEM 13 PT-1 4-20mA OU2-I				
							MIN SETTING [BAR]	MAX SETTING [BAR]	MIN SETTING [BAR]	MAX SETTING [BAR]			
		[N]	[BAR]	[BAR]	ITEM 12 [BAR]	ITEM 7 [BAR]					ITEM 18 [Nm]/[BAR]		ITEM 19 [TURNS]
OLC-1	VKSD-VR119	119,000	116	138	116	160	135	145	0	160	60,000/1	32/1	1 OPEN
OLC-1A	VKSD-VR96	96,000	92	116	92	140	115	125	0	140	40,000/8.5	55/8.5	1 OPEN
OLC-2	VCSD-VR62	62,000	63	83	63	110	80	90	0	110	20,000/14.5	32/14.5	1 OPEN

- Note:
- 1) All pressures in bar.
 - 2) Pressures based on maximum caliper spring tolerance.
 - 3) The adjustable orifice, item 19 is opened approximately 1 turn or set so the bleed down rate to 0 bar of residual pressure after accumulator exhaustion is within 10 seconds.
Open too far and the counter pressure setting cannot be held.
Closed too far and the pressure fall from the counter pressure setting after accumulator exhaustion will be greater than 10 seconds

2. Settings for digital pressure switch, item (13)

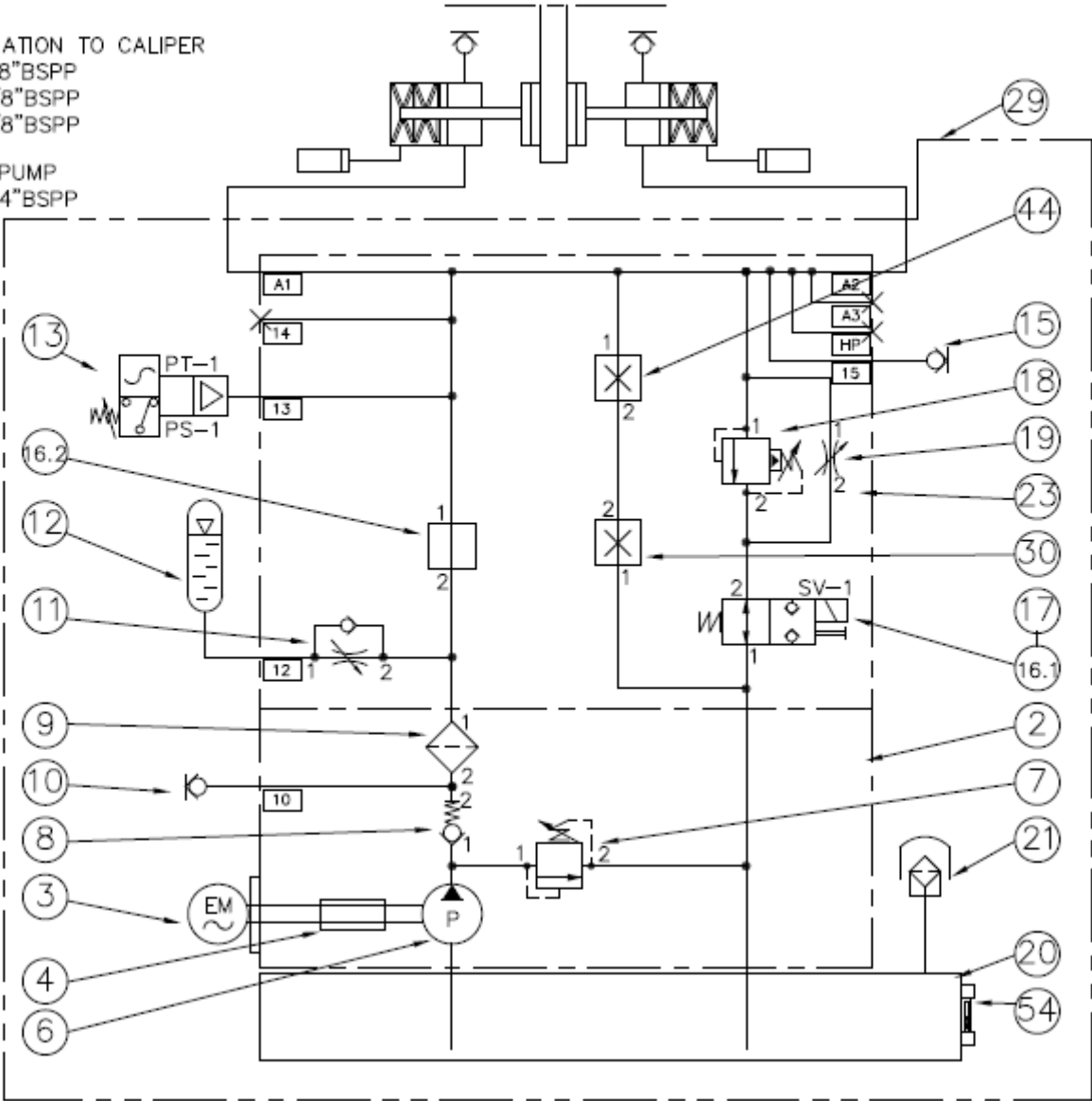
Symbol	Parameter	Setting
<i>SP1</i>	Switching on point output 1	SEE 9.1
<i>rP1</i>	Switching off point output 1	SEE 9.1
<i>ASP</i>	Analogue 4mA setting (min)	SEE 10.1
<i>AEP</i>	Analogue 20mA setting (max)	SEE 10.1
<i>OU1</i>	Configuration of output 1	Hnc
<i>OU2</i>	Configuration of output 2	I
<i>HI LO</i>	Min-Max memory for system pressure	Default -----
<i>COF</i>	Calibration offset	Default - 0%
<i>dS dr</i>	Delay time for switching outputs, on & off	Default - 0
<i>P-n</i>	Switching polarity	PnP
<i>dAP</i>	Damping of switching outputs	6
<i>DiS</i>	Setting of the display	d2
<i>Uni</i>	Display Units	Bar / mbar

3. Hydraulic schematic

SPECIFIC APPLICATION SETTINGS BASED ON MINIMUM SPRING
TOLERANCE
DISC DIAMETER: 1200mm, THICKNESS: 40mm
NOMINAL CO-EFFICIENT OF FRICTION: 0.4

PORTS
TERMINATION TO CALIPER
A1-3/8"BSPP
A2-3/8"BSPP
A3-3/8"BSPP

HAND PUMP
HP-1/4"BSPP

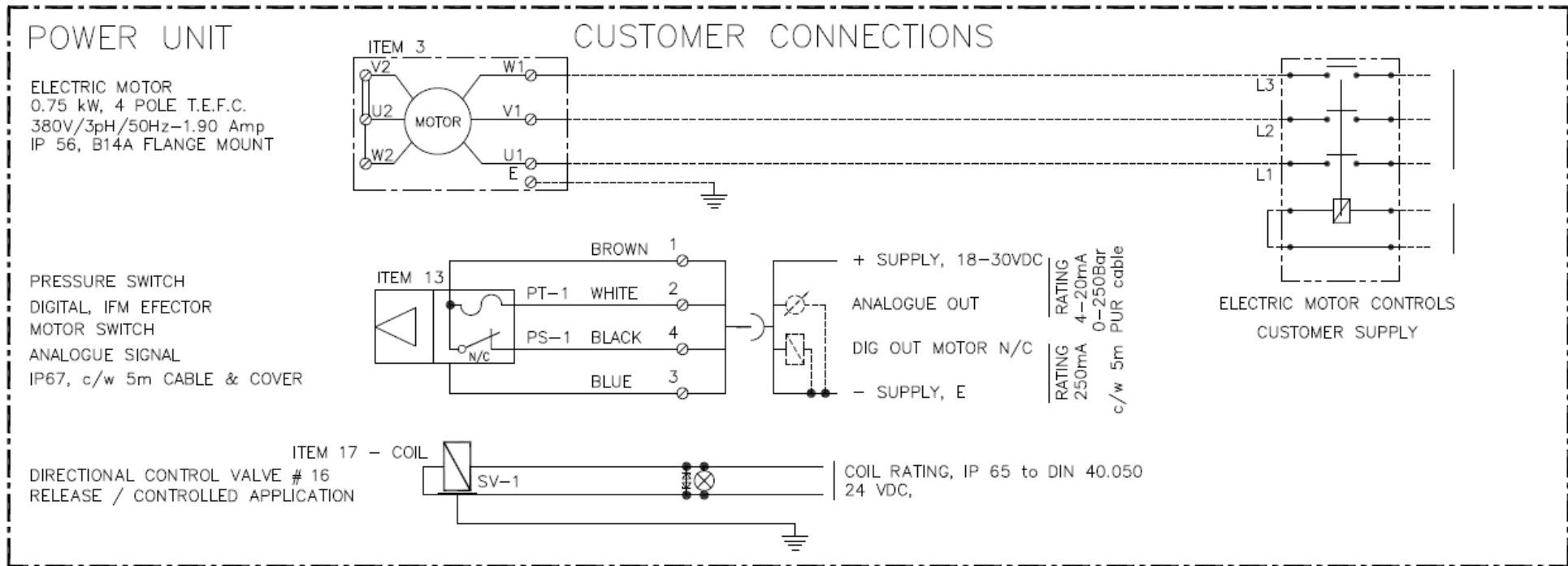


4. Parts list

54	1	SIGHT GLASS	800-1054-010
44	1	CAVITY PLUG (N/C)	800-1007-008
30	1	CAVITY PLUG (N/C)	800-1016-100
29	1	ENCLOSURE	800-1029-101
16.2	1	CAVITY PLUG (N/O)	800-1025-001
23	1	CONTROL MANIFOLD	800-1023-104
21	1	SPIN ON BREATHER 10um	800-1021-008
20	1	RESERVOIR – 5.0L	800-1020-100
19	1	NEEDLE VALVE	800-1019-002
18	1	COUNTER PRESSURE VALVE (OCL-2)	800-1018-002
18	1	COUNTER PRESSURE VALVE (OCL-1&1A)	800-1018-019
17	1	SOLENOID COIL 24V DC	800-1017-006
16.1	1	DIRECTIONAL CONTROL VALVE N/O	800-1016-002
15	1	TEST POINT – POST FILTER	800-1010-003
13	1	PRESSURE SWITCH MOTOR + RELEASE	800-1013-012
12	1	ACCUMULATOR 0.75L	800-1012-015
11	1	FLOW CONTROL VALVE	800-1011-001
10	1	TEST POINT – PRE FILTER	800-1010-003
9	1	FILTER 10um [ELEMENT 800-1009-009]	800-1009-007
8	1	CHECK VALVE	800-1008-001
7	1	MAIN RELIEF VALVE	800-1007-001
6	1	PUMP 1.5L/min	800-1006-100
4	1	MOTOR COUPLING	800-1004-100
3	1	ELECTRIC MOTOR 0.75Kw 380/3/50	800-1003-048
2	1	BELLHOUSING	800-1002-100
ITEM	QTY	DESCRIPTION	PART NUMBER

Items 13 & 17 are available in 24 or 48 VDC, please ensure correct voltage when ordering spares.

5. Electrical schematic



PRESSURE SWITCH (ITEM 13)

MOTOR CONTROL - OUTPUT 1 [PS-1]

ELECTRIC MOTOR OPERATION IS CONTROLLED BY THIS SWITCH. NORMALLY CLOSED AND MOTOR RUNNING UNTIL OPERATING PRESSURE IS REACHED, WHEN SWITCH OPENS AND MOTOR STOPS. ON FALLING PRESSURE, SWITCH CLOSSES AND MOTOR RESTARTS. CLOSING PRESSURE IS SET ABOVE FULL RELEASE PRESSURE OF CALIPER. THE ELECTRIC MOTOR, ITEM 3 MUST NOT RUN CONTINUOUSLY FOR LONGER THAN 3 MINUTES. AT SUCH AN EVENT AN ALARM IS RAISED AND SHUTDOWN SEQUENCE INITIATED. ADDITIONALLY IF THE ELECTRIC MOTOR STARTS MORE THAN 3 OCCASIONS PER MINUTE, AN ALARM IS RAISED. BOTH THE ABOVE LIMITS PROTECT THE POWER UNIT FROM OVERHEATING AND ARE SIGNS OF A PROBLEM I.E LEAKAGE OR OTHER PROBLEM, AND MUST BE INVESTIGATED.

BRAKE POSITION TRANSDUCER - OUTPUT 2 [PT-1]

PROVIDES INDICATION OF CALIPER PRESSURE, EITHER WHEN RELEASED BY POWER UNIT SUPPLIED PRESSURE OR HANDPUMP SUPPLIED PRESSURE (4mA=0BAR, 20mA=PT-1 MAX ON HYDRAULIC SCHEMATIC)

LOGIC SEQUENCE

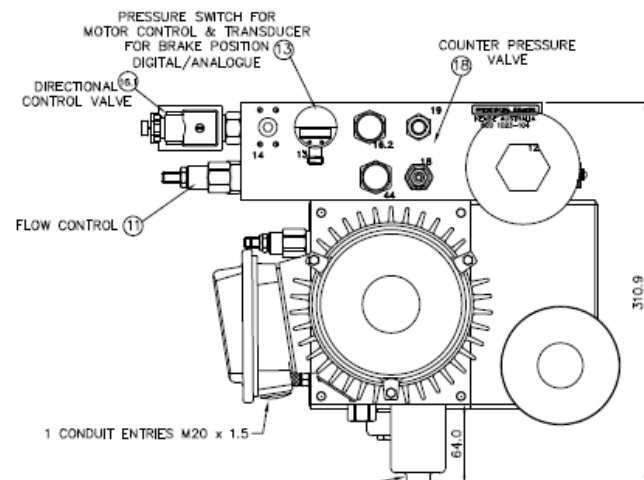
- BRAKE RELEASE IS BY ENERGISING MOTOR ITEM 3 THROUGH SWITCH ITEM 13 OUTPUT 1 [PS-1] & ENERGISING DIRECTIONAL CONTROL VALVE 16 [SV-1] AT THE SAME TIME.
- EMERGENCY OR POWER FAIL BRAKE APPLICATION IS BY DE-ENERGISING MOTOR ITEM 3 & DIRECTIONAL VALVE ITEM 16 [SV-1]. THE BRAKE WILL THEN APPLY AT PRESET CONTROLLED TORQUE SETTING.

NOTES:

- ALL CONNECTIONS ARE BY THE CUSTOMER
- SVENDBORG BRAKES AUSTRALIA SUPPLY COVERS ONLY HARDWARE WITH ITEM NUMBERS, ALL ELSE INCLUDING TERMINAL BOXES, CONTROLS AND WIRING IS THE CUSTOMERS SCOPE OF SUPPLY.
- ITEM NUMBERS LISTED CAN BE FOUND ON HYDRAULIC SCHEMATIC.

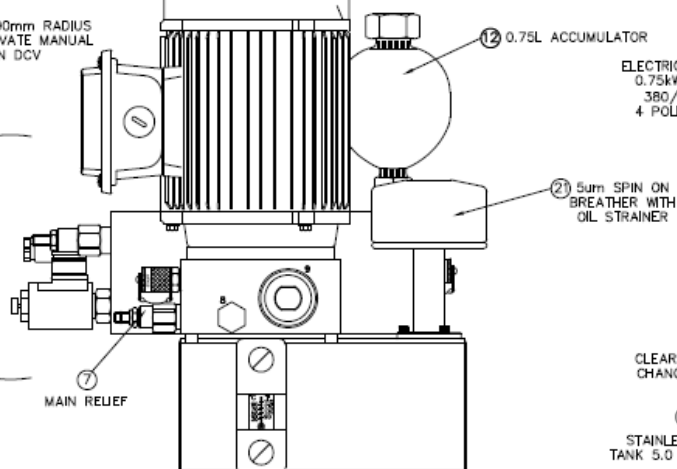
6. Arrangement drawing

DO NOT SCALE - IF IN DOUBT ASK

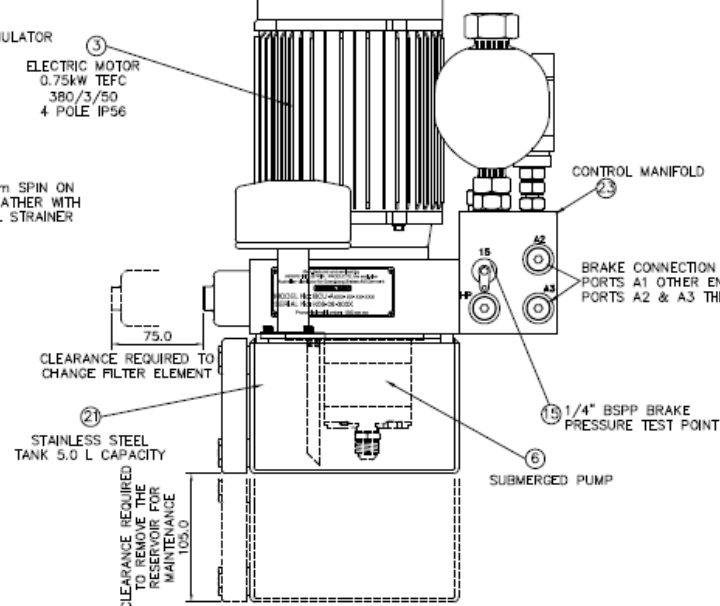


PLAN VIEW

CLEARANCE ON 100mm RADIUS REQUIRED TO ACTIVATE MANUAL OVERRIDE ON DCV

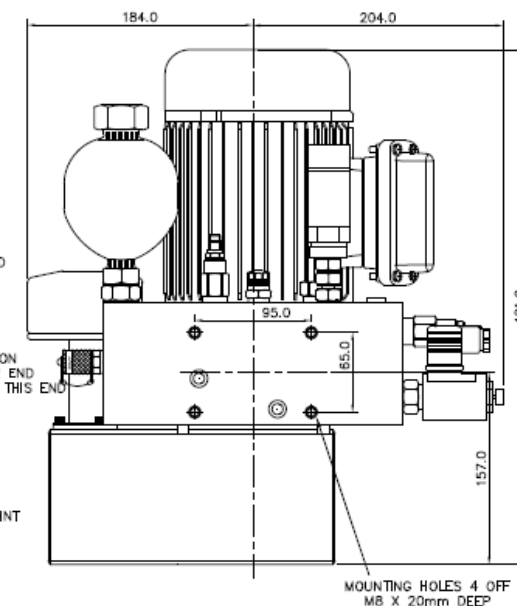


FRONT ELEVATION



SIDE ELEVATION

CLEARANCE ON 120mm RADIUS REQUIRED FROM ACCUMULATOR TOP TO PRE-CHARGE



REAR ELEVATION

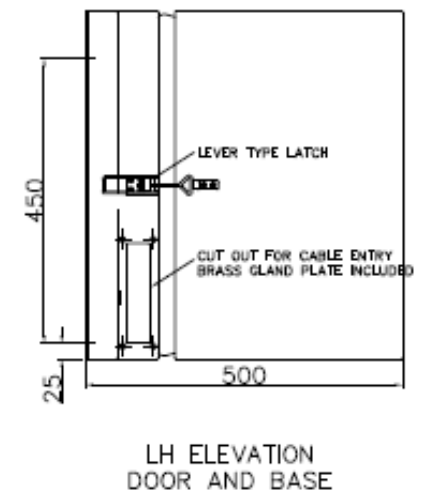
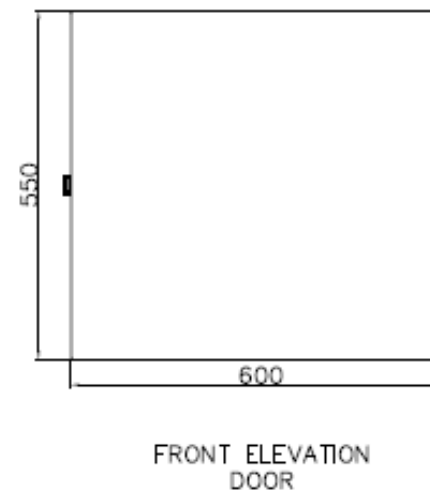
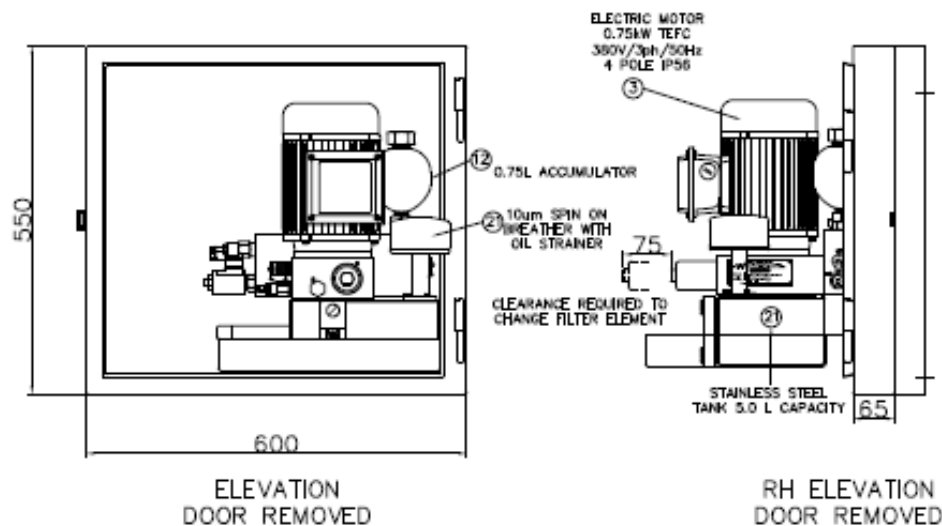
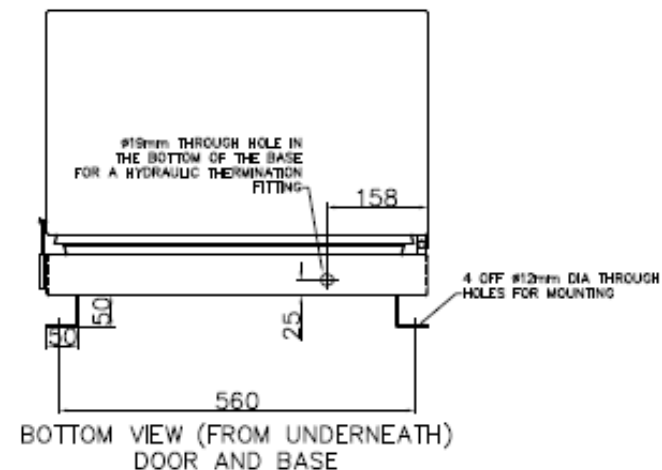
NOTES:-

- 1) REFER TO HYDRAULIC SCHEMATIC FOR ITEM NUMBERS
- 2) WEIGHT: APPROX. 31.5kg DRY
- 3) OIL CAPACITY: APPROXIMATELY 5 LITRES
- 4) MOTOR: 0.75kW 380VAC 3PH 50Hz 1.9Amp T.E.F.C
- 5) CONTROL VOLTAGE: 24VDC
- 6) CONNECTION PORTS: TO CALIPER
A1 - 3/8"BSPP
A2 - 3/8"BSPP
A3 - 3/8"BSPP

7. Enclosure Arrangement drawing

NOTES:

- 1 MATERIAL TO BE STAINLESS STEEL, GRADE 316 2B BRUSHED FINISH.
- 2 PROTECTION RATING IP56
- 3 GENERAL THICKNESS 1.5mm UNLESS NOTED OTHERWISE.
- 4 ENCLOSURE BASE AND BCU SUPPORT BRACKET THICKNESS 2.0mm
- 5 WEIGHT FOR THE BASIC ENCLOSURE APPROX 50Kg DRY.
- 6 THE BASE PRICE ENCLOSURE INCLUDES
 - BASE
 - DOOR
 - POWER UNIT MOUNTING BRACKET
 - GLAND PLATE
 - DRIP TRAY FOR POWER UNIT



10. NOTES